The Consultative Committee for Space Data Systems

Draft Recommendation for
Space Data System Standards

SPACE COMMUNICATIONS SESSION CONTROL

PROPOSED RECOMMENDED STANDARD

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FOREWORD

Through the process of normal evolution, it is expected that expansion, deletion, or modification of this document may occur. This Recommended Standard is therefore subject to CCSDS document management and change control procedures, which are defined in the *Organization and Processes for the Consultative Committee for Space Data Systems* (CCSDS A02.1-Y-4). Current versions of CCSDS documents are maintained at the CCSDS Web site:

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PREFACE

This document is a proposed CCSDS Recommended Standard. Its 'White Book' status indicates that its contents are not stable, and several iterations resulting in substantial technical changes are likely to occur before it is considered to be sufficiently mature to be released for review by the CCSDS Agencies.

Implementers are cautioned **not** to fabricate any final equipment in accordance with this document's technical content.

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1 INTRODUCTION

1.1 PURPOSE

The purpose of this Recommended Standard is to specify the Space Communications Session Control (SC^2) data services operations for use with CCSDS Space Communications Protocols.

1.2 SCOPE

This Recommended Standard defines data services operations, expedited and sequenced-controlled data transfer, and the procedures for establishing and terminating a session between a caller and responder.

This Recommended Standard does not specify a) individual implementations or products, b) implementation of service interfaces within real systems, c) the methods or technologies required to perform the procedures, or d) the management activities required to configure and control the protocol.

1.3 APPLICABILITY

This Recommended Standard applies to the creation of Agency standards and to future data communications over space links between CCSDS Agencies in cross-support situations. It applies also to internal Agency links for which no cross support is required. It includes specification of the services and protocols for inter-Agency cross support. It is neither a specification of, nor a design for, systems that may be implemented for existing or future missions.

The Recommended Standard specified in this document is to be invoked through the normal standards programs of each CCSDS Agency and is applicable to those missions for which cross support based on capabilities described in this Recommended Standard is anticipated. Where mandatory capabilities are clearly indicated in sections of this Recommended Standard, they must be implemented when this document is used as a basis for cross support. Where options are allowed or implied, implementation of these options is subject to specific bilateral cross-support agreements between the Agencies involved.

1.4 RATIONALE

The CCSDS believes it is important to document the rationale underlying the recommendations chosen so that future evaluations of proposed changes or improvements will not lose sight of previous decisions. Concepts and rationale behind the decisions that formed the basis of SC^2 are documented SC^2 Green Book (planned).

1.5 CONVENTIONS AND DEFINITIONS

1.5.1 DEFINITIONS

1.5.1.1 Terms from the Open Systems Interconnection Basic Reference Model

This Recommended Standard makes use of a number of terms defined in the Open Systems Interconnection (OSI) Basic Reference Model (reference [1]). In this Recommended Standard, those terms are used in a generic sense, that is, in the sense that those terms are generally applicable to any of a variety of technologies that provide for the exchange of information between real systems. Those terms are as follows:

a) Data Link Layer;

b) entity;

c) Physical Layer;

d) Session Layer;

e) protocol data unit;

f) real system;

g) service;

h) service data unit.

1.5.1.2 Terms Defined in This Recommended Standard

For the purposes of this Recommended Standard, the following definitions also apply. Many other terms that pertain to specific items are defined in the appropriate sections.

asynchronous data link: A data link consisting of a sequence of variable-length data units that are not necessarily concatenated.

caller and responder: Initiator and receiver, respectively, in a space link session. A **caller transceiver** is the initiator of the link establishment process and manager of negotiation (if required) of the session. A **responder transceiver** typically receives link establishment parameters from the caller. The caller initiates communication between itself and a responder on a prearranged communications channel with predefined controlling parameters. As necessary, the caller and responder may negotiate the controlling parameters for the session (at some level between fully controlled and completely adaptive).

Communication Operations Procedure-P, COP-P: Procedure to enable the delivery of service data units to the receiving end of the layer above, correct and without omission or duplication, and in the same sequential order in which they were received from the layer above at the sending end. The COP-P includes both the Frame

Acceptance and Reporting Mechanism-P (FARM-P) and Frame Operation Procedure-P **(**FOP-P) of the caller and responder unit.

enterprise: A project or undertaking, especially one of some scope, complexity, and risk.

Frame Acceptance and Reporting Mechanism-P, FARM-P: Procedure for returning reports on the status of Transfer Frame acceptance for the Sequence Controlled service carried out within the receiver in the space link.

Frame Operation Procedure-P, FOP-P: Procedure for ordering the output frames for Sequence Controlled service carried out in the transmitter in the space link.

forward link: That portion of a space link in which the caller transmits and the responder receives (typically a command link).

hailing: The persistent activity used to establish a link by a caller to a responder in either full or half duplex. It does not apply to simplex operations.

hailing channel: The forward and return frequency pairs that a caller and responder use to establish physical link communications.

Protocol Link Control Word, PLCW: The protocol data unit for reporting Sequence Controlled service status via the return link from the responder back to the caller.

P-frame: Protocol (transfer) frame. A Version-3 or Version-4 Transfer Frame that contains only self-identified and self-delimited supervisory protocol data units (compare U-frame).

physical channel: The Radio Frequency (RF) channel upon which the stream of symbols is transferred over a space link in a single direction.

reconnect: Process in which the caller attempts to rehail the responder (because of lack of communication progress) during the data services phase within the ongoing session. Upon entering this state, the FARM-P and FOP-P variables of the caller and responder are not reset (in particular, their frame sequence counters).

resynchronization: (COP-P) Process in which a sequence count anomaly is detected by the caller and the caller forces the responder to readjust its Sequence Controlled frame numbers via the SET V(R) activity.

return link: That portion of a space link in which the responder transmits and the caller receives (typically a telemetry link).

session: A dialog between two or more communicating link transceivers. A session consists of three distinct operational phases: session establishment, data services (which may include resynchronization and/or reconnect subphases), and session termination. Session termination may be coordinated (through the exchange of no-more-data-to-

send directives), or if communication is lost (inability to resynchronize or reconnect), the transceivers should eventually independently conclude the dialog is over.

space link: A communications link between transmitting and receiving entities, at least one of which is in space.

Supervisory Protocol Data Unit, SPDU: Protocol Data Unit (PDU) used by the local transceiver either to control or to report status to the remote partnered transceiver. Consists of one or more directives, reports, or PLCWs.

synchronous channel: A data channel where the symbol data are continuously modulated onto the channel at a fixed data rate. When no PLTU is available for transmission, Idle data is transmitted to maintain the continuous symbol stream.

Sync-Marked Transfer Frame, SMTF: The data unit that consists of the ASM and the Transfer Frame

Transfer Frame: The protocol data unit of a Space Data Link Protocol. In this document it can be either a Version-3 or a Version-4 Transfer Frame.

U-frame: User (transfer) frame. Any CCSDS version Transfer Frame that contains user data information (compare P-frame).

vehicle controller: The entity (e.g., spacecraft control computer) that receives the notifications defined in annex G and potentially acts upon them.

1.5.2 NOMENCLATURE

1.5.2.1 NORMATIVE TEXT

The following conventions apply for the normative specifications in this Recommended Standard:

a) the words ‘shall’ and ‘must’ imply a binding and verifiable specification;

b) the word ‘should’ implies an optional, but desirable, specification;

c) the word ‘may’ implies an optional specification;

d) the words ‘is’, ‘are’, and ‘will’ imply statements of fact.

NOTE – These conventions do not imply constraints on diction in text that is clearly informative in nature.

1.5.2.2 INFORMATIVE TEXT

In the normative sections of this document, informative text is set off from the normative specifications either in notes or under one of the following subsection headings:

- Overview;
- Background;
- Rationale;
- Discussion.

1.5.3 CONVENTIONS

In this document, the following convention is used to identify each bit in an *N*-bit field. The first bit in the field to be transmitted (i.e., the most left-justified when drawing a figure) is defined to be 'Bit 0', the following bit is defined to be 'Bit 1', and so on up to 'Bit *N*-1'. When the field is used to express a binary value (such as a counter), the Most Significant Bit (MSB) is the first transmitted bit of the field, that is, 'Bit 0', as shown in figure 1-1.

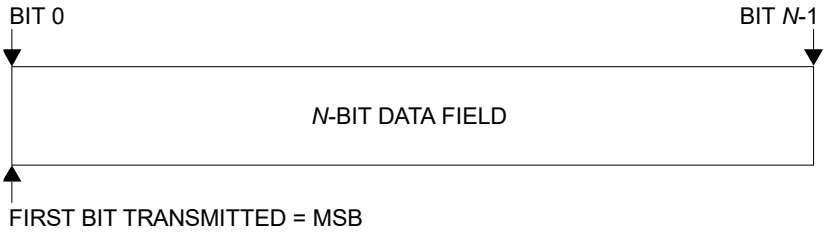


Figure 1-1: Bit Numbering Convention

In accordance with standard data-communications practice, data fields are often grouped into 8-bit 'words' that conform to the above convention. Throughout this Recommended Standard, such an 8-bit word is called an 'octet'.

The numbering for octets within a data structure begins with zero. Octet zero is the first octet to be transmitted.

By CCSDS convention, all 'spare' bits are permanently set to value 'zero'.

1.6 REFERENCES

The following publications contain provisions which, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All publications are subject to revision, and users of this document are encouraged to investigate the possibility of applying the most recent editions of the

publications indicated below. The CCSDS Secretariat maintains a register of currently valid CCSDS publications.

[1] *Information Technology—Open Systems Interconnection—Basic Reference Model: The Basic Model*. 2nd ed. ISO/IEC 7498-1:1994. Geneva: ISO, 1994.

[2] *TM Synchronization and Channel Coding*. Issue 5. Recommendation for Space Data System Standards (Blue Book), CCSDS 131.0-B-5. Washington, D.C.: CCSDS, September 2023.

[3] *Time Code Formats*. Issue 4. Recommendation for Space Data System Standards (Blue Book), CCSDS 301.0-B-4. Washington, D.C.: CCSDS, November 2010.

[4] *Proximity-1 Space Link Protocol—Coding and Synchronization Sublayer*. Issue 3. Recommendation for Space Data System Standards (Blue Book), CCSDS 211.2-B-3. Washington, D.C.: CCSDS, October 2019.

[5] *Proximity-1 Space Link Protocol—Physical Layer*. Issue 4. Recommendation for Space Data System Standards (Blue Book), CCSDS 211.1-B-4. Washington, D.C.: CCSDS, December 2013.

[6] *Proximity-1 Space Link Protocol—Data Link Layer*. Issue 6.0. Proposed Draft Recommendation for Space Data System Standards (Proposed Pink Book), CCSDS 211.0-P-6.0. Washington, D.C.: CCSDS, Under development.

[7] *Unified Space Data Link Protocol*. Issue 3. Recommendation for Space Data System Standards (Blue Book), CCSDS 732.1-B-3. Washington, D.C.: CCSDS, June 2024.

[8] *TM Synchronization and Channel Coding*. Issue 5. Recommendation for Space Data System Standards (Blue Book), CCSDS 131.0-B-5. Washington, D.C.: CCSDS, September 2023.

[9] *Pseudo-Noise (PN) Ranging Systems*. Issue 3. Recommendation for Space Data System Standards (Blue Book), CCSDS 414.1-B-3. Washington, D.C.: CCSDS, January 2022.

2 OVERVIEW

2.1 SUPERVISORY PROTOCOL DATA UNITS

Supervisory Protocol Data Units (SPDUs) defined in section 3 are used by the local transceiver either for local control within the transceiver or for reporting status to and controlling the remote transceiver.

SPDUs carry protocol directives and reports. Either a Version-3 or Version-4 transfer frame with SPDUs in its data field is called a Protocol frame (P-frame).

Protocol directives are used for configuring and controlling the protocol processor at the receiving node; for the establishment, maintenance, and termination of a communications session; and for the transfer of time-correlation and synchronization data. Protocol reports are used for reporting the configuration and status of the transmitting node.

SPDUs are of either fixed or variable length based upon the value of the SPDU format ID. Currently, there are only two fixed-length SPDU defined, for a 16 bit and 32 bit PLCW. Variable-length SPDUs provide the capability for concatenating and multiplexing protocol objects, that is, directives and status reports.

SPDUs are transmitted by setting the QoS Indicator Flag in the Version-3 transfer frames to "Expedited" or by setting the Bypass/Sequence Control Flag in Version-4 transfer frames to "Expedited QoS".

SPDUs are all self-identifying and self-delimiting. Only variable-length SPDUs further decompose into specific types of supervisory directives or reports. (See annexes B–E for detailed specification of variable-length SPDUs.)

2.2 DATA SERVICES OPERATIONS

2.2.1 GENERAL

Data Services Operations are detailed in section 5 in a comprehensive set of session state tables, state variable descriptions, and state diagrams for data services operations. Table 2-1 below provides a roadmap to help navigate this section. The states are dependent on four state-controlling variables: MODE, DUPLEX, TRANSMIT (T), and SUB-STATE (SS). The Receive and Send State Descriptions consist of the values *off*, *on*, *synchronous* (channel), and *asynchronous* (channel). Currently, Proximity-1 is defined solely for asynchronous data links. (See 1.5.1.2 for these definitions.)

Data Services Operations control the order of transfer of the user data (including user-supplied directives) that are to be transmitted within the session. It provides the following two qualities of service:

a) Expedited service ensures transmission without errors of Expedited frame data in the order received;

b) Sequence Controlled service guarantees that data within a communication session are delivered in order without errors, gaps, or duplications.

The guarantee of reliable data delivery by the Sequence Controlled service is constrained to a single communication session without COP-P resynchronization. Sessions in which COP-P resynchronization occurs may result in duplicate or lost data because of factors outside the scope of the protocol.

The mechanisms provided will not eliminate duplicate data associated with the transition between the end of one session and the beginning of the next. Elimination of this problem is left to the controlling data system.

2.2.2 FUNCTIONALITY

2.2.2.1 Send Side Functionality

The send side:

a) runs the FOP-P process;

b) processes received PLCWs;

c) acknowledges delivery of complete SDUs to the I/O Sublayer;

d) provides frame accountability to the I/O Sublayer;

e) accepts either an Expedited or a Sequence Controlled frame from the I/O Sublayer.

2.2.3 RECEIVE SIDE FUNCTIONALITY

The receive side:

a) runs the FARM-P process;

b) accepts U-frames from the Frame Sublayer.

Table 2-1: Data Services Operations Roadmap

Operations	Applicable State Tables	Applicable State Transition Tables	Applicable State Transition Diagram
Full Duplex	Tables 5-1, 5-2	Session Establishment and Data Services: table 5-6 COMM_CHANGE: table 5-7 Session Termination: table 5-8	Full Duplex Operations: figure 5-1
Half Duplex	Tables 5-1, 5-3	Session Establishment and Data Services: table 5-9 COMM_CHANGE: table 5-10 Session Termination: table 5-11	Half Duplex Operations: figure 5-2
Simplex	Tables 5-1, 5-4	Simplex State Transition Table: table 5-12	Simplex Operations: figure 5-3

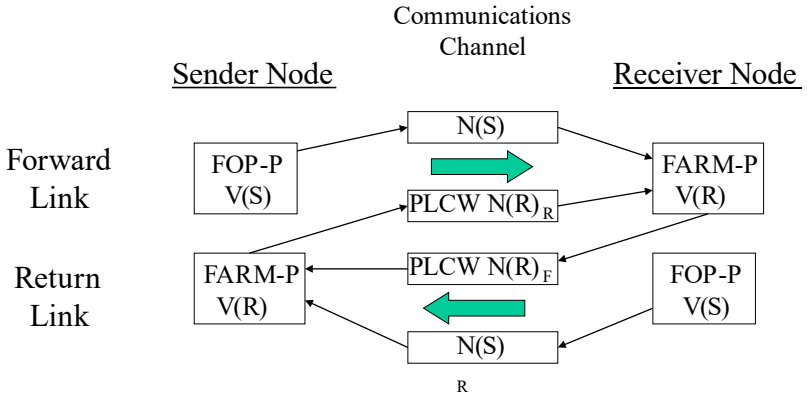
2.3 COMMUNICATIONS OPERATIONS PROCEDURE FOR PROXIMITY LINKS

2.3.1 GENERAL

The COP-P protocol is used with one Sender Node, one Receiver Node, and a direct link between them. The sender delivers frames to the receiver. The receiver accepts all valid Expedited frames, and valid Sequence Controlled frames that arrive in sequence. The receiver provides feedback to the sender in the form of a PLCW. The sender uses this feedback to retransmit Sequence Controlled frames when necessary. Expedited frames are never retransmitted by the COP-P protocol.

Concurrent bi-directional data transfer is a capability. In this case, each node has both sender and receiver functionality as shown in figure 2-1, COP-P Process.

COP-P provides two qualities of service (Sequence Controlled and Expedited) that determine how reliably data from the sending user are delivered to the receiving user. On the send side, COP-P defines the FOP-P. On the receive side, COP-P defines the FARM-P.



Notes:

1 The User data frames (U-frames) in the forward link contain the frame sequence number $N(S)_F$. The U-frames in the return link contain the frame sequence number $N(S)_R$.

2 The PLCW Supervisory protocol frames (P-frames) in the forward link are reporting return link progress and contain the frame sequence number $N(R)_R$. The P-frames in the return link are reporting forward link progress and contain the frame sequence number, $N(R)_F$.

Figure 2-1: COP-P Process

Both the Sender Node and the Receiver Node contain two types of procedures: the send side procedures (i.e., the FOP-P) and the receive side procedures (i.e., the FARM-P).

The FOP-P drives the Expedited and Sequence Controlled services. It is responsible for ordering and multiplexing the user-supplied data and maintaining synchronization with the FARM-P. It initiates a retransmission when required. If a valid PLCW is not received in a reasonable time period (defined by the MIB parameter, Synch_Timeout—cf. annex F), the Sender Node's FOP-P notifies the local controller that it is not synchronized with the Receiver Node's FARM-P. If the MIB parameter Resync_Local equals *false* (cf. annex F), it is the responsibility of the local controller to decide how synchronization will be re-established. Otherwise, the Sender Node's FOP-P forces a resynchronization by executing the SET V(R) activity.

The FARM-P is data-driven; that is, it simply reacts to what it receives from the FOP-P and provides appropriate feedback via the PLCW. The FARM-P utilizes the services of the C&S Sublayer to verify that the frame was received error free. The FARM-P depends upon the Frame Sublayer to verify that the frame is a valid Version-3 Transfer Frame and that it should be accepted for processing by Data Services. When Version-4 Transfer Frames are used, the applicability of the FOP-P and FARM-P is specified in reference [7].

The FOP-P and FARM-P procedures control both Expedited and Sequence Controlled qualities of service.

FOP-P provides frame level accounting, that is, V(S) and VE(S) to the I/O Sublayer for every Sequence Controlled and Expedited frame it numbers.

2.3.2 SERVICE QUALITIES

2.3.2.1 General

The COP-P procedure provides two qualities of service (Sequence Controlled and Expedited) that determine how reliably SDUs supplied by the sending user are delivered to the receiving user. The procedure consists of a FOP-P, used on the sending side of the service, and a FARM-P, used on the receiving side of the service.

Each data transfer with one of the data transfer services has a service quality associated with it. Packetized data units that are larger than the maximum frame size in asynchronous frames can be transferred only by using the segmentation process, utilizing either the Sequence Controlled service or the Expedited service.

2.3.2.2 Sequence Controlled Service

The Sequence Controlled service ensures that data are reliably transferred across the space link and delivered in order, without gaps, errors, or duplications within a single communication session without COP-P resynchronization during the session. This service is based on a 'go-back-n' type of Automatic Repeat Queuing (ARQ) procedure that utilizes sequence-control mechanisms of both sending and receiving ends and a standard report (PLCW) returned from the receiving end to the sending end.

SDUs supplied by a sending user for transfer with the Sequence Controlled quality of service are inserted into Transfer Frames as required and transmitted on a physical channel in the order in which they are presented. SDUs are passed to the receiving user via the identified port. The retransmission mechanism ensures with a high probability of success that:

a) no SDU is lost;

b) no SDU is duplicated;

c) no SDU is delivered out of sequence.

2.3.2.3 Expedited Service

The Expedited service is nominally used with upper-layer protocols that provide their own retransmission features, or in exceptional operational circumstances such as during spacecraft recovery operations.

Expedited SDUs supplied by the sending user are transmitted without ARQ. At the sending end, Expedited SDUs are transmitted on specified physical channels independently of the Sequence Controlled SDUs waiting to be transmitted on the same physical channel. At the receiving end, the SDUs are passed to the receiving user via the identified port. It should be noted that Expedited SDUs may be sent once or multiple times, but they are not sent again as

a result of a request for retransmission. If such a request occurs, it is performed outside the purview of the protocol.

There is no guarantee that all Expedited SDUs will be delivered to the receiving user. If the packet delivery service is used to transfer packets using segmentation in the Expedited service, then the service delivers only complete packets to the user.

NOTE – In Expedited service, the capability is provided to deliver portions of user-defined data units that are greater than the maximum frame size allowed for the link.

3 SUPERVISORY PROTOCOL DATA UNITS

3.1 GENERAL

Constraints on the use of the transfer frames over the Proximity-1 Data Link:

a) SPDUs (P-frames) shall only be transmited over Version-3 or Version-4 frames.

b) AOS (Version-2) transfer frames can only be transmitted as U-frames.

c) Transfer frames in general with different transfer frame version numbers cannot be multiplexed across the Proximity-1 data link, once the link has been established.

3.1.1 Fixed-length SPDUs shall consist of the following fields positioned contiguously in the following sequence:

a) SPDU Header (2 bits) consisting of:

   1) SPDU Format ID (1 bit),

   2) SPDU Type Identifier (1 bit);

b) SPDU Data field (14 or 30 bits) consisting of either a 16 bit or a 32 bit fixed-length PLCW.

3.1.2 Variable-length SPDUs shall consist of the following fields positioned contiguously in the following sequence:

a) SPDU Header (1 octet) consisting of:

   1) SPDU Format ID (1 bit),

   2) SPDU Type Identifier (3 bits),

   3) Data Field Length (4 bits) (this represents the actual number of octets in the data field of the SPDU);

NOTE – Data Field Length is not a ‘length minus one’ field.

b) Supervisory Data field (variable length, i.e., 0 to 15 octets) consisting of one or more supervisory directives or status reports of the same SPDU type.

3.1.3 Implementations shall include a Directive Decoder function for processing supervisory protocol directives defined in this section and in annexes B–E.

NOTE – The Directive Decoder is a function that decodes supervisory protocol directives received either from the local link controller or from the remote vehicle controller. The directive decoder processes the received directives, setting the configuration (state and parameters) of both the Physical and Data Link Layers.

3.2 FIXED-LENGTH SPDU

3.2.1 GENERAL

A '1' in the SPDU Format ID field shall identify a fixed-length SPDU. This format provides for only two fixed SPDUs (see table 3-1), which shall be differentiated by the SPDU Type Identifier field:

a) a '0' in the SPDU Type Identifier field identifies the SPDU as a type F1, 16 bit PLCW;

b) a '1' in the SPDU Type Identifier field identifies the SPDU as a Type F2, 32 bit PLCW.

Table 3-1: Fixed-Length Supervisory Protocol Data Unit

Fixed-Length SPDU (16 or 32 bits)	**SPDU Header (2 bits)**		**SPDU Data (14 or 30 bits)**
	SPDU Format ID (Bit 0)	SPDU Type Identifier (Bit 1)	(Contains 1 protocol object, i.e., directive or report or PLCW) (Bits 2 through 15 or 31)
Type F1	'1'	'0'	14 bit Fixed Length PLCW Data (see 3.2.2.1)
Type F2	'1'	'1'	30 bit Fixed Length PLCW Data

3.2.2 FIXED-LENGTH SPDU DIRECTIVES

3.2.2.1 16-Bit PLCW

3.2.2.1.1 General

3.2.2.1.1.1 A '1' in the SPDU Format ID field and a 0' in the SPDU Type Identifier field shall identify the SPDU as a 16-bit PLCW.

NOTE – The 16-bit PLCW is currently used for proximity operations at Mars, which has currently lower data rate applications.

3.2.2.1.1.2 The PLCW shall consist of seven fields positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0—see figure 3-1):

a) Report Value (8 bits);

b) Expedited Frame Counter (3 bits);

c) Reserved Spare (1 bit);

d) PCID (1 bit);

e) Retransmit Flag (1 bit);

f) SPDU Type Identifier (1 bit);

g) SPDU Format ID (1 bit).

3.2.2.1.1.3 The PLCW shall be transmitted using the Expedited QoS.

NOTE – The structural components of the PLCW are shown in figure 3-1.

Bit 0 Bit 15

SPDU Header		SPDU Data Field				
SPDU Format ID	SPDU Type Identifier	Retrans-mit Flag	PCID	Reserved Spare	Expedited Frame Counter	Report Value (FSN)
1 bit	1 bit	1 bit	1 bit	1 bit	3 bits	8 bits

Figure 3-1: Type F1 Proximity Link Control Word Fields

3.2.2.1.2 Report Value

3.2.2.1.2.1 Bits 8–15 of the PLCW shall contain the Report Value.

3.2.2.1.2.2 The Report Value field shall contain the value of V(R) (see 6.3.2).

3.2.2.1.2.3 Separate Report Values shall be reported for each physical channel independent of the I/O port.

3.2.2.1.3 Expedited Frame Counter

3.2.2.1.3.1 Bits 5–7 of the PLCW shall contain the EXPEDITED_FRAME_COUNTER.

3.2.2.1.3.2 The EXPEDITED_FRAME_COUNTER shall provide a modulo-8 counter indicating that Expedited frames have been received.

3.2.2.1.4 Reserved Spare

3.2.2.1.4.1 Bit 4 of the PLCW shall contain a Reserved Spare bit.

3.2.2.1.4.2 The Reserved Spare bit field shall be set to '0'.

3.2.2.1.5 Physical Channel Identification

3.2.2.1.5.1 Bit 3 of the PLCW shall contain the PCID field.

3.2.2.1.5.2 The 1-bit PCID field shall contain the PCID of the physical channel with which this report is associated. (See 5.1.2.10, 'RECEIVING_PCID_BUFFER'.)

NOTE – Each PCID in use has its own PLCW reporting activated.

3.2.2.1.6 PLCW Retransmit Flag

3.2.2.1.6.1 Bit 2 of the PLCW shall contain the PLCW Retransmit Flag.

3.2.2.1.6.2 A setting of '0' in the PLCW Retransmit Flag shall indicate that there are no outstanding frame rejections in the sequence received so far, and thus retransmissions are not required.

3.2.2.1.6.3 A setting of '1' in the PLCW Retransmit Flag shall indicate that a received frame left a frame sequence number gap and that a retransmission of the expected frame is required.

3.2.2.2 32-Bit PLCW

3.2.2.2.1 A '1' in the SPDU Format ID field and a 1' in the SPDU Type Identifier field shall identify the SPDU as a 32-bit PLCW.

NOTE – The 32-bit PLCW is currently envisioned for proximity operations at Lunar, which is envisioned to have higher data rate applications than Mars.

3.2.2.2.2 The 32-bit PLCW including the SPDU header shall consist of seven fields positioned contiguously in the following sequence (described from least significant bit, bit 15, to most significant bit, bit 0—see figure 3-2):

a) Report Value (16 bits);

b) Expedited Frame Counter (3 bits);

c) PCID (1 bit);

d) Retransmit Flag (1 bit);

e) Reserved Spares (9 bits);

f) SPDU Type Identifier (1 bit);

g) SPDU Format ID (1 bit).

NOTE – This PLCW shall be transmitted using the Expedited QoS.

Bit 0 Bit 31

SPDU Header		SPDU Data Field				
SPDU Format ID	SPDU Type Identifier	Reserved Spares	Retransmit Flag	PCID	Expedited Frame Counter	Report Value (FSN)
1 bit	1 bit	9 bits	1 bit	1 bit	3 bits	16 bits

Figure 3-2: Type F2 Proximity Link Control Word Fields

3.3 VARIABLE-LENGTH SPDU

3.3.1 GENERAL

A '0' in the SPDU Format ID field shall identify a variable-length SPDU data field, which may contain from 0 to 15 octets of supervisory data.

NOTE – This form of SPDU uses bits 1 through 3 of the SPDU header to identify one of eight possible SPDU types, summarized in table 3-2.

Table 3-2: Variable-Length Supervisory Protocol Data Unit

Variable-Length SPDU	**SPDU Header (1 octet, fixed)**			**SPDU Data Field (0–15 octets)**
	Format ID (Bit 0)	SPDU Type Identifier (Bits 1,2,3)	Length of SPDU Data Field (Bits 4,5,6,7)	(Contains one or more protocol objects, i.e., directives, reports)
Type 1	'0'	'000'	Length in Octets	Directives/Reports for UHF operations
Type 2	'0'	'001'	"	TIME DISTRIBUTION PDU
Type 3	'0'	'010'	"	Status Reports
Type 4	'0'	'011'	"	First Generation Lunar Use
Type 5	'0'	'100'	"	Second Generation Lunar Use
Type 6	'0'	'101'	"	Reserved for CCSDS Use
Type 7	'0'	'110'	"	Reserved for CCSDS Use
Type 8	'0'	'111'	"	Reserved for CCSDS Use
NOTE – Directives and Reports can be multiplexed within the SPDU Data Field.				

3.3.2 TYPE 1 SPDU—FIRST GENERATION MARS USE (UHF BAND)

An SPDU Type Identifier equal to '000' shall identify a Type 1 SPDU with a data field containing from 0 to 15 octets containing first-generation UHF directives.

NOTE – The formats of the Type 1 SPDU and its directives are defined in annex B.

3.3.3 TYPE 2 SPDU—TIME DISTRIBUTION DIRECTIVES

3.3.3.1 An SPDU Type Identifier equal to '001' shall identify a Type 2 SPDU with a data field containing from 1 to 15 octets of TIME DISTRIBUTION supervisory data.

3.3.3.2 Octet 0 of the data field shall contain the TIME DISTRIBUTION directive type, followed by the actual time field value (1 to 14 octets).

NOTE – The format of the Type 2 SPDU is defined in annex C.

3.3.4 TYPE 3 SPDU—STATUS REPORT DIRECTIVES

An SPDU Type Identifier equal to '010' shall identify a Type 3 SPDU with a data field containing from 0 to 15 octets of Status Report information.

NOTES

1 The format of these reports is enterprise specific and is left up to the implementation.

2 Provision is made in the protocol to identify when a status report is required (NEED_STATUS_REPORT) and when a status report is requested (see Type 1 SPDU Report Request Directive, B6).

3.3.5 TYPE 4 SPDU—FIRST GENERATION LUNAR USE

An SPDU Type Identifier equal to '011' shall identify a Type 4 SPDU with a data field containing from 0 to 15 octets.

NOTE – The format of the Type 4 SPDU is defined in annex D.

3.3.6 TYPE 5 SPDU—SECOND GENERATION LUNAR USE

An SPDU Type Identifier equal to '100' shall identify a Type 5 SPDU with a data field containing from 0 to 15 octets containing the second generation Lunar directives.

NOTE – The formats of the Type 5 SPDU and its directives are defined in annex E.

3.3.7 TYPE 6 THROUGH 8 SPDUS—RESERVED FOR CCSDS USE

3.3.7.1 An SPDU Type Identifier equal to '101', '110', and '111' shall identify, respectively, a Type 6, a Type 7, and a Type 8 SPDU with a data field containing from 0 to 15 octets.

3.3.7.2 Type 6 through Type 8 SPDUs are reserved for CCSDS use.

4 PERSISTENCE

4.1 OVERVIEW

A persistent activity is a process for ensuring reliable communication between a caller and a responder using the Expedited QoS while transmitting from the MAC queue (reference [6]). Because of the potential for frame loss due to corruption across the space link, these control activities require a persistence process to ensure that supervisory protocol directives are received and acted upon correctly.

Persistence activities can be linked in series to accomplish a task, but persistence applies to only one activity at a time. Each persistent activity is named and consists of one or more actions (e.g., issuing selective directives), followed by a WAITING_PERIOD during which a specific RESPONSE is expected. The protocol defines three persistent activities: hailing (i.e., session establishment; see 5.1.3 and tables 5-7 and 5-10), COMM_CHANGE (see 5.1.3 and tables 5-8 and 5-11), and resynchronization (see 6.2.3.2 and 6.2.3.3).

4.2 PERSISTENCE ACTIVITY PARAMETERS

The parameters associated with a persistent activity are described below; their values vary based on the activity to be performed and are defined per activity in the MIB (see annex F):

a) ACTIVITY: the name of the persistent activity;

b) WAITING_PERIOD: the amount of time specified for the RESPONSE to be received before the process declares that the activity is to be either repeated or aborted;

c) RESPONSE: the acknowledgement by the responder that the persistent activity has been accepted;

d) NOTIFICATION: the message provided to the local vehicle controller, for example, spacecraft C&DH by the caller and responder upon success or failure of the persistent activity;

e) LIFETIME: the time period during which the persistent activity will be repeated until the expected RESPONSE is detected.

NOTE – The LIFETIME can be locally defined in terms of a duration or a maximum number of times this activity will be repeated before the activity is aborted.

4.3 PERSISTENT ACTIVITY PROCESS

4.3.1 Upon initiation of a persistent activity, the PERSISTENCE signal shall be set to *true*.

4.3.2 While PERSISTENCE is *true* the Frame Sublayer shall select frames only from the MAC queue for output (see reference [6]).

4.3.3 The success or failure of the activity shall be determined by the detection of the expected RESPONSE within the activity's LIFETIME.

4.3.4 If the RESPONSE is not detected within the activity's LIFETIME, the activity shall be deemed failed and aborted.

4.3.5 NOTIFICATION of the activity's success or failure shall be communicated back to the vehicle controller, at which time the PERSISTENCE signal shall be set to *false*.

5 DATA SERVICES OPERATIONS

5.1 STATE TABLES

Table 5-1: Session States Independent of the DUPLEX Variable

State Name	State Description	Receive State Desc.	Send State Desc.	MODE	T	SS	Description
S1	Inactive	*off*	*off*	*inactive*	*off*	0	The only actions that are permitted in state S1 are those in response to local directives. In this state, the Data Services operational variables and MIB parameter values can be modified and their status read via local directives from the local controller. When the protocol enters this state, the variables identified in table 5-5 are initialized. The Local SET INITIALIZE MODE directive forces entry to this state.
S2	Waiting for Hail	*on*	*off*	*connecting-L*	*off*	0	In this state, receiving operations are enabled. FARM-P operations are enabled but only for processing received supervisory directives; that is, Transfer Frame Version-3 header PDU TYPE ID or Version-4 header PROTOCOL CONTROL COMMAND FLAG = '1'. It should be noted that only receiving operations are enabled so that transmission is not permitted.
S80	Reconnect	*on*	*off*	*active*	off	0	In this state, the caller attempts to maintain the current session by reconnecting with the responder as follows: the caller's transmitter is turned off (for Drop_Carrier_Duration) to force the responder to drop carrier lock and transition into State S2: Waiting for Hail. The FARM-P and FOP-P variables of the caller and responder are not reset.

Table 5-2: States When DUPLEX = Full

State Name	State Description	Receive State Desc.	Send State Desc.	MODE	T	SS	Description
S31	Start Hail Action	*on*	*async*	*connecting-T*	*on*	1	In this state, the Hail activity starts with the radiation of the carrier signal.
S32	Send Hail Acquisition	*on*	*async*	*connecting-T*	*on*	2	In this state, the idle pattern is radiated to achieve symbol lock with the hailed remote unit.
S33	Send Hail Directives	*on*	*async*	*connecting-T*	*on*	3	In this state, the Hail Directives: (if present) (SET_TRANSMITTER_PARAMETERS, SET_RECEIVER_PARAMETERS) or SET_PL EXTENSIONS (TX and RX) or LINK ESTABLISHMENT & CONTROL (TX and RX) are radiated in one Transfer Frame to initiate a session with the hailed remote unit, that is, the responder.
S34	Send Hail Tail	*on*	*async*	*connecting-T*	*on*	4	In this state, the idle pattern is radiated to allow the HAIL directives to be received and processed through the decoding chain of the responder.
S35	Wait for Hail Response	*on*	*async*	*connecting-T*	*off*	5	In this state, the transmitter is turned off and the receiver awaits a response from the hailed remote unit.
S41	Radiate Carrier Only	*on*	*sync*	*active*	*on*	1	In this state, the receiver is *on* and ready to process all received data while the transmission process is started with carrier radiation only.
S42	Radiate Acquisition Idle	*on*	*sync*	*active*	*on*	2	In this state, the receiver is *on* and processing all received data while the transmission process is trying to achieve bit lock with a potential partnered transceiver, that is, the caller transceiver.
S40	Data services	*on*	*sync*	*active*	*on*	0	In this state, data transfer services controlled by the COP-P protocol are conducted with a partnered transceiver.

State Name	State Description	Receive State Desc.	Send State Desc.	MODE	T	SS	Description
S48	COMM_CHANGE	*on*	*sync*	*active*	*on*	6	This state is involved with the protocol actions required to perform a data rate or frequency change with a partnered transceiver. This state contains numerous sub-states whose transitions are described in table 5-7.
S45	Terminating Tail	*on*	*sync*	*active*	*on*	4	In this state, the receiver is *on* and processing all received data while the transmission process is terminating. (See table 5-8.)

Table 5-3: States When DUPLEX = Half

State Name	State Description	Receive State Desc.	Send State Desc.	MODE	T	SS	Description
S11	Start Hail Action	*off*	*async*	*connecting-T*	*on*	1	In this state, the Hail activity starts with the radiation of the carrier signal.
S12	Send Hail Acquisition	*off*	*async*	*connecting-T*	*on*	2	In this state, the idle pattern is radiated to achieve symbol lock with the hailed remote unit.
S13	Send Hail Directives	*off*	*async*	*connecting-T*	*on*	3	In this state, the HAIL directives: (if present): (SET_TRANSMITTER_PARAMETERS, SET_RECEIVER_PARAMETERS) or SET_PL EXTENSIONS (TX and RX) or LINK ESTABLISHMENT & CONTROL (TX and RX) are radiated in one Transfer Frame to initiate a session with the hailed remote unit, that is, the responder.
S14	Send Hail Tail	*off*	*async*	*connecting-T*	*on*	4	In this state, the idle pattern is radiated to allow the HAIL directives to be received and processed through the decoding chain of the responder.
S36	Wait for Hail Response	*on*	*off*	*connecting-T*	*off*	5	In this state, the transceiver awaits a response from the called remote unit.
S51	Radiate Carrier Only	*off*	*sync*	*active*	*on*	1	In this state, the transmission process is started with carrier radiation only.
S52	Radiate Acquisition Idle	*off*	*sync*	*active*	*on*	2	In this state, the transmission process is trying to achieve symbol lock with a potential partnered transceiver.
S50	Data Services (send)	*off*	*sync*	*active*	*on*	0	In this state, the user data transmission process functions.
S54	Terminate Reply	*off*	*sync*	*active*	*on*	3	In this state, the transmission process is sending the termination directive.

State Name	State Description	Receive State Desc.	Send State Desc.	MODE	T	SS	Description
S55	Tail before Quit	*off*	*sync*	*active*	*on*	7	In this state, the transmission process is sending the terminating tail sequence bits.
S56	Token Pass or COMM_CHANGE	*off*	*sync*	*active*	*on*	6	In this state, the transmission process is sending either a token or the COMM_CHANGE directive.
S58	Tail before Switch	*off*	*sync*	*active*	*on*	4	In this state, the transmission process is sending the terminating tail sequence bits.
S60	Data Services (receive)	*on*	*off*	*active*	*off*	0	In this state, the receiver is processing received data.
S61	Awaiting First Frame	*on*	*off*	*active*	*off*	1	In this state, the receiver is *on*, waiting receipt of the first frame for processing.
S62	Wait for Carrier	*on*	*off*	*active*	*off*	2	In this state, the receiver is *on*, waiting for the CARRIER_ACQUIRED Physical Layer signal to transition to *true*.

Table 5-4: States When DUPLEX = Simplex (Receive or Transmit)

State Name	State Description	Receive State Desc.	Send State Desc.	MODE	T	SS	Description
S71	Simplex Transmit	*off*	*on*	*active*	*on*	0	In this state, only the transmission operations are enabled while receiving operations are inhibited.
S72	Simplex Receive	*on*	*off*	*active*	*off*	0	In this state, only the receiving operations are enabled while transmission operations are inhibited.

5.1.1 STATE CONTROL VARIABLES

NOTE – These variables are contained within the State Tables: MODE, DUPLEX, TRANSMIT, and SUB-STATE.

5.1.1.1 MODE

The MODE parameter provides control information for operations within the Data Link Layer, and control operations within the Physical Layer. The following states of MODE (set via the local SET MODE directive) shall be allowed:

a) ***inactive***: In the Inactive state, the transceiver's transmitter and receiver are both turned off.

b) ***connecting-T***: In the Physical Layer, the connecting-transmit state in full duplex dictates that the receiver (sequentially in half duplex) and transmitter are powered on and enabled to process received frames, and that the transmitter is enabled for asynchronous channel operations. (In half duplex, only the transmitter is powered on.) The Hail activity is conducted while MODE is *connecting-T*.

c) ***connecting-L***: In the connecting-listen state, the receiver is powered on and enabled to process received frames while the transmitter is turned off.

d) ***active***: In the Active state, when DUPLEX is 'full', the receiver is powered on and enabled to process received frames; the transmitter is enabled for synchronous channel operations responding to the control of the TRANSMIT parameter.

NOTE – The local SET INITIALIZE MODE directive puts MODE into the inactive state and initializes the COP-P variables described in 5.2.3.1.1.2, SET INITIALIZE MODE.

5.1.1.2 DUPLEX

The DUPLEX parameter identifies the physical channel communications characteristics so that the protocol can perform within the transceiver's operational constraints. The following values of DUPLEX (set via the local SET DUPLEX directive) shall be allowed:

a) ***full*****:** Both the receiver and transmitter are simultaneously enabled.

b) ***half*****:** Operation switches between receiving and transmitting within a communications session, with only the receiver or the transmitter enabled at one time.

c) ***simplex transmit*****:** The transmitter is enabled but not the receiver.

d) ***simplex receive:*** The receiver is enabled but not the transmitter.

5.1.1.3 TRANSMIT

The TRANSMIT parameter is used to control Physical Layer operations when MODE is not equal to *inactive*. The following two states for TRANSMIT shall be allowed:

a) ***off***: The Physical Layer is signaled to transition the transmitter to *off*.

b) ***on***: The Physical Layer is signaled to transition the transmitter to *on*.

5.1.1.4 SS (SUB-STATE)

The SS variable shall be used

a) to keep track of sequencing through protocol states in response to events in order to uniquely identify these states;

b) to determine what data to load into the output FIFO (see table 5-13).

5.1.2 OPERATIONAL CONTROL VARIABLES

5.1.2.1 X (Session Termination)

X (Session Termination) shall be used to track the sub-states of the full and half duplex session termination process. In half duplex, it is shared between receive and transmit functionality. The values and definitions of the states of X are as follows:

a) X=0: Bi-directional data passing in progress. Neither transceiver has declared that it is out of data to send. Used in full and half duplex.

b) X=1: Local transceiver informed that there is locally no more data to send, that is, LOCAL_NO_MORE_DATA (LNMD). Used in half duplex only.

c) X=2: Local transceiver has received the LNMD local directive and is sending the REMOTE_NO_MORE_DATA (RNMD) directive to the remote transceiver. When an RNMD directive is received in this state, session termination begins. Used in full and half duplex.

d) X=3: Local transceiver has data to send and it has received an RNMD directive from the remote transceiver. Used in half duplex only.

e) X=4: Local transceiver has received the RNMD directive. When there is locally no more data to send, an RNMD directive is sent to the remote transceiver. Used in full and half duplex.

f) X=5: Both local and remote transceivers have no more data to send. Once the RNMD directive is sent, the session is terminated and X is reset to 0. Used in full and half duplex.

5.1.2.2 Y (COMM_CHANGE)

Y (COMM_CHANGE) shall be used to track the sub-states during the commanding of a Physical Layer communications change. In half duplex, it is set on the transmit side and reset on the receive side. The values and the states of Y are as follows:

a) Y=0: No COMM_CHANGE in progress.

b) Y=1: Local directive received to initiate the COMM_CHANGE.

c) Y=2: COMM_CHANGE directive being sent across the link.

d) Y=3: COMM_CHANGE directive sent, and now waiting for the COMM_CHANGE acknowledgement.

e) Y=4: Receive the Remote COMM_CHANGE Directive (RCCD). Used only in full duplex.

f) Y=5: Act upon the received RCCD. Used only in full duplex.

5.1.2.3 Z (SYMBOL_INLOCK_STATUS)

Z (SYMBOL_INLOCK_STATUS) shall be used during a Physical Layer communications change to track nondeterministic events within State 48 (COMM_CHANGE in Data Services), as follows:

a) Z=0: SYMBOL_INLOCK_STATUS has not transitioned to *false*;

b) Z=1: SYMBOL_INLOCK_STATUS has transitioned to *false*.

5.1.2.4 MODULATION

MODULATION is an interface variable with the Physical Layer which shall control the modulation of the transmitted carrier. When MODULATION=*on*, the coded symbols are modulated onto the radiated carrier; when MODULATION=*off*, the radiated output is not modulated (i.e., carrier only).

5.1.2.5 Persistence

NOTE – Persistence is specified in 4.

5.1.2.6 NEED_PLCW and NEED_STATUS_REPORT

5.1.2.6.1 NEED_PLCW and NEED_STATUS_REPORT shall be used in the data selection for output process to determine if a PLCW or status report should be sent. Where applicable, these variables shall be set to *true*:

a) at initialization;

b) by events in the state transition processes;

c) by PLCW TIMER; and

d) by actions within the COP-P.

5.1.2.6.2 NEED_PLCW shall be set to *false* when a PLCW is selected for output. NEED_STATUS_REPORT shall be set to *false* when a status report is selected for output.

5.1.2.7 REMOTE_SCID_BUFFER

REMOTE_SCID_BUFFER holds the value of the spacecraft ID that shall be used in testing all frames whose Source-or-Destination ID is set to *destination*.

5.1.2.8 COMMUNICATION_VALUE_BUFFER

COMMUNICATION_VALUE_BUFFER shall be used to hold the communication values for the HAIL and COMM_CHANGE directives and operations.

5.1.2.9 RECEIVING_SCID_BUFFER

RECEIVING_SCID_BUFFER shall be used in the frame validation process to compare a received spacecraft ID value with that held within this buffer. This buffer may be loaded by a directive from the local vehicle controller, or it may be loaded with the spacecraft ID contained in the first valid received frame.

5.1.2.10 RECEIVING_PCID_BUFFER

RECEIVING_PCID_BUFFER shall be used in the frame reception process. This buffer shall be loaded with the PCID contained in the first valid received frame.

5.1.3 MIB PARAMETERS

5.1.3.1 Local_Spacecraft_ID

Local_Spacecraft_ID shall contain the value of the spacecraft ID for this local spacecraft.

5.1.3.2 Test_Source

The Test_Source parameter shall be used to determine whether the SCID field in received frames whose Source-or-Destination IDs are set to *source* is to be validated:

a) Test_Source=*false*: no test shall be performed;

b) Test_Source=*true*:

   1) if the RECEIVING_SCID_BUFFER is non-zero, that is, contains a valid SCID, a test shall be performed;

   2) if the RECEIVING_SCID_BUFFER is zero, the value of the SCID field in the header of the first received frame whose Source-or-Destination ID is *source* shall be loaded into RECEIVING_SCID_BUFFER and thereafter used for validation for the remainder of the session. (See also related subsection 5.6.2.)

5.1.3.3 Carrier_Only_Duration

Carrier_Only_Duration represents the time that shall be used to radiate an unmodulated carrier at the beginning of a transmission.

5.1.3.4 Acquisition_Idle_Duration

Acquisition_Idle_Duration represents the time that shall be used to radiate the idle sequence pattern after carrier only to enable the receiving transceiver to achieve symbol synchronization and decoder lock.

5.1.3.5 Tail_Idle_Duration

Tail_Idle_Duration represents the time that shall be used to radiate the idle sequence pattern at the end of a transmission to enable the receiving transceiver to process the last transmitted frame (i.e., push the data through the decoders).

NOTE – The time value for the Tail_Idle_Duration parameter can be calculated from the number of idle bits that need to be sent.

5.1.3.6 Carrier_Loss_Timer_Duration

Carrier_Loss_Timer_Duration shall contain the value loaded into the CARRIER_LOSS_TIMER based upon the conditions defined in 5.2.1.3 (CARRIER_LOSS_TIMER and Associated Events).

5.1.3.7 Comm_Change_Waiting_Period

Comm_Change_Waiting_Period represents the time that the caller shall wait for the Comm_Change_Response to the COMM_CHANGE directive.

5.1.3.8 Comm_Change_Response

Comm_Change_Response shall be used by the responder to acknowledge that the persistent activity has been accepted. (For the full-duplex Comm_Change_Response see table 5-7, Events E17 and E20. For the Half Duplex Comm_Change_Response, see table 5-10, Event E68.)

5.1.3.9 Comm_Change_Notification

Comm_Change_Notification shall be provided to the local vehicle controller, e.g., spacecraft C&DH, by the caller and/or responder upon success or failure of the COMM_CHANGE activity. (See annex G, Notifications to Vehicle Controller.)

5.1.3.10 Comm_Change_Lifetime

5.1.3.10.1 Comm_Change_Lifetime shall represent the time period during which the COMM_CHANGE activity shall be repeated until the expected Comm_Change_Response is detected.

5.1.3.10.2 The Comm_Change_Lifetime may be locally defined in terms of a duration or a maximum number of times this activity is to be repeated before the activity is aborted.

5.1.3.11 Hail_Wait_Duration

Hail_Wait_Duration shall represent the time that the initiating transceiver (caller) waits for a response to the Hail.

5.1.3.12 Hail_Response

Hail_Response shall be used by the responder to acknowledge that the persistent activity has been accepted. In this case, either a valid Transfer Frame has been received or SYMBOL_INLOCK_STATUS (Physical Layer) = *true* (implementation option). (For full duplex, see table 5-6, Event 9; for half duplex, see table 5-9, Event 37.)

5.1.3.13 Hail_Notification

Hail_Notification shall be provided to the local vehicle controller (e.g., spacecraft C&DH) by the caller and/or responder upon success or failure of the persistent activity. (See also annex G, Notifications to Vehicle Controller).

5.1.3.14 Hail_Lifetime

5.1.3.14.1 Hail_Lifetime represents the time period during which the persistent activity shall be repeated until the expected Hail_Response is detected.

5.1.3.14.2 The Hail_Lifetime may be locally defined in terms of a duration or a maximum number of times this activity is to be repeated before the activity is aborted.

5.1.3.15 Discussion—Hailing_Channel, Working Channel and Link Negotiation

The hailing channel is enterprise specific. The default configuration of the Physical Layer parameters (frequencies, duplex/simplex, polarization, modulation index, modulation type, encoding, symbol rate), which is established by the enterprise defines the hailing channel managed parameters that enable two transceivers to initially communicate (via a demand or negotiation query process) so that they can establish a configuration for the data services portion of the session on the working channel. Frequency assignments are defined in the Physical Layer.

Hailing may be carried out as a demand or as a negotiation between partnered radios. A demand hail means that the caller expects the responder to accept the parameters in this directive 'as is' (i.e., no negotiation). A query/response hail enables the caller and responder to negotiate the working channel parameters during link establishment. For example, the response to a query might be an ACK, indicating acceptance of all the parameters by the responder. In this case the caller proceeds with moving onto the working channel with the initial proposal. However, if the response to a query is a NACK, then the responder modifies the original directive based upon the configuration it can support i.e., a new proposed configuration, which the caller might in turn NACK with a counter proposal, continuing until one of the transceivers sends an ACK that they both agree. This specification does not define the details of such a negotiation process including the number of NACKs bounding the duration of the negotiation.

The SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS directives as well as the SET PL_EXTENSIONS directives (configured for transmit or receive) defined in annex B establish the link using a demand hail process only.

The LINK ESTABLISHMENT & CONTROL directive (configured for transmit or receive) defined in annex E provides the option to establish the link via either a demand or a negotiated (via query/response) hail process.

All of the directives for link establishment (hailing) and transitioning to a working channel are defined in annex B (SPDU Type 1), annex D (SPDU Type 4) and annex E (SPDU Type 5).

5.1.3.16 Hailing_Data_Rate/Hailing_Symbol_Rate

Hailing_Data_Rate shall represent the data rate assigned during the Hail activity. Similarly, the Hailing_Symbol_Rate shall represent the symbol rate assigned during the Hail activity.

NOTES

1 Data rates are defined in the Physical Layer.

2 The LINK ESTABLISHMENT & CONTROL directive is defined in terms of symbol rates (see annex E).

5.1.3.17 Send_Duration

Send_Duration represents the maximum time that the half-duplex transmitter shall transmit data before it relinquishes the token (transfers to receive).

5.1.3.18 Receive_Duration

Receive_Duration represents the maximum time that the half-duplex receiver is anticipating that the sending side shall be transmitting.

5.1.3.19 PLCW_Repeat_Interval

5.1.3.19.1 PLCW_Repeat_Interval shall represent the maximum transmission time between successive PLCWs, even if PLCWs are not required for Sequence Control operations.

5.1.3.19.2 A zero value shall represent an infinite time period.

5.2 ELEMENTS AND EVENTS THAT AFFECT STATE STATUS

5.2.1 TIMERS

5.2.1.1 General

5.2.1.1.1 All timers shall use the MIB parameter Interval_Clock.

NOTE – Interval_Clock is a frequency (e.g., 100 Hz) that is used for interval timing.

5.2.1.1.2 The following behavior applies to all timers:

- the timer shall count down when it is not equal to ‘zero’;
- when the timer equals ‘1’, the event associated with the timer shall occur;
- when the timer equals ‘zero’ it shall be in an inactive state;
- the timer may be reset to ‘zero’ by specific actions identified in the state transition tables.

5.2.1.2 Wait Timer (WT) and Its Associated Events

The values loaded into the timer shall represent a desired time value consistent with the Interval Clock frequency. The timer shall be loaded with the required MIB parameter value (see state tables), and shall be counted down using the Interval_Clock. The value in the timer shall be counted down until underflow.

5.2.1.3 CARRIER_LOSS_TIMER and Associated Events

5.2.1.3.1 The CARRIER_LOSS_TIMER contains the duration during which the session shall be maintained even though the carrier is no longer present.

NOTE – This mechanism is intended to reduce complexities from momentary (short term) carrier loss due to multipath or obstacles in the communications path. When CARRIER_LOSS_TIMER counts down to 1, signaling that either the spacecraft is no longer in view or the RF null was larger than expected, the vehicle controller (see annex G) is notified and decides whether to terminate or continue the session via the reconnect process (see 1.5.1.2).

5.2.1.3.2 The CARRIER_LOSS_TIMER shall be loaded with the value contained in the MIB parameter Carrier_Loss_Timer_Duration and down counting shall be enabled when the following conditions are simultaneously satisfied:

a) the CARRIER_ACQUIRED (Physical Layer) signal is *false*;

b) the CARRIER_LOSS_TIMER value is '0';

c) MODE = *active*;

d) either [DUPLEX = *full* or (DUPLEX = *half* .AND. TRANSMIT = *off*)].

5.2.1.3.3 The CARRIER_LOSS_TIMER shall be reset to 'zero' when the CARRIER_ACQUIRED (Physical Layer) signal is *true*.

5.2.1.4 PLCW Timer and Associated Events

5.2.1.4.1 The PLCW TIMER shall be used periodically to request the issuance of a PLCW.

5.2.1.4.2 When the PLCW TIMER='1', the NEED_PLCW variable shall be set *true*.

5.2.1.4.3 The timer shall be loaded with the value in the MIB parameter PLCW_Repeat_Interval whenever a PLCW is transmitted. (See the COP-P state tables in section 6 for when the NEED_PLCW variable is set to *true.*)

NOTE – The PLCW TIMER does not appear in the state transition tables.

5.2.2 OUTPUT FIFO

5.2.2.1 Overview

The Output FIFO is a FIFO cache for the storage of bits that are serially output to the C&S Sublayer for encoding. The channel coding options of the C&S Sublayer (e.g., convolutional coding) are applied to the data from the output FIFO before transmission by the Physical Layer.

5.2.2.2 General

The FIFO shall be filled with data per the specification defined in table 5-13. The 'Output FIFO = empty' signals that no data are contained within the FIFO, and more data needs to be input to the FIFO to keep the output bitstream synchronous.

5.2.2.3 No_Frames_Pending

The No_Frames_Pending event shall occur when the Output FIFO becomes empty and there are no frames selectable for output.

5.2.3 DIRECTIVES

5.2.3.1 Local Directives

NOTE – Local directives are sent internally, that is, not across the link.

5.2.3.1.1 SET MODE

5.2.3.1.1.1 SET MODE shall assign values to the MODE parameter (5.1.1.1) as follows:

a) ***connecting-L***;

b) ***connecting-T***;

c) ***active***;

d) ***inactive*** (see 5.2.3.1.1.2).

5.2.3.1.1.2 SET MODE *inactive* shall initialize the variables in table 5-5 to the values indicated.

Table 5-5: Session Control Variable Initialization Table

Variables	Value
TRANSMIT, MODULATION, PERSISTENCE	*off*, *off*, *false*
SS, X, Y and Z	0
WAIT TIMER (WT), CARRIER_LOSS_TIMER, PLCW TIMER	0
SEQUENCE CONTROLLED (SEQ_CTRL_FSN) AND EXPEDITED FRAME SEQUENCE COUNTERS (EXP_FSN)	0

5.2.3.1.2 SET INITIALIZE MODE

SET INITIALIZE MODE shall use SET MODE *inactive* to place MODE into the *inactive* state and trigger the COP-P events SE0 and RE0 described in the state tables in 6.2.3.3 and 6.3.1.

5.2.3.1.3 LOCAL COMM_CHANGE (LCCD)

Local COMM_CHANGE shall use SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS (and possibly SET PL EXTENSIONS for transmit and receive or LINK ESTABLISHMENT & CONTROL for transmit and receive) to initiate a change in one or more communications channel parameters.

NOTES

1 Equivalently, one can utilize the Link Establishment & Control directive.

2 The Remote COMM_CHANGE Directive (RCCD) uses the SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS directives (or possibly the SET PL EXTENSIONS for transmit and receive or LINK ESTABLISHMENT & CONTROL directives for transmit and receive) and is sent across the link.

5.2.3.1.4 LOAD COMMUNICATIONS VALUE BUFFER

LOAD COMMUNICATIONS VALUE BUFFER shall load the values for the remote transmitter and receiver associated with either the Hail, COMM_CHANGE, or half-duplex receiver/transmitter switching activities.

5.2.3.1.5 LOCAL_NO_MORE_DATA

LOCAL_NO_MORE_DATA shall inform the transceiver that the local data source has no more data to send.

NOTE – This directive initiates the session termination process.

5.2.3.1.6 SET DUPLEX

SET DUPLEX shall configure the local transmitter and/or receiver for either full-duplex, half-duplex, or simplex operations.

5.2.3.1.7 SET RECEIVING SCID BUFFER

SET RECEIVING SCID BUFFER shall be used by the vehicle controller to load the RECEIVING_SCID_BUFFER.

5.2.3.1.8 READ STATUS

READ STATUS shall selectively read the local status registers and buffers (including timing services) within the transceiver.

5.2.3.2 Remote Directives

5.2.3.2.1 Overview

Remote directives are sent over the link from the local transceiver to the remote transceiver. SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS, or SET PL EXTENSIONS for transmit and receive, or LINK ESTABLISHMENT & CONTROL for transmit and receive can be used for both the HAIL directive and for Physical Layer communication changes. SET CONTROL PARAMETERS or LINK ESTABLISHMENT & CONTROL/REPORT REQUEST directives are used to 1) swap receive/transmit functionality in half-duplex, 2) terminate the session, 3) change duplex, and 4) set up the timing service.

5.2.3.2.2 SET TRANSMITTER PARAMETERS

5.2.3.2.3 SET TRANSMITTER PARAMETERS shall be used to set the transmission parameters that control the data rate, encoding, modulation, and frequency in the transceiver receiving the directive.

5.2.3.2.4 Upon receipt, the transceiver shall use the local directive SET MODE *active* to put the receiver's MODE parameter into the *active* state.

NOTE – This directive is formulated using the values contained in the sender's COMMUNICATION VALUE BUFFER. (See annex B for a complete definition.)

5.2.3.2.5 SET RECEIVER PARAMETERS

5.2.3.2.5.1 SET RECEIVER PARAMETERS shall be used to set the receiver parameters that control the data rate, decoding, modulation, and frequency in the transceiver receiving the directive.

5.2.3.2.5.2 Upon receipt, the transceiver shall use the local directive SET MODE *active* to put the receiver's MODE parameter into the *active* state.

NOTE – This directive is formulated using the values contained in the sender's COMMUNICATION VALUE BUFFER. (See annex B for a complete definition.)

5.2.3.2.6 SET PL EXTENSIONS

SET PL EXTENSIONS directive shall be used to enable or disable additional parameters for the receive and send sides of a radio.

NOTE – This directive is provided for compatibility between transceivers with Physical Layer and C&S Sublayer extensions.

5.2.3.2.7 SET CONTROL PARAMETERS

NOTE – This directive provides the capability of changing zero or more session control parameters at a time. (See annex B for a complete definition.)

SET CONTROL PARAMETERS shall be used to transmit operational control information during a session. It includes the following fields:

a) **Token Field**: When this field is non-zero, it shall notify the recipient that the sender is relinquishing the half-duplex 'Send Token' and is switching to receive.

b) **RNMD Field**: When this field is non-zero, it shall notify the recipient that the sending spacecraft has no more data to send, and that the session may be terminated when the recipient also has no more data to send.

c) **Duplex Field**: When this field is non-zero, it shall notify the recipient to change communication directionality (*full*, *half*, *simplex-transmit*, *simplex-receive*).

d) **Time Sample Field**: When this field is non-zero it shall notify the recipient to capture the time, sequence number, direction, and QoS Indicator (for Version-3 frames)/Bypass Flag (for Version-4 frames) for the next n frames received and transmitted (where 'n' is the value, i.e., number of frames contained within the Time Sample Field).

5.2.3.2.8 LINK ESTABLISHMENT & CONTROL

The LINK ESTABLISHMENT & CONTROL directive enables hailing and moving onto a working channel, as well as additional communication configuration changes for the receive and send sides of a radio during a session. (See annex E for a complete definition.)

5.2.3.2.9 REPORT_REQUEST

The REPORT REQUEST directive is the mechanism by which either (1) a status report, (2) time-tag sample collection, or (3) a PLCW per PCID can be requested of a Proximity-1 node.

5.3 STATE TRANSITION TABLES AND DIAGRAMS

5.3.1 OVERVIEW

The following subsections contain State Transition Tables and State Transition Diagrams that should be read in conjunction with one another for completeness.

The State Transition Diagrams are intended to illustrate transitions from one state to another, and the events that trigger them. States are shown in boxes. Events that cause transitions from one state to a resultant state are given in italic text beside arrows that indicate the transition between states.

States, which have a descriptive title, are assigned the letter S and a number in the State Transition Tables. These tables are organized by event number in column 1. Column 2 describes the event that causes the state to transition from the starting state (column 3) to the resulting state (column 4). Column 5 contains any additional actions (in addition to what is described in tables 5-1 through 5-4) that take place as a result of entering that state.

The diagrams do not show all possible states for reasons of simplicity and clarity. For completeness, the State Transition Tables and accompanying text contain a description of all states and events not included in the diagrams.

5.3.2 FULL DUPLEX OPERATIONS

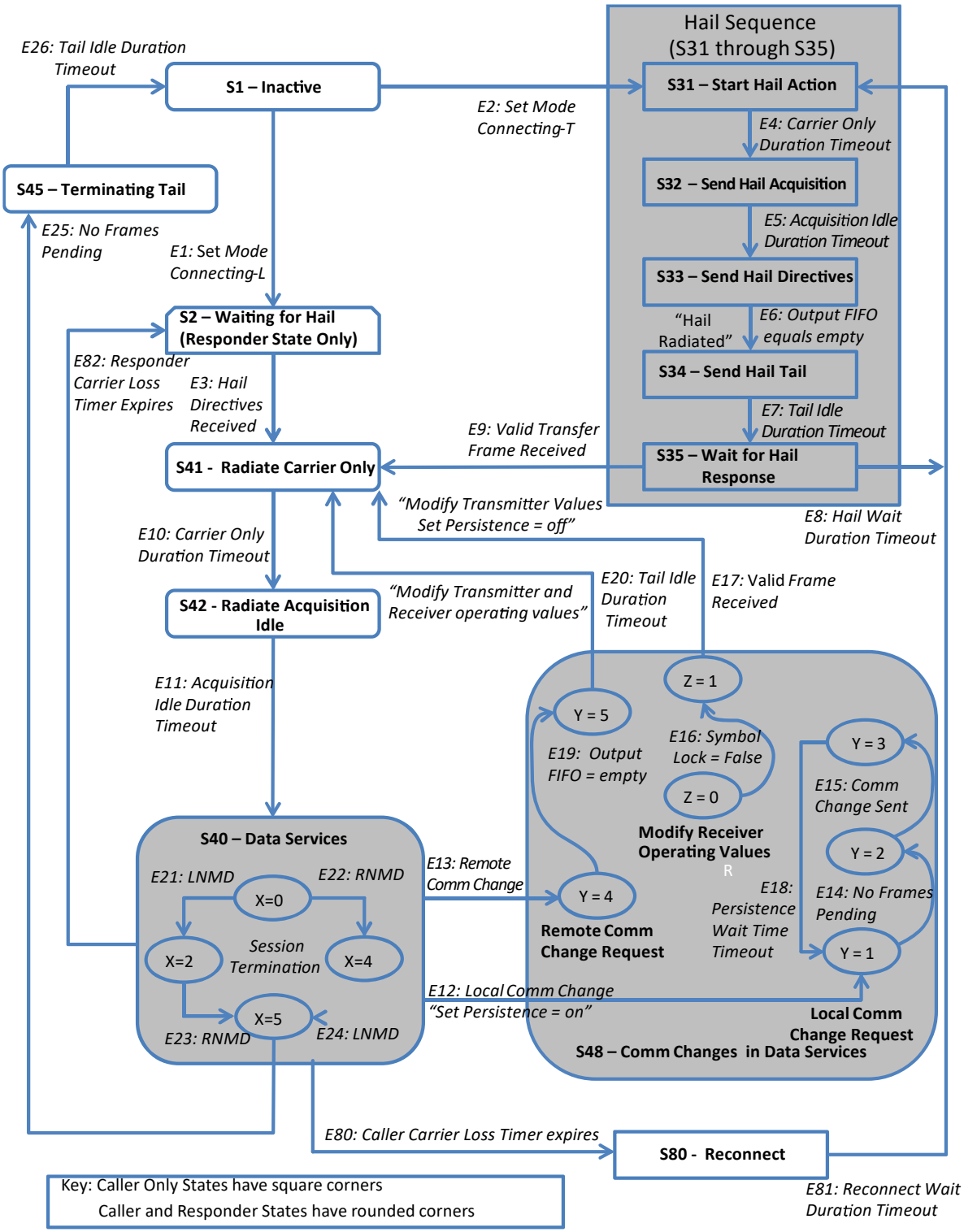


Figure 5-1: Full Duplex State Transition Diagram

Table 5-6: Full Duplex Session Establishment/Data Services State Transition Table

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-2, and *Comments*
E1	Local Directive - SET MODE *connecting-L*	S1	S2	
E2	Local Directive - SET MODE *connecting-T*	S1	S31	- WT = Carrier_Only_Duration - Set PERSISTENCE = *true* - Form and load HAIL directives into Comm Value Buffer
E3	HAIL Directives Received	S2	S41	- WT = Carrier_Only_Duration - Set TRANSMIT = *on* - Set NEED_PLCW = *true* - Set Receiver and Transmitter parameters per HAIL directives. See 5.1.3.15 - Send Hail_Notification to C&DH
E4	WT = 1 Carrier_Only_Duration Timeout	S31	S32	- WT = Acquisition_Idle_Duration - Set MODULATION = *on*
E5	WT = 1 Acquisition_Idle_Duration Timeout	S32	S33	- Radiate Hail - Transmit the appropriate Receiver and Transmitter Hail directives – see 5.1.3.15
E6	Output FIFO = empty	S33	S34	*Hail Radiated* - WT = Tail_Idle_Duration
E7	WT = 1 Tail_Idle_Duration Timeout	S34	S35	- WT = Hail_Wait_Duration (see 5.1.3.11) - Set TRANSMIT = *off*
E8	WT = 1 Hail_Wait_Duration Timeout	S35	S31	- WT = Carrier_Only_Duration - Set MODULATION = *off* - Set TRANSMIT = *on* - If CARRIER ONLY RECEIVED, send carrier received notification to C&DH
E9	Valid Transfer Frame Received (or SYMBOL_INLOCK_STATUS=*true*—implementation option. See Hail_Response MIB parameter.)	S35	S41	- Set Transmitter values from Comm Value Buffer - WT = Carrier_Only_Duration - Set MODULATION = *off* - Set TRANSMIT = *on* - Set PERSISTENCE = *false* - Send Hail_Notification to C&DH
E10	WT = 1 Carrier_Only_Duration Timeout	S41	S42	- WT = Acquisition_Idle_Duration - Set MODULATION = *on*
E11	WT = 1 Acquisition_Idle_Duration Timeout	S42	S40	*Data Service begins*

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-2, and *Comments*
E80	Caller's CARRIER_LOSS_TIMER expires	S40	S80	- WT = Reconnect_Wait_Duration - Set TRANSMIT = *off* - Reconfigure caller's Transceiver back to initial Hail settings as detailed in the initial hailing directives. – See 5.1.3.15 - Utilize (do not reset) existing FARM-P and FOP-P variable values
E81	WT = 1 Reconnect_Wait_Duration Timeout	S80	S31	- Set TRANSMIT = *on* WT = Carrier_Only_Duration - Set PERSISTENCE = *true* - Form and load HAIL directives into Comm Value Buffer - Implementations may consider alternative values for: Carrier_Only_Duration, Acquisition_Idle_Duration, Tail_Idle_Duration, and Hail_Wait_Duration for reconnections
E82	Responder's CARRIER_LOSS_TIMER expires	S40	S2	- Set TRANSMIT = *off* - Reconfigure responder's transceiver back to initial Hail settings as detailed in the initial hailing directives. – See 5.1.3.15 - Utilize (do not reset) existing FARM-P and FOP-P variable values
NOTE – FOP-P Data operations (6.2) occur within State 40. FARM-P operations (6.3) occur in States 40, 41, 42, and 48 whenever MODE is active and the receiver is on. Comm Value Buffer is the local buffer used for staging the transmit and receive parameters in support of the hailing and COMM_CHANGE directives. Values can be sent in locally or remotely.				

Table 5-7: Full Duplex Communication Change State Table

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-2, and *Comments*
E12	Local COMM_CHANGE Request	S40(Y = 0)	S48(Y = 1)	- Set Y = 1 - Set PERSISTENCE = *true*
E13	Remote COMM_CHANGE Request	S40(Y = 0)	S48(Y = 4)	- Set Y = 4 - Set PERSISTENCE = *true*
E14	No Frames Pending	S48(Y = 1)	S48(Y = 2)	- Form and Send Remote COMM_CHANGE Directive (RCCD) - Set Y = 2
E15	Output FIFO = empty (COMM_CHANGE sent)	S48(Y = 2)	S48(Y = 3)	- WT = Persistence_Wait_Time - Set Y = 3
E16	Symbol Lock = *false*	S48(Y = 1 or 2 or 3)	S48(Z = 1)	- Set Z = 1 - SET_RECEIVER_PARAMETERS or equivalent directive – see 5.1.3.15 from Comm Value Buffer
E17	Valid Frame Received and Z = 1	S48(Z = 1)	S41	- Set Y = 0 - Set PERSISTENCE = *false*, - Set Z = 0 - SET TRANSMITTER PARAMETERS or equivalent directive – see 5.1.3.15 from Comm Value Buffer - WT = Carrier_Only_Duration - Set MODULATION = *off*
E18	WT = 1 Persistence_Wait_Time Timeout	S48(Y = 3)	S48(Y = 1)	- Set Y = 1 *No Response to RCCD received yet*
E19	Output FIFO = empty	S48(Y = 4)	S48(Y = 5)	- WT = Tail_Idle_Duration - Set Y = 5
E20	WT = 1 Tail_Idle_Duration Timeout	S48(Y = 5)	S41	- Set Y = 0 - Set PERSISTENCE = *false* - SET TRANSMITTER PARAMETERS & SET_RECEIVER_PARAMETERS or equivalent directives – see 5.1.3.15 into Comm Value Buffer - Set NEED_PLCW = *true* - WT = Carrier_Only_Duration, Set MODULATION = *off*
NOTE – X, Y, Z are sub-state variables used in the process of session termination (X) and COMM_CHANGE (Y, Z).				

Table 5-8: Full Duplex Session Termination State Table

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-2 and *Comments*
E21	Receive LNMD (X = 0)	S40(X = 0)	S40(X = 2)	- Form and Load RNMD directive into the MAC queue (see reference [6]) - Set X = 2 - Send RNMD
E22	Receive RNMD (X = 0)	S40(X = 0)	S40(X = 4)	- Set X = 4
E23	Receive RNMD (X = 2)	S40(X = 2)	S40(X = 5)	- Set X = 5 *Begin Termination Process*
E24	Receive LNMD (X = 4)	S40(X = 4)	S40(X = 5)	- Form and Load RNMD directive into MAC queue - Set X = 5 - Send RNMD
E25	No_Frames_Pending (X = 5)	S40(X = 5)	S45	- WT = Tail_Idle_Duration
E26	WT = 1 Tail_Idle_Duration Timeout	S45	S1	- Local directive SET MODE *inactive* - Notify vehicle controller: End of Session (# octets received)
E28	Receive a Local SET MODE = *Inactive* Directive or SET INITIALIZE MODE Directive	Any state	S1	SET MODE *inactive* - Notify vehicle controller: End of Session (# octets received) NOTES 1 Not Shown on full-duplex transition diagram. 2 E28 initializes session control variables. (See table 5-5.)
NOTE – LNMD = LOCAL_NO_MORE_DATA Directive received from the local controller; RNMD is the REMOTE_NO_MORE_DATA Directive received over the link.				

5.3.3 HALF DUPLEX OPERATIONS

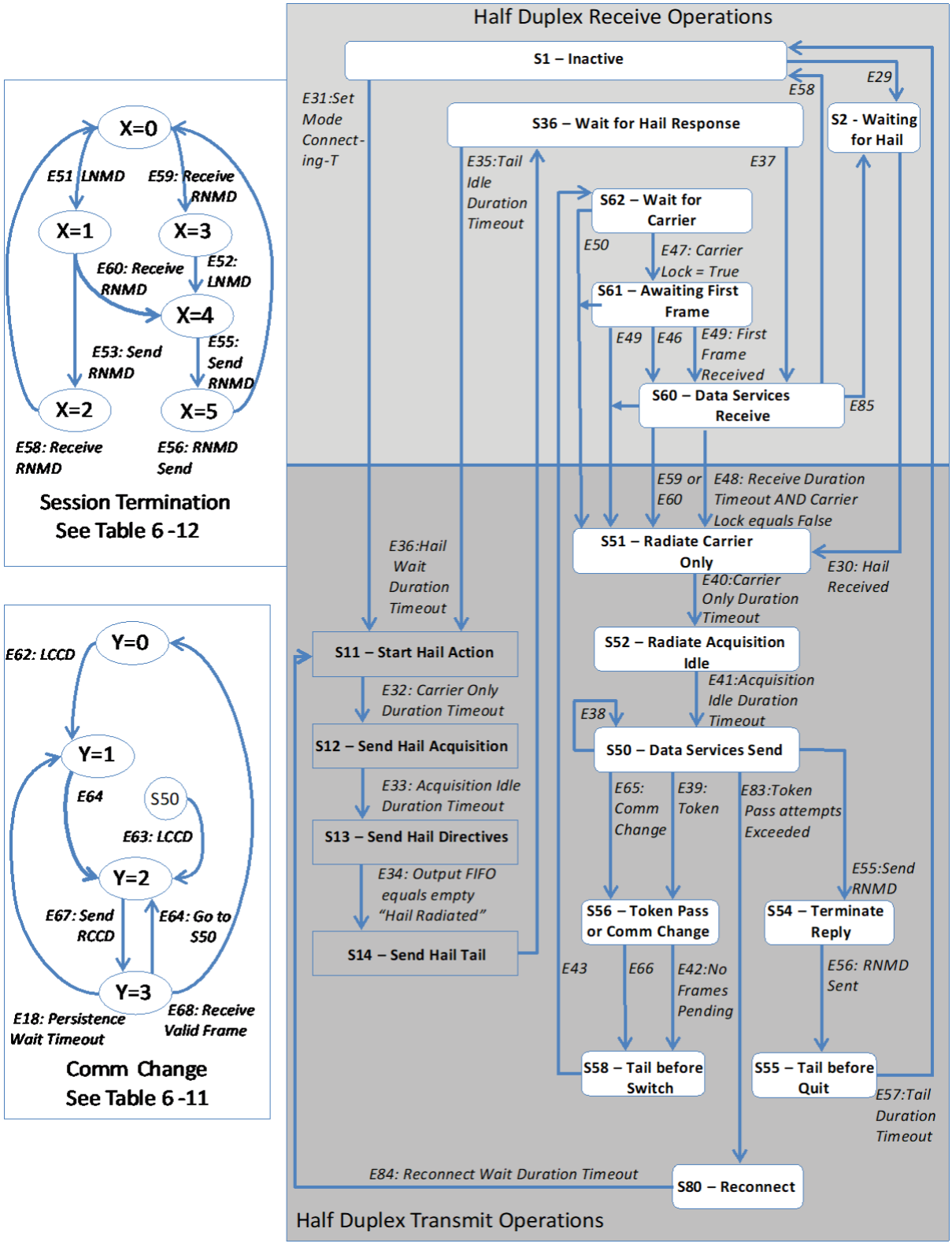


Figure 5-2: Half Duplex State Transition Diagram

Table 5-9: Half Duplex Session Establishment and Data Services

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-3, and *Comments*
E29	Local Directive – SET MODE *connecting-L*	S1	S2	- Set NEED_PLCW = *true*
E30	Hail Received	S2	S51	- WT = Carrier_Only_Duration - Set Receiver and Transmitter values per HAIL directives - Set TRANSMIT = *on* - Send Hail_Notification to C&DH
E31	Local Directive – SET MODE *connecting-T*	S1	S11	- WT = Carrier_Only_Duration - Load HAIL directives to Comm Value Buffer - Set PERSISTENCE = *true* - Set Receiver Values per HAIL directive
E32	WT = 1 Carrier_Only_Duration Timeout	S11	S12	- WT = Acquisition_Idle_Duration - Set MODULATION = *on*
E33	WT = 1 Acquisition_Idle_Duration Timeout	S12	S13	- Radiate Hail - Transmit the appropriate Receiver and Transmitter Hail directives – see 5.1.3.15
E34	Output FIFO=empty	S13	S14	*Hail Radiated* - WT = Tail_Idle_Duration
E35	WT = 1 Tail_Idle_Duration Timeout	S14	S36	- WT = Hail_Wait_Duration (see 5.1.3.11) - Set MODULATION = *off* - Set TRANSMIT = *off*
E36	WT = 1 Hail_Wait_Duration Timeout	S36	S11	- Set TRANSMIT = *on* - WT = Carrier_Only_Duration - Set Load HAIL directives to Comm Value Buffer - If CARRIER ONLY RECEIVED, send carrier received notification to C&DH
E37	Valid Transfer Frame Received (or SYMBOL_INLOCK_STATUS = *true*)—implementation option. (See Hail_Response MIB parameter.)	S36	S60	- Set Transmitter values per Comm Value Buffer - Set PERSISTENCE = *false* *(get ready for next transmit contact)* - WT = Receive_Duration - Send Hail_Notification to C&DH

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-3, and *Comments*
E38	(Transmit Timer Event – End of Send Period) WT = 1 Send_Duration Timeout	S50	S50 Y = 0	- Set PERSISTENCE = *true* *Setting* PERSISTENCE *blocks the transmission of data from data services. Now only send from the MAC queue. Therefore, no frames pending occurs.*
E39	No Frames Pending AND. Y = 0 .AND. NEED_PLCW is *false*	S50 Y = 0	S56	- Form and load the token via the SET CONTROL PARAMETERS or LINK ESTABLISHMENT & CONTROL Directive into the MAC queue – See 5.1.3.15
E40	WT = 1 Carrier_Only_Duration Timeout	S51	S52	- WT = Acquisition_Idle_Duration - Set MODULATION = *on*
E41	(End of Acquire) WT = 1 Acquisition_Idle_Duration Timeout	S52	S50	- WT = Send_Duration
E42	No Frames Pending (Y = 0)	S56, Y = 0	S58	- WT = Tail_Idle_Duration
E43	WT = 1 Tail_Idle_Duration Timeout .AND. Y ≠ 2	S58 Y ≠ 2	S62	- WT = Receive_Duration - Set PERSISTENCE = *false* - Set MODULATION = *off* - Switch transmit to receive
E44	WT = 1 Receive_Duration Timeout .AND. Carrier Lock = *true*	S60	S60	- WT = Receive_Duration - Notify vehicle controller: Sender exceeded prescribed transmission interval
E45	WT=1 Receive_Duration Timeout .AND. Carrier Lock = *true*	S61	S61	- WT = Receive_Duration - Notify vehicle controller: No data transferred during contact period
E46	Receive Valid frame .AND. Y ≠ 3	S61 Y≠3	S60	
E47	Carrier Lock = *true*	S62	S61	
E48	WT = 1 Receive_Duration Timeout .AND. Carrier Lock = *false*	S60	S51	- WT = Carrier_Only_Duration *back-up action for missed token* - Switch receive to transmit
E49	Receive Token - SET CONTROL PARAMETERS Directive or LINK ESTABLISHMENT & CONTROL directive – See 5.1.3.15	S60/S61	S51	- WT = Carrier_Only_Duration - Switch receive to transmit
E50	WT = 1 Receive_Duration Timeout .AND. Carrier Lock = *false*	S62/S61	S51	- WT = Carrier_Only_Duration - Notify vehicle controller: S62: No carrier received for contact period S61: No data transferred during contact period - Switch receive to transmit

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Action(s) in Addition to Tables 5-1, 5-3, and *Comments*
E83	Token Pass attempts exceeded (Maximum_Failed_Token_Passes)	S50	S80	- WT = Reconnect_Wait_Duration - Set TRANSMIT = *off* - Reconfigure caller's Transceiver back to initial Hail settings in the initial hailing directives. – See 5.1.3.15 - Utilize (do not reset) existing FARM-P and FOP-P variable values
E84	WT = 1 Reconnect_Wait_Duration Timeout	S80	S11	- Set TRANSMIT = *on* WT = Carrier_Only_Duration - Set PERSISTENCE = *true* - Form and load HAIL directives into Comm Value Buffer - Implementations may consider alternative values for: Carrier_Only_Duration, Acquisition_Idle_Duration, Tail_Idle_Duration, Hail_Wait_Duration for reconnections.
E85	Responder's CARRIER_LOSS_TIMER expires	S60	S2	- Set TRANSMIT = *off* - Utilize (do not reset) existing FARM-P and FOP-P variable values
NOTE – FOP-P Data operations occur within State 50 and are described in 6.2. FARM-P operations occur within States 60 and 61 and are described in 6.3.1.				

Table 5-10: Half Duplex Communication Change State Table

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Additional Action(s) and *Comments*
E62	Receive Local COMM_CHANGE Directive (LCCD)	Any State other than State S50 Y = 0	No State Change Y = 1	- Load SET TRANSMITTER/SET RECEIVER PARAMETERS or equivalent Directives values – see 5.1.3.15 into Comm Value Buffer
E63	Receive Local COMM_CHANGE Directive (LCCD)	S50	S50 Y = 2	- Set PERSISTENCE = *true* - SET RECEIVER PARAMETERS or equivalent directive – see 5.1.3.15 from Comm Value Buffer
E64	Transition to State 50	Y = 1 .OR. Y = 3	S50 Y = 2	- Set Y = 2, Set PERSISTENCE = *true* - SET RECEIVER PARAMETERS or equivalent directive – see 5.1.3.15 from Comm Value Buffer
E65	No Frames Pending .AND. Y = 2	S50 Y = 2	S56	- Form and load into the Comm Value Buffer the COMM_CHANGE Directives
E66	No Frames Pending	S56 Y = 2	S58 Y = 2	- WT = Tail_Idle_Duration COMM_CHANGE *Sent*
E67	WT = 1 Tail_Idle_Duration Timeout .AND. Y = 2	S58 Y = 2	S62 Y = 3	-WT = Receive_Duration - Switch transmit to receive
E47	Carrier Lock = *true*	S62	S61	*Same event - provided for clarity*
E68	Receive Valid Frame	S61 Y = 3	S60 Y = 0	- SET TRANSMITTER PARAMETERS or equivalent directive – see 5.1.3.15 from Comm Value Buffer - Set Y = 0 - PERSISTENCE = *false*
E69	Receive COMM_CHANGE (Not Shown in State Transition Diagram)	S60/S61	S51	- Set Transmitter and Receiver Parameters or equivalent directives – see 5.1.3.15 into Comm Value Buffer - Set NEED_PLCW = *true*

Table 5-11: Half Duplex Session Termination State Table

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Additional Action(s) and *Comments*
E51	Receive LNMD (can be received at any time)	X = 0	X = 1	- Set X = 1
E52	Receive LNMD (can be received at any time)	X = 3	X=4	- Set X = 4

Event No.	Event Causing the Transition (Description)	Starting State (from)	Resulting State (to)	Additional Action(s) and *Comments*
E53	No Frames Pending .AND. X = 1	S50 X = 1	S50 X = 2	- Form and Load RNMD into the MAC queue, - Set X = 2 - Send RNMD
E55	No Frames Pending .AND. X = 4	S50 X = 4	S54 X = 5	- Form and Load RNMD into the MAC queue, - Set X = 5; Send RNMD
E56	No Frames Pending .AND. X = 5	S54 X = 5	S55 X = 0	- WT = Tail_Idle_Duration - Set X = 0 *Transmission of* RNMD *complete*
E57	WT = 1 Tail_Duration Timeout	S55	S1	- SET MODE Inactive - Notify vehicle controller: End of Session(# octets received)
E58	Receive RNMD .AND. X = 2	S60/S61 X = 2	S1 X = 0	- SET MODE *Inactive* - Set X = 0 *Both nodes have no more data to send; Notify vehicle controller: End of Session(# octets received)*
E59	Receive RNMD .AND. X = 0	S60/S61 X = 0	S51 X = 3	- Set X = 3 *WT = Carrier_Duration_Only*
E60	Receive RNMD .AND. X = 1	S60/S61 X = 1	S51 X = 4	- Set X = 4 *WT = Carrier_Duration_Only*
E61	Receive a Local SET MODE *Inactive* directive	any	S1	- SET MODE *Inactive* - Notify vehicle controller: End of Session(# octets received) *Not shown on half duplex state transition diagram*

5.3.4 SIMPLEX OPERATIONS

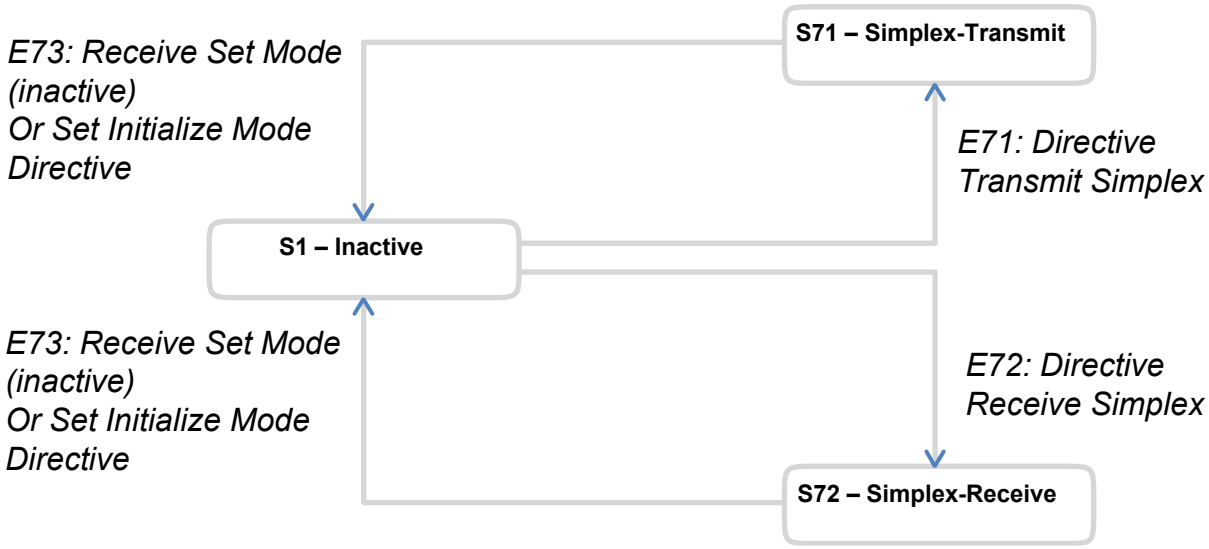


Figure 5-3: Simplex Operations

Table 5-12: Simplex State Transition Table

Event No	Starting State (from)	Resulting State (to)	Event Causing the Transition (Description)	Action(s) in addition to Tables 5-1, 5-4
E71	S1	S71	Receive Directive - Transmit Simplex	- Set DUPLEX = *Simplex transmit* - Set TRANSMIT = *on* - Local Directive SET MODE = *active*
E72	S1	S72	Receive Directive - Receive Simplex	- Set DUPLEX = *Simplex receive* - Set TRANSMIT = *off* - Local Directive SET MODE = *active*
E73	S71 or S72	S1	Receive a Local SET MODE = *Inactive* Directive	- Notify vehicle controller: End of Session(# octets received)

5.4 INTERFACES WITH THE PHYSICAL LAYER (VIA CODING AND SYNCHRONIZATION SUBLAYER)

5.4.1 OUTPUT INTERFACES

5.4.1.1 The TRANSMIT parameter (5.1.1.3) shall be set to *on* to signal the transceiver to turn its transmitter ‘on’.

5.4.1.2 The Frame Sublayer shall output frames via the C&S Sublayer ChannelAccess.request service primitive.

5.4.1.3 The MODULATION parameter (5.1.2.4) shall be set to *on* to signal the transceiver to modulate the carrier with the coded symbols provided on the Output coded symbol stream. (See 5.5.3.)

5.4.2 INPUT INTERFACES

The Physical Layer provides information about carrier and symbol lock status via the CARRIER_ACQUIRED (see 5.2.1.3) and SYMBOL_INLOCK_STATUS (see 5.1.2.3) parameters, respectively. The values of these parameters shall be interpreted as follows:

a) CARRIER_ACQUIRED = *true* indicates that the receiver has acquired a carrier signal;

b) SYMBOL_INLOCK_STATUS = *true* indicates that symbol synchronization has been acquired.

NOTE – The C&S layer receives a coded symbol stream from the Physical Layer and delivers frames to the Frame Sublayer.

5.5 SENDING OPERATIONS

5.5.1 SESSION ESTABLISHMENT

5.5.1.1 Local SET TRANSMITTER PARAMETERS and Local SET RECEIVER PARAMETERS directives (and, if needed, Local SET PL EXTENSIONS directives) shall be used to set the local transceiver to the desired physical configuration.

5.5.1.2 If required for the session, the Test_Source MIB parameter shall be loaded.

5.5.1.3 A Local SET MODE (*connecting-T*) directive shall initiate the Hail activity and start the session establishment process (see 5.3.2 for full-duplex operation and 5.3.3 for half-duplex operation).

5.5.2 DISCUSSION

Once a frame is ready for output, at the C&S Sublayer, an ASM is prepended, and a CRC is computed and appended to the frame. The output coded symbol stream is formulated for radiation in accordance with table 5-13.

An idle pattern generated by an Idle Pattern Generator (described in reference [4]) is used for acquisition periods, that is, periods when no frames are available for transmission, as well as for providing a tail stream (which provides the added bits required to push the data through the receiving and decoding processes at the remote terminus of the link).

5.5.3 OUTPUT CODED SYMBOL STREAM FORMULATION

Table 5-13: Data Source Selection for Output Coded Symbol Stream with TRANSMIT = *on* and MODULATION = *on*

Based Upon the Values Below, Take the Following Action					Action
SS (SUB-STATE)	SPDU Pending	PERSISTENCE	NEED_ PLCW or Status	SDU Pending	Data to be loaded into output FIFO when output FIFO is empty
2, 4, or 7	X	X	X	X	IDLE (Acquisition or Tail)
0, 3, or 6	***true***	X	X	X	ASM + P-frame(SPDU) + CRC
0, 3, or 6	*false*	***true***	X	X	IDLE
0, 3, or 6	*false*	*false*	***true***	X	ASM + PLCW/Status + CRC
0, 3, or 6	*false*	*false*	*false*	***true***	ASM + U-frame(SDU) + CRC
0, 3, or 6	*false*	*false*	*false*	*false*	IDLE

NOTES

1 X means the value is not important.

2 SPDU Pending is *true* if there is a supervisory protocol data unit available to send.

3 SDU Pending is *true* if there is a service data unit (user data) available to send.

4 When NEED_PLCW is *true*, NEED_STATUS_REPORT can optionally be set to *true*, enabling the generation and transmission of a status report as well.

5 PERSISTENCE is a variable used for selected Supervisory protocol activities. (See 4.)

6 The selection of an SDU issues an extract data unit request to the FOP-P. (See FOP-P data selection, described in 6.2.)

7 SS = 1 indicates transmit carrier only.

8 SS = 5 indicates TX off, RX waiting for response.

5.5.4 DISCUSSION—PROVISION OF U-FRAME FOR SELECTION

The provision of a U-frame for selection through use of the procedures contained in table 5-13 is defined in the FOP-P portion of the COP-P specification (see 6.2). Operations on a single PCID are described in this specification. The simultaneous use of multiple PCIDs is possible, but concurrent COP-P procedures are required and the reporting is then required to contain the status for each PCID. Data prioritization and its multiplexing for selection into the output coded symbol stream of simultaneous multiple PCID operations as specified above is outside the scope of this document.

5.5.5 DISCUSSION—EVENTS RELATED TO DATA HANDLING ACTIVITIES

NEED_PLCW is set to *false* whenever a PLCW is chosen for output (5.1.2.6.2). NEED_STATUS_REPORT is set to *false* whenever a status report is chosen for output (5.1.2.6.2). No_Frames_Pending (5.2.2.3) occurs when none of the conditions for selecting an SPDU (including a PLCW) or U-frame is satisfied. Output FIFO = empty (5.2.2.2) is *true* when the last bit contained within the Output FIFO is extracted.

5.6 RECEIVING OPERATIONS

5.6.1 FRAME RECEPTION

5.6.1.1 Establishment of physical channel characteristics and initialization of receiving procedures shall be accomplished via a SET MODE (*connecting-L*) or SET MODE (*connecting-T*) local directive (see 5.2.3.1.1 and 5.1.1.1).

5.6.1.2 The Frame Sublayer shall accept for validation frames delivered via the C&S Sublayer ChannelAccess.indication service primitive (see reference [4], annex B).

NOTES

1 When the Receive State is *on,* the received (optionally decoded) bitstream is processed by the C&S Sublayer to delimit the contained frames (this process requires frame synchronization and frame length determination using the frame header length field).

2 The delimited frame and the attached CRC-32 are processed by the C&S Sublayer to determine if the frame contains errors. Erred frames are rejected as invalid.

5.6.2 FRAME VALIDATION

Frame Validation Criteria shall be as follows:

a) If the first two bits of the Transfer Frame header are

   1) ‘00’, then the transfer frame is rejected as invalid;

   2) ‘01’, then a Version-2 (AOS) Transfer Frame has been received from the C&S sublayer. (See reference[2], subsection 2.2.3). Ignore points b) and c) below.

   3) ‘10’, then a valid Version-3 (Prox-1)Transfer Frame has been received from the C&S sublayer. (See reference [4], subsection 3.6.4.)

   4) ‘11’, then a valid Version-4 (USLP) transfer frame has been received from the C&S sublayer. (See reference [4], subsection 3.6.4.)

b) If the Spacecraft ID field in the Version-3 or Version-4 Transfer Frame header does not contain the value of the Local_Spacecraft_ID (MIB parameter) when the Source-or-

Destination Identifier value equals '1', that is, *destination*, the frame is rejected as invalid.

c) If the Spacecraft ID field in the Version-3 or Version-4 Transfer Frame header does not contain the value equal to the RECEIVING_SCID_BUFFER for all frames received (i.e., Remote_Spacecraft_ID, MIB parameter) when the RECEIVING_SCID_BUFFER is non-zero and the Source-or-Destination Identifier value equals '0', that is, *source*, and Test_Source is true, a session violation has occurred and the vehicle controller is notified (see annex G). When the RECEIVING_SCID_BUFFER is zero and Test_Source is *true*, the value of the SCID field in the header of the first received frame whose Source-or-Destination ID is *source* is loaded into RECEIVING_SCID_BUFFER and thereafter used for validation for the remainder of the session. (See also related 5.1.3.2.)

NOTE – The EXPEDITED_FRAME_COUNTER increments for each validated Expedited frame received for Version-3 or Version-4 Transfer Frames.

5.6.3 VALIDATED FRAME PROCESSING

5.6.3.1 Validated U-Frames

Validated received U-frames shall be processed per the COP-P process described in section 6.

5.6.3.2 Validated P-Frames

5.6.3.2.1 Validated P-frames shall be processed by first delimiting the contained SPDUs.

5.6.3.2.2 PLCWs contained within SPDUs shall be transferred to the COP-P processor.

5.6.3.2.3 All other reports or directives shall be processed for protocol actions.

6 COMMUNICATIONS OPERATIONS PROCEDURE FOR PROXIMITY LINKS

6.1 OVERVIEW

This section applies to both Version-3 and Version-4 Transfer Frames. An important overview of the COP-P protocol is provided in 2.3.

NOTE –Version-2 transfer frames (AOS) can only be transmitted using the Expedited QoS.

The sending and receiving procedures for COP-P use single-octet variables that are modulo-256 counters. When subtracting or comparing any two of these variables, special handling is required:

– Subtraction: The difference, A–B, is the number of times B needs to be incremented to reach A.

– Comparison: B < A is true if the difference, A–B, is between 1 and 127.
B > A is true if the difference, A–B, is between 128 and 255.
B = A is true if the difference, A–B, is 0.

6.2 SENDING PROCEDURES (FOP-P)

6.2.1 QUEUE

The FOP-P shall maintain a single output queue.

NOTES

1 The **Sent Frame queue** contains Sequence Controlled frames that have been sent but not yet acknowledged by the receiver. (This name is abbreviated to 'Sent queue' in the state table).

2 The local directive CLEAR QUEUE (*Queue Type*) allows for the clearing of frames within a specified queue.

6.2.2 FOP-P VARIABLES

FOP-P variables are:

a) VE(S): an 8-bit positive integer whose value shall represent the sequence number (modulo 256) of the next Expedited frame to be sent.

b) V(S): an 8-bit positive integer whose value shall represent the sequence number (modulo 256) of the next new Sequence Controlled frame to be sent.

c) VV(S): an 8-bit positive integer whose value shall represent the sequence number (modulo 256) to be assigned to the next Sequence Controlled frame to be sent. It equals V(S) unless a retransmission is in-progress.

d) N(R): an 8-bit positive integer that is a copy of the Report Value (see subsection 3.2.4, Supervisory PDU, in reference [6]) from the current PLCW. It shall represent the sequence number plus one (modulo 256) of the last Sequence Controlled frame acknowledged by the receiver.

e) NN(R): an 8-bit positive integer system variable known both inside and outside of the FOP-P state table. It shall contain a copy of the Report Value from the previous valid PLCW.

f) R(R): a Boolean variable (i.e., its value is either *true* or *false*) that is a copy of the Retransmit Flag from the current PLCW. It shall indicate whether or not Sequence Controlled frame(s) need to be retransmitted.

g) RR(R): a Boolean variable that is a copy of the Retransmit Flag from the previous valid PLCW.

h) NEED_PLCW/NEED_STATUS_REPORT: Boolean system variables known both inside and outside of the FOP-P state table. (See 5.1.2.6 for complete usage.) They shall indicate whether or not a new PLCW or status report needs to be sent (the PLCW needs to be sent whenever its contents change).

i) SYNCH_TIMER: a countdown timer that contains the time a sender shall wait to receive a valid PLCW from a receiver before taking action to synchronize with the receiver. The MIB parameter associated with this timer, Synch_Timeout (cf. annex F), represents a desired time value consistent with the Interval Clock frequency (see 5.2). The SYNCH_TIMER counts down when its value is non-zero. When the SYNCH_TIMER counts down to 1, the SYNCH_TIMER expires and the Start Local Resync Event, SE6 is triggered. Subsequently, the timer then underflows to zero, which is the inactive state for the timer.

j) RESYNC: a Boolean variable that tracks the status of resynchronization within the COP-P. This variable is set to *true* when the SET V(R) activity occurs, indicating that FOP-P is resynchronizing. It is reset to *false* by the End Local Resync event, SE7.

6.2.3 FOP-P STATE TABLE EVENTS

6.2.3.1 General Procedures

6.2.3.1.1 'Initialize' shall set the following variables to the indicated values:

a) V(S) = VE(S) = VV(S) = NN(R) = N(R) = 0;

b) R(R) = RR(R) = RESYNC = *false*;

c) NEED_PLCW = NEED_STATUS_REPORT = *true*;

d) CLEAR QUEUE (*Sent queue*); CLEAR QUEUE (*Seq queue*).

6.2.3.1.2 'Remove acknowledged frames from Sent queue' shall remove *n* frames from the Sent queue, where $n = N(R) - NN(R)$ (i.e., the number of times that NN(R) has to be incremented to reach N(R)).

6.2.3.1.3 'Start SYNCH_TIMER' shall, if SYNCH_TIMER is equal to '0', set SYNCH_TIMER to the value of the MIB parameter Synch_Timeout (cf. annex F).

NOTE – If the value of Synch_Timeout is '0', then the SYNCH_TIMER never expires.

6.2.3.1.4 'Clear SYNCH_TIMER' shall set the SYNCH_TIMER value to '0'.

NOTE – This does not trigger a resynchronization.

6.2.3.1.5 'Store this PLCW' shall

a) assign the value of N(R) to NN(R);

b) assign the value of R(R) to RR(R).

6.2.3.1.6 'Send EXP (Expedited) Frame' shall

a) remove frame from EXP queue;

b) assign VE(S) to the frame;

c) increment VE(S);

d) report VE(S) to the I/O Sublayer;

e) transfer this frame to the Frame Sublayer.

6.2.3.1.7 'Resend SEQ (Sequence Controlled) Frame' shall

a) copy frame number VV(S) from the Sent queue;

b) increment VV(S);

c) transfer this frame to the Frame Sublayer.

6.2.3.1.8 'Send New SEQ Frame' shall

a) remove frame from SEQ queue;

b) assign V(S) to the frame;

c) insert a copy of the frame to the end of the Sent queue;

d) increment V(S);

e) increment VV(S);

f) report V(S) to the I/O Sublayer;

g) transfer this frame to the Frame Sublayer.

6.2.3.2 SET V(R) Persistent Activity

NOTE – The SET V(R) persistent activity (including setup of the MIB parameters before the execution of this activity) is defined below. (See 4**, 'Persistence',** for a general overview of how the MIB parameters relate to the persistent activity.)

6.2.3.2.1 Configure the SET V(R) persistent activity. The MIB Parameters that shall be used for setup are (cf. annex F):

a) Activity is SET V(R), that is, Resync;

b) Resync_Waiting_Period (implementation specific) is the amount of time specified for the Resync_Response to be received before the process declares that this activity is to be either repeated or aborted;

c) Resync_Response is the acknowledgement from the Receiver Node that the SET V(R) directive has been accepted, that is, a valid PLCW with report value of N(R) = NN(R) and R(R) = *false* has been received, when RESYNC = *true* (see State S2, Event SE2, FOP-P State table);

d) Resync_Notification is a notification to inform the vehicle controller of success or failure of resynchronization;

e) Resync_Lifetime (implementation specific) is the time period during which the Resynchronization activity shall be repeated until the Resync_Response is detected;

f) Resync_Local is set to *true*.

6.2.3.2.2 Execute the SET V(R) persistent activity. The following shall occur:

a) the FOP-P requests a SET V(R) Persistent Activity by setting RESYNC = *true* (see state S1, event SE4 in the FOP-P state table);

b) the MAC Sublayer builds a SET V(R) directive by: 1) copying NN(R) into the SEQ_CTRL_FSN field within the SET V(R) directive and 2) setting the PCID field in the SET V(R) directive to the value associated with this FOP-P;

c) the MAC Sublayer loads this directive into the MAC queue for transmission and sets MAC_FRAME_PENDING = *true*;

d) the MAC Sublayer sets PERSISTENCE = *true*;

NOTE – This initiates the SET V(R) persistent activity.

e) the FOP-P terminates the SET V(R) Persistent Activity when the SET V(R) Resync_Response is received or when the Resync_Lifetime expires. (See 6.2.3.2.1 c).)

6.2.3.3 FOP-P State Table

Event Number/Name	Resulting Action in State S1 Active	Resulting Action in State S2 Resync
SE0 At Session Startup (see note 1)	*Initialize* (see note 2)	N/A

SE1
Frame Sublayer needs frame to transmit
(see note 3)
 	
(see note 4)
	If Expedited_Frame_Available = true
	Send EXP Frame
	Else if VV(S) < V(S)
	# Continue the in-progress retransmission:
	Resend SEQ Frame
	Else if (SEQUENCE_CONTROLLED_FRAME_AVAILABLE = *true* and V(S)-NN(R)<Transmission_Window)
	Send New SEQ Frame
	Else if NN(R) <V(S)
	# Initiate a Progressive Retransmission:
	VV(S) = NN(R)
	Resend SEQ Frame
	Else
	# There is no Data Services frame to send
	Endif
		N/A

SE2 Valid PLCW Received (see note 5)	If N(R) > NN(R) *Remove acknowledged frames from Sent queue* Endif If R(R) = *true* or N(R)>VV(S) VV(S) = N(R) Endif *Store this PLCW* *Clear SYNCH_TIMER*	If R(R) = false and N(R) = NN(R) Resync = false Persistence=false **State = S1** Endif
SE3 Invalid PLCW Received (see note 5)	*Start SYNCH_TIMER* VV(S) = NN(R)	Ignore
SE4 Synch-timer Expired (see note 6)	Notify vehicle controller that *SYNCH_TIMER* expired (see annex G) If Resync_Local (MIB parameter) = *true* RR(R) = *false* Resync = *true* **State = S2** Endif	N/A
SE5 Set Transmission_Window Request	Accept Set	Accept Set
SE6 Set Synch_Timeout Request	Accept Set	Accept Set

Event Number/Name	Resulting Action in State S1 Active	Resulting Action in State S2 Resync
SE7 Reset Request	Accept *Initialize*	Accept *Initialize* **State = S1**
SE8 Invalid Request	Reject	Reject

NOTES

1 At each session startup, the FOP-P enters state S1 and triggers event SE0 before allowing any other events to occur. If subsequent reconnection is desired without starting a new session, the FOP-P can continue data services by maintaining and using the current state of the FOP-P variables.

2 Procedures are in italics and are described in 6.2.3.1; comments are preceded by the '#' sign.

3 'Progressive Retransmission' causes the frames on the Sent queue to be retransmitted.

4 Transmission_Window (MIB parameter—cf. annex F): The maximum number of Sequence Controlled frames that can be unacknowledged at any given time. For example, if the Transmission_Window is 10 and the sender sends 10 Sequence Controlled frames, the sender must wait for at least one of those frames to be acknowledged by the receiver before it can send any additional Sequence Controlled frames. The value of Transmission_Window cannot exceed 127. When selecting a value for this parameter, the system designer should consider the latency involved whenever frames are required to be retransmitted from the Sent queue before a new Sequence Controlled frame can be transmitted.

5 An incoming PLCW is invalid if any of these conditions is true:

a) PLCW does not match PLCW format;

b) N(R) < NN(R) 'Invalid N(R)—too small';

c) N(R) > V(S) 'Invalid N(R)—too large';

d) R(R) = *true* and N(R) = V(S) 'Retransmit is set though all frames are acknowledged';

e) R(R) = *false* and RR(R) = *true* and N(R) = NN(R) 'Retransmit has cleared though no new frames are acknowledged'.

Otherwise, the PLCW is valid.

6 Setting Resync = *true* causes a Set V(R) persistent activity to be initiated. (See 6.2.3.2, 4, and annex G.)

6.3 RECEIVING PROCEDURES (FARM-P)

6.3.1 FARM-P STATE TABLE

Events	Event #/Name	Action
'Entered this state' at each session startup	**RE0** **Initialization**	R(S) = *false* V(R) = 0 EXPEDITED_FRAME_COUNTER = 0 NEED_PLCW AND NEED_STATUS_REPORT = *true*
Invalid frame arrives	**RE1** **Invalid Frame**	Discard the frame
Valid 'SET V(R)' directive arrives	**RE2** **SET V(R)**	R(S) = *false* Set V(R) to the SEQ_CTRL_FSN in the directive NEED_PLCW = *true*
Valid Expedited frame arrives	**RE3** **Valid Expedited Frame**	Accept/Pass the frame to I/O Sublayer Increment EXPEDITED_FRAME_COUNTER
Valid Sequence Controlled frame arrives, N(S)= V(R)	**RE4** **Sequence Frame 'in-sequence'**	Accept/Pass the frame to I/O Sublayer R(S) = *false* Increment V(R) NEED_PLCW = *true*
Valid Sequence Controlled frame arrives, N(S)>V(R)	**RE5** **Sequence Frame 'gap detected'**	Discard the frame R(S) = *true* NEED_PLCW = *true*
Valid Sequence Controlled frame arrives, N(S)<V(R)	**RE6** **Sequence Frame 'already received'**	Discard the frame
Frame Sublayer requests content for PLCW	**RE7** **Report PLCW contents**	Report value of R(S), V(R), and EXPEDITED_FRAME_COUNTER

6.3.2 INTERNAL FARM-P VARIABLES

The internal FARM-P variables shall be:

a) V(R): an 8-bit positive integer whose value represents the sequence number plus one (modulo 256) of the last Sequence Controlled frame acknowledged by the receiver;

b) R(S): a Boolean variable (i.e., its value is either *true* or *false*) that is copied to the PLCW, indicating whether or not Sequence Controlled frames need to be retransmitted;

c) N(S): an 8-bit positive integer whose value represents the sequence number (modulo 256) contained in the Transfer Frame header of the Transfer Frame;

d) EXPEDITED_FRAME_COUNTER: a 3-bit positive integer whose value represents the number of Expedited frames received (modulo 8). This counter can be used by the receiver to keep track of the number of Expedited frames received over a communications session.

6.3.3 INTERFACE TO THE I/O LAYER

FARM-P shall pass valid Expedited and valid in-sequence U-frames to the I/O Sublayer (reference [6] subsection 4.4).

NOTE – At the I/O Sublayer, these frames are buffered, assembled into packets as required, and then delivered via the specified output port.

ANNEX A

PROTOCOL IMPLEMENTATION CONFORMANCE STATEMENT PROFORMA

(NORMATIVE)

A1 INTRODUCTION

A1.1 OVERVIEW

This annex provides the Protocol Implementation Conformance Statement (PICS) Requirements List (RL) for an implementation of *Space Communications Session Control* (CCSDS 235.1-W-0.4). The PICS for an implementation is generated by completing the RL in accordance with the instructions below. An implementation claiming conformance must satisfy the mandatory requirements referenced in the RL.

The RL support column in this annex is blank. An implementation's completed RL is called the PICS. The PICS states which capabilities and options have been implemented. The following can use the PICS:

- the implementer, as a checklist to reduce the risk of failure to conform to the standard through oversight;
- a supplier or potential acquirer of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma;
- a user or potential user of the implementation, as a basis for initially checking the possibility of interworking with another implementation (it should be noted that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICSes);
- a tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A1.2 ABBREVIATIONS AND CONVENTIONS

The RL consists of information in tabular form. The status of features is indicated using the abbreviations and conventions described below.

Item Column

The item column contains sequential numbers for items in the table.

NOTE – The item-number prefix 'SC' = 'Session Control'.

Feature Column

The feature column contains a brief descriptive name for a feature. It implicitly means 'Is this feature supported by the implementation?'

Status Column

The status column uses the following notations:

M	mandatory.
O	optional.
O.<n>	optional, but support of at least one of the group of options labeled by the same numeral <n> is required.

Support Column Symbols

The support column is to be used by the implementer to state whether a feature is supported by entering Y, N, or N/A, indicating:

Y	Yes, supported by the implementation.
N	No, not supported by the implementation.
N/A	Not applicable.

The support column should also be used, when appropriate, to enter values supported for a given capability.

A1.3 INSTRUCTIONS FOR COMPLETING THE RL

An implementer shows the extent of compliance to the Recommended Standard by completing the RL; that is, the state of compliance with all mandatory requirements and the options supported are shown. The resulting completed RL is called a PICS. The implementer shall complete the RL by entering appropriate responses in the support or values supported column, using the notation described in A1.2. If a conditional requirement is inapplicable, N/A should be used. If a mandatory requirement is not satisfied, exception information must be supplied by entering a reference X*i*, where *i* is a unique identifier, to an accompanying rationale for the noncompliance.

A2 PICS PROFORMA FOR SPACE COMMUNICATIONS SESSION CONTROL (CCSDS 235.1-W-0.4)

A2.1 GENERAL INFORMATION

A2.1.1 Identification of PICS

Date of statement (DD/MM/YYYY)	
PICS serial number	
System conformance statement cross-reference	

A2.1.2 Identification of Implementation Under Test (IUT)

Implementation name	
Implementation version	
Special configuration	
Other information	

A2.1.3 Identification of Supplier

Supplier	
Contact point for queries	
Implementation name(s) and version(s)	
Other information necessary for full identification, for example, name(s) and version(s) for machines and/or operating systems; System name(s)	

A2.1.4 Identification of Specification

CCSDS 235.1-W-0.4	
Have any exceptions been required? NOTE – A YES answer means that the implementation does not conform to the Recommended Standard. Non-supported mandatory capabilities are to be identified in the PICS, with an explanation of why the implementation is non-conforming.	Yes [ ] No [ ]

A2.2 REQUIREMENTS LIST

State Control Variables					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
1	Mode	5.1.1.1	M	inactive, *connecting-T*, *connecting-L*, active	
2	DUPLEX	5.1.1.2	M	full, half, simplex transmit, simplex receive	
3	TRANSMIT	5.1.1.3	M	on, off	
4	SUB-STATE	5.1.1.4	M	0-7	

Operational Control Variables					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
5	Session Termination	5.1.2.1	M	0–5	
6	COMM_CHANGE	5.1.2.2	M	0–5	
7	SYMBOL_INLOCK_STATUS	5.1.2.3	M	0, 1	
8	Modulation	5.1.2.4	M	on, off	
9	NEED_PLCW and NEED_STATUS_REPORT	5.1.2.6	M	true, false	
10	REMOTE_SCID_BUFFER	5.1.2.7	M	0–1023	
11	COMMUNICATION_VALUE_BUFFER	5.1.2.8	M		
12	RECEIVING_SCID_BUFFER	5.1.2.9	M	0–1023	
13	RECEIVING_PCID_BUFFER	5.1.2.10	M	0, 1	

Elements and Events That Affect State Status					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
14	Interval_Clock	5.2.1.1.1	M		
15	WAIT TIMER (WT) and Associated Events	5.2.1.2	M		
16	CARRIER_LOSS_TIMER and Associated Events	5.2.1.3	M		
17	PLCW TIMER and Associated Events	5.2.1.4	M		
18	OUTPUT FIFO	5.2.2.2	M		
19	NO_FRAMES_PENDING	5.2.2.3	M		

Directives					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
20	Local Directives	5.2.3.1	M		
21	SET MODE	5.2.3.1.1	M		
22	SET INITIALIZE MODE	5.2.3.1.1.2	M		
23	Local COMM_CHANGE (LCCD)	5.2.3.1.3	M		
24	LOAD COMMUNICATIONS VALUE BUFFER	5.2.3.1.4	M		
25	LOCAL_NO_MORE_DATA (LNMD)	5.2.3.1.5	O		
26	SET DUPLEX	5.2.3.1.6	M		
27	SET RECEIVING SCID BUFFER	5.2.3.1.7	M		
28	READ STATUS	5.2.3.1.8	O		
29	Remote directives	5.2.3.2	M		
30	SET TRANSMITTER PARAMETERS directive	5.2.3.2.2	M		
31	SET RECEIVER PARAMETERS directive	5.2.3.2.5	M		
32	SET PL_EXTENSIONS directive	5.2.3.2.6	M		
33	The Set CONTROL PARAMETERS directive (Time Sample Field)	5.2.3.2.7	M		
34	Initialized values of session control variables	Table 5-5	M		

Duplex/Simplex Operations					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
35	Full duplex operations	5.3.2, Tables 5-6, 5-7, 5-8	M		
36	Half duplex operations	5.3.3, Table 5-9, 5-10, 5-11	O		
37	Simplex operations	5.3.4, Table 5-12	O		

Interfaces with the Physical Layer					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
38	Output Interfaces	5.4.1	M		
39	The TRANSMIT parameter	5.4.1.1	M	On, Off	
40	Frame output	5.4.1.2	M		
41	The MODULATION parameter	5.4.1.3	M	On, Off	
42	Input interfaces	5.4.2	M		
Sending Operations					
43	Session Establishment	5.5.1	M		
44	Reset NEED_PLCW or NEED_STATUS_REPORT	5.1.2.6.2–5.1.2.6.2	M	True, False	
45	Setting No_Frames_Pending	5.2.2.3	M	True, False	
46	Setting OUTPUT_FIFO=empty	5.2.2.2	M	True, False	
Receiving Operations					
47	Frame Reception	5.6.1	M		
48	Frame Validation Criteria	5.6.2	M		
49	Validated Frame Processing	5.6.3	M		

Variable-length Supervisory Protocol Data Field Formats					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
50	SET TRANSMITTER PARAMETERS Directive	B2	M		
51	SET CONTROL PARAMETERS Directive	B3	M		
52	SET RECEIVER PARAMETERS Directive	B4	M		
53	SET V(R) Directive	B5	M		
54	REPORT REQUEST Directive	B6	O		
55	SET PL EXTENSIONS Directive	B7	M		
56	REPORT SOURCE SPACECRAFT ID Directive	B7.13	O		

Communications Operations Procedure for Proximity Links (COP-P)					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
57	Implement COP-P Protocol	6	M		
58	DATA SERVICES OPERATIONS (COP-P): Modulo-256 counters will be used for sequence number in the COP-P protocol	6	M		
59	SENDING PROCEDURES (FOP-P)	6.2	M		
60	The FOP-P Sent Frame Queue	6.2.1	M		
FOP-P Variables					
61	VE(S)	6.2.2 a)	M		
62	V(S)	6.2.2 b)	M		
63	VV(S)	6.2.2 c)	M		
64	N(R):	6.2.2 d)	M		
65	NN(R):	6.2.2 e)	M		
66	R(R):	6.2.2 f)	M		
67	RR(R):	6.2.2 g)	M		
68	NEED_PLCW/NEED_STATUS_REPORT	6.2.2 h)	M		
69	SYNCH_TIMER	6.2.2 i)	M		
70	RESYNC	6.2.2 j)	M		
71	FOP-P General Procedures	6.2.3.1	M		
72	Set V(R) persistent activity	6.2.3.2	M		
73	Configure the SET V(R) persistent activity	6.2.3.2.1	M		
74	Execute the SET V(R) persistent activity	6.2.3.2.2	M		
75	FOP-P State Table	6.2.3.3	M		
76	DATA SERVICES RECEIVING OPERATIONS	6.3	M		
77	FARM-P STATE TABLE	6.3.1	M		
INTERNAL FARM-P PARAMETERS					
78	V(R):	6.3.2 a)	M		
79	R(S):	6.3.2 b)	M		
80	N(S):	6.3.2 c)	M		
81	EXPEDITED_FRAME_COUNTER	6.3.2 d)	M		
82	INTERFACE TO THE I/O SUBLAYER	6.3.3	M		

Management Information Base Parameters (alphabetical)					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
83	Acquisition_Idle_Duration	5.1.3.4	M		
84	Carrier_Loss_Timer_Duration	5.1.3.6	M		
85	Carrier_Only_Duration	5.1.3.3	M		
86	Comm_Change_Lifetime	5.1.3.10	M		
87	Comm_Change_Notification	5.1.3.9	M		
88	Comm_Change_Response	5.1.3.8	M		
89	Comm_Change_Waiting_Period	5.1.3.7	M		
90	Drop_Carrier_Duration	table 5-1	M		
91	Hail_Lifetime	5.1.3.14	M		
92	Hail_Notification	5.1.3.13	M		
93	Hail_Response	5.1.3.12	M		
94	Hail_Wait_Duration	5.1.3.11	M		
95	Hailing_Channel	5.1.3.15	M		
96	Hailing_Data_Rate	5.1.3.15	M		
97	Hailing_Symbol/_Rate	5.1.3.15	M		
98	Interval_Clock	5.2.1.1.1	M		
99	Maximum_Failed_Token_Passes	5.3.3, table 5-9	O		
100	Persistence_Wait_Time	5.3.2, table 5-7	M		
101	PLCW_Repeat_Interval	5.1.3.19	M		
102	Receive_Duration	5.1.3.18	O		
103	Resync_Lifetime	6.2.3.2	M		
104	Resync_Local	6.2.3.2.1, 6.2.3.3	M		
105	Resync_Notification	6.2.3.2	M		
106	Resync_Response	6.2.3.2	M		
107	Resync_Waiting_Period	6.2.3.2	M		
108	Send_Duration	5.1.3.17	O		
109	Synch_Timeout	6.2.2	M		
110	Tail_Idle_Duration	5.1.3.5	M		
111	Test_Source	5.1.3.2	M		
112	Transmission_Window	6.2.3.3	M		

Notifications to Vehicle Controller					
Item	**Description**	**Reference**	**Status**	**Values Allowed**	**Support**
113	1 RESULT OF PERSISTENT ACTIVITY	Annex G #1	M		
114	2 STATE CONTROL STATUS	Annex G #2	M		
115	3 INVALID FRAME SOURCE	Annex G #3	M		
116	4 TIMING SERVICES INSTANCE	Annex G #4	M		
117	5 NO CARRIER RECEIVED - HALF DUPLEX	Annex G #5	O		
118	6 NO DATA TRANSFERRED - HALF DUPLEX	Annex G #6	O		
119	7 SENDER EXCEEDED PRESCRIBED TRANSMISSION PERIOD - HALF DUPLEX	Annex G #7	O		
120	8 COP-P LOSS OF SYNCHRONIZATION	Annex G #8	M		
121	9 CARRIER_LOSS_TIMER UNDERFLOWS	Annex G #9	M		
122	10 END OF SESSION (# octets received)	Annex G #10	M		
123	11 CARRIER ONLY RECEIVED	Annex G #11	M		

ANNEX B

TYPE 1 SPDU: DIRECTIVE/REPORT/PLCW SPDU DATA FIELD

(NORMATIVE)

<table>
<tr><td colspan="12"></td><td>Directive Type
3 Bits (13, 14, 15)</td></tr>
<tr><td colspan="3">Mode
(0,1,2)</td><td colspan="3">Data Rate
(3,4,5,6)</td><td>Modulation
(7)</td><td colspan="2">Data Encoding
(8,9)</td><td colspan="2">Frequency
(10,11,12)</td><td>‘000’ = SET TRANSMITTER PARAMETERS</td></tr>
<tr><td colspan="5">Time Sample
(1,2,3,4,5)</td><td colspan="3">Duplex
(6,7,8)</td><td colspan="2">Reserved
(9,10)</td><td>Remote No More Data
(11)</td><td>Token
(12)</td><td>‘001’ = SET CONTROL PARAMETERS</td></tr>
<tr><td colspan="3">Mode
(0,1,2)</td><td colspan="3">Data Rate
(3,4,5,6)</td><td>Modulation
(7)</td><td colspan="2">Data Encoding
(8,9)</td><td colspan="2">Frequency
(10,11,12)</td><td>‘010’ = SET RECEIVER PARAMETERS</td></tr>
<tr><td colspan="8">Receiver Frame Sequence Number (SEQ_CTRL_FSN)
(0,1,2,3,4,5,6,7)</td><td colspan="4">Reserved
(8,9,10,11,12)</td><td>‘011’ = SET V(R)</td></tr>
<tr><td colspan="3">Reserved
(0,1,2)</td><td colspan="4">Status Report Request
(3,4,5,6,7)</td><td colspan="2">Time-Tag Request
(8,9,10)</td><td>PCID 0 PLCW Request
(11)</td><td>PCID 1 PLCW Request
(12)</td><td>‘100’ = REPORT REQUEST</td></tr>
<tr><td colspan="12"></td><td>‘101’ = Reserved</td></tr>
<tr><td>Direction
(0)</td><td>Freq. Table
(1)</td><td>Rate Table
(2)</td><td colspan="2">Carrier Mod.
(3,4)</td><td colspan="2">Data Mod.
(5,6)</td><td colspan="2">Mode Select
(7,8)</td><td colspan="2">Scrambler
(9,10)</td><td>Diff. Encoding
(11)</td><td>R-S Code
(12)</td><td>‘110’ = SET PL EXTENSIONS</td></tr>
<tr><td colspan="10">Source Spacecraft ID
(0.1.2.3.4.5.6.7.8.9)</td><td colspan="2">Reserved
(10,11,12)</td><td>‘111’ = REPORT SOURCE SCID</td></tr>
</table>

Figure B-1: Type 1 SPDU Data Field Contents

B1 GENERAL

B1.1 The Directive/Report/PLCW SPDU shall be used for space link supervisory configuration and control of the transceiver and its operation.

B1.2 The SPDU data field shall be a container that can hold up to seven 16-bit discrete self-delimiting and self-identifying directives:

a) each directive shall have a specific functionality;

b) each directive shall be 16 bits in length and shall be self identified by the value in the Directive Type field (contained in bits 13, 14, and 15 of the directive);

c) the directives shall be concatenated without intervening bits within the data field.

NOTE – Figure B-1 shows the Type 1 SPDU Data Field Contents.

B2 SET TRANSMITTER PARAMETERS DIRECTIVE

B2.1 GENERAL

The SET TRANSMITTER PARAMETERS directive shall consist of six fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) Transmitter Frequency (3 bits);

c) Transmitter Data Encoding (2 bits);

d) Transmitter Modulation (1 bit);

e) Transmitter Data Rate (4 bits);

f) Transmitter (TX) Mode (3 bits).

NOTE – The structural components of the SET TRANSMITTER PARAMETERS directive are shown in figure B-2.

Bit 0 Bit 15

TX Mode 3 bits	TX Data Rate 4 bits	TX Modulation 1 bit	TX Data Encoding 2 bits	TX Frequency 3 bits	Directive Type 3 bits
0,1,2	3,4,5,6	7	8,9	10,11,12	13,14,15

Figure B-2: SET TRANSMITTER PARAMETERS Directive

B2.2 DIRECTIVE TYPE

B2.2.1 Bits 13–15 of the SET TRANSMITTER PARAMETERS directive shall contain the Directive Type.

B2.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value '000' for the SET TRANSMITTER PARAMETERS directive.

B2.3 TRANSMITTER FREQUENCY

B2.3.1 General

Bits 10–12 of the SET TRANSMITTER PARAMETERS directive shall be used to set the transmitter frequency of the partnered transceiver to the desired value.

B2.3.2 Return Transmitter Frequency (e.g., Orbiter as Initiator; Landed Asset as Responder)

In the context of the forward link, this 3-bit field shall define the transmit frequency of the responder. Actual frequency assignments are given in the Physical Layer. (See section 5.)

'000'	'001'	'010'	'011'	'100'	'101'	'110'	'111'
Ch0R	Ch1R	Ch2R	Ch3R	Ch4R	Ch5R	Ch6R	Ch7R

B2.4 TRANSMITTER DATA ENCODING

Bits 8–9 of the SET TRANSMITTER PARAMETERS directive shall contain the following coding options:

a) '00' = LDPC(2048,1024) rate 1/2 code (see reference [2]);

b) '01' = Convolutional Code(7,1/2) (G2 vector inverted) with attached CRC-32 (see reference [4]);

c) '10' = Bypass all codes;

d) '11' = Concatenated (R-S(204,188), CC(7,1/2)) Codes.

NOTE – R-S(204,188) with CC(7,1/2) code is an ETSI standard. This option is not required for cross support. (See reference [J1] for more details.)

B2.5 TRANSMITTER MODULATION

Bit 7 of the SET TRANSMITTER PARAMETERS directive shall contain the transmission modulation options:

a) '0' = Coherent frequency PSK;

b) '1' = Non-coherent frequency PSK.

B2.6 TRANSMITTER DATA RATE

B2.6.1 Bits 3–6 of the SET TRANSMITTER PARAMETERS directive shall contain one of the following transmission data rates (rates in kb/s, e.g., 4 = 4000 b/s) prior to encoding.

NOTE – Because of the NASA Mars Surveyor Project 2001 Odyssey implementation, there is an added constraint on the use of the values in the Data Rate field for 8, 32, 128, 256 kb/s. Data rate selection is linked to the modulation field value as shown in the tables below. NC indicates non-coherent PSK, and C indicates coherent PSK. R1 through R4 indicate the field is reserved for future definition by the CCSDS. 1, 512, 1024, and 2048 kb/s data rates can only be selected using the SET PL EXTENSIONS directive (see B7).

B2.6.2 Ordered by Data Rate:

‘1000’	‘1001’	‘0000’	‘0001’	‘1100’	‘0010’	‘0011’	‘1101’	‘0100’	‘0101’	‘0110’	‘0111’	‘1010’	‘1011’	‘1110’	‘1111’
2	4	8 NC	8 C	16	32 NC	32 C	64	128 NC	128 C	256 NC	256 C	R1	R2	R3	R4

B2.6.3 Ordered by Bit pattern:

‘0000’	‘0001’	‘0010’	‘0011’	‘0100’	‘0101’	‘0110’	‘0111’	‘1000’	‘1001’	‘1010’	‘1011’	‘1100’	‘1101’	‘1110’	‘1111’
8 NC	8 C	32 NC	32 C	128 NC	128 C	256 NC	256 C	2	4	R1	R2	16	64	R3	R4

B2.6.4 Coded symbol (R_{cs}) and data rate (R_d) table:

Coded Symbol Rates (R_{cs})	Uncoded Data Rates (R_d) $R_d = R_{cs}$	Convolutionally Coded Data Rates (R_d) $R_d = .5 * R_{cs}$	LDPC computed data rates (R_d) $R_d = .48484 * R_{cs}$
1000	1000	N/A	N/A
2000	2000	1000	969.6969697
4000	4000	2000	1939.393939
8000	8000	4000	3878.787879
16000	16000	8000	7757.575758
32000	32000	16000	15515.15152
64000	64000	32000	31030.30303
128000	128000	64000	62060.60606
256000	256000	128000	124121.2121
512000	512000	256000	248242.4242
1024000	1024000	512000	496484.8485
2048000	2048000	1024000	992969.697
4096000	N/A	2048000	1985939.394

B2.6.5 1, 512, 1024, and 2048 kb/s data rates can only be selected using the SET PL EXTENSIONS directive (see B7).

B2.7 TRANSMITTER MODE

Bits 0–2 of the SET TRANSMITTER PARAMETERS directive shall contain the Transmitter Mode options. This field identifies the operating mode of the transmitter:

a) '000' = Mission Specific;

b) '001' = Proximity-1 Protocol;

c) '010' = Mission Specific;

d) '011' = Mission Specific;

e) '100' = Mission Specific;

f) '101' = Mission Specific;

g) '110' = Reserved by CCSDS;

h) '111' = Reserved by CCSDS.

B3 SET CONTROL PARAMETERS

B3.1 GENERAL

B3.1.1 The SET CONTROL PARAMETERS directive shall consist of six fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) Token (1 bit);

c) Remote No More Data (1 bit);

d) Reserved (2 bits);

e) Duplex (3 bits);

f) Time Sample (6 bits).

B3.1.2 This directive is used to set from zero to four control parameters at a time: 1) setting the token for half-duplex operations; 2) setting the Remote No More Data condition for session termination in full or half duplex; 3) setting the Duplex parameter; 4) setting the number of time samples to be taken during timing services.

NOTE – The structural components of the SET CONTROL PARAMETERS directive are shown in figure B-3.

Bit 0 Bit 15

Time Sample 6 bits	Duplex 3 bits	Reserved 2 bits	Remote No More Data 1 bit	Token 1 bit	Directive Type 3 bits
0,1,2,3,4,5	6,7,8	9,10	11	12	13,14,15

Figure B-3: SET CONTROL PARAMETERS Directive

B3.2 DIRECTIVE TYPE

B3.2.1 Bits 13–15 of the SET CONTROL PARAMETERS directive shall contain the Directive Type.

B3.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value '001' to identify the SET CONTROL PARAMETERS directive.

B3.3 TOKEN

Bit 12 of the SET CONTROL PARAMETERS directive shall contain the value of the Token field as follows:

a) '0' = No Change;

b) '1' = Transmit.

NOTE – This field either notifies the remote node that there is no change in who has permission to transmit (i.e., ignore this field) or commands the remote node to the transmit state.

B3.4 REMOTE NO MORE DATA

Bit 11 of the SET CONTROL PARAMETERS directive shall contain the Remote No More Data field as follows:

a) '0' = No Change;

b) '1' = Remote Node has No More Data to Send (RNMD).

NOTE – This field either notifies the recipient node that there is no change in the remote node's data state (i.e., ignore this field) or notifies the recipient node that the remote node has no more data to send, in which case, the session may be terminated when the recipient node locally has no more data to send.

B3.5 RESERVED

Bits 9–10 of the SET CONTROL PARAMETERS directive shall contain spares and be set to 'all zero'.

B3.6 DUPLEX

Bits 6–8 of the SET CONTROL PARAMETERS directive shall contain the Duplex field as follows:

a) '000' = No Change;

b) '001' = Full Duplex;

c) '010' = Half Duplex;

d) '011' = Simplex Transmit;

e) '100' = Simplex Receive;

f) '101' = Reserved;

g) '110' = Reserved;

h) '111' = Reserved.

NOTE – This field either notifies the recipient node that there is no change in the remote node's Duplex state (i.e., ignore this field) or notifies the recipient node to change the directionality of communication accordingly.

B3.7 TIME SAMPLE

Bits 0–5 of the SET CONTROL PARAMETERS directive shall contain the Time Sample field. When this field is non-zero, it notifies the recipient to capture the time and frame sequence number (associated with the protocol timing service; see reference [6], section 5) for the next *n* frames received, where *n* is the number of Transfer Frames contained within the Time Sample Field.

B4 SET RECEIVER PARAMETERS DIRECTIVE

B4.1 GENERAL

The SET RECEIVER PARAMETERS directive shall consist of six fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) Receiver Frequency (3 bits);

c) Receiver Data Decoding (2 bits);

d) Receiver Modulation (1 bit);

e) Receiver Data Rate (4 bits);

f) Receiver (RX) Mode (3 bits).

NOTE – The structural components of the SET RECEIVER PARAMETERS directive are shown in figure B-4.

Bit 0 Bit 15

RX Mode 3 bits	RX Rate 4 bits	RX Modulation 1 bit	RX Data Decoding 2 bits	RX Frequency 3 bits	Directive Type 3 bits
0,1,2	3,4,5,6	7	8,9	10,11,12	13,14,15

Figure B-4: SET RECEIVER PARAMETERS Directive

B4.2 DIRECTIVE TYPE

B4.2.1 Bits 13–15 of the SET RECEIVER PARAMETERS directive shall contain the Directive Type.

B4.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘010’ for the SET RECEIVER PARAMETERS directive.

B4.3 RECEIVER FREQUENCY

B4.3.1 General

Bits 10–12 of the SET RECEIVER PARAMETERS directive shall be used to set the receiver frequency of the partnered transceiver to the desired value.

B4.3.2 Forward Receive Frequency (e.g., Orbiter as Initiator, Landed Asset as Responder)

In the context of the forward link, this 3-bit field shall define the receive frequency of the responder. Actual frequency assignments are given in the Physical Layer (see section 5).

'000'	'001'	'010'	'011'	'100'	'101'	'110'	'111'
Ch0F	Ch1F	Ch2F	Ch3F	Ch4F	Ch5F	Ch6F	Ch7F

B4.4 RECEIVER DATA DECODING

Bits 8–9 of the SET RECEIVER PARAMETERS directive shall contain the following coding options:

'00' = LDPC (2048,1024) rate 1/2 code (see reference [2]);

'01' = Convolutional Code(7,1/2) (G2 vector inverted) with attached CRC-32 (see reference [4]);

'10' = Bypass all codes;

'11' = Concatenated R-S(204,188), CC(7,1/2).

NOTE – R-S(204,188) with CC(7,1/2) code is an ETSI standard. This option is not required for cross support. It is the R-S(255,239) code shortened by q = 51. (See reference [J1] for more details.)

B4.5 RECEIVER MODULATION

Bit 7 of the SET RECEIVER PARAMETERS directive shall contain the following transmission modulation options:

a) '0' = Coherent frequency PSK;

b) '1' = Non-coherent frequency PSK.

B4.6 RECEIVER DATA RATE

B4.6.1 Bits 3–6 of the SET RECEIVER PARAMETERS directive shall contain one of the following receiver data rates (rates in kb/s, e.g., 4 = 4000 b/s) after decoding.

NOTE – Because of the NASA Mars Surveyor Project 2001 Odyssey implementation, there is an added constraint on the use of the values in the Data Rate field for 8, 32, 128, and 256 kb/s. Data rate selection is linked to the modulation field value as shown in the tables below ('NC' indicates non-coherent, and c indicates coherent). R1 through R4 indicates the field is reserved for future definition by the CCSDS. 1, 512, 1024, and 2048 kb/s data rates can only be selected using the SET PL EXTENSIONS directive (see B7).

B4.6.2 Ordered by data rate:

'1000'	'1001'	'0000'	'0001'	'1100'	'0010'	'0011'	'1101'	'0100'	'0101'	'0110'	'0111'	'1010'	'1011'	'1110'	'1111'
2	4	8 NC	8 C	16	32 NC	32 C	64	128 NC	128 C	256 NC	256 C	R1	R2	R3	R4

B4.6.3 Ordered by bit pattern:

'0000'	'0001'	'0010'	'0011'	'0100'	'0101'	'0110'	'0111'	'1000'	'1001'	'1010'	'1011'	'1100'	'1101'	'1110'	'1111'
8 NC	8 C	32 NC	32 C	128 NC	128 C	256 NC	256 C	2	4	R1	R2	16	64	R3	R4

B4.6.4 Coded symbol (R_{cs}) and data rate (R_d) table:

Coded Symbol Rates (R_{cs})	Uncoded Data Rates (R_d) $R_d = R_{cs}$	Convolutionally Coded Data Rates (R_d) $R_d = .5 * R_{cs}$	LDPC Computed Data Rates (R_d) $R_d = .48484 * R_{cs}$
1000	1000	N/A	N/A
2000	2000	1000	969.6969697
4000	4000	2000	1939.393939
8000	8000	4000	3878.787879
16000	16000	8000	7757.575758
32000	32000	16000	15515.15152
64000	64000	32000	31030.30303
128000	128000	64000	62060.60606
256000	256000	128000	124121.2121
512000	512000	256000	248242.4242
1024000	1024000	512000	496484.8485
2048000	2048000	1024000	992969.697
4096000	N/A	2048000	1985939.394

NOTE – 1, 512, 1024, and 2048 kb/s data rates can only be selected using the SET PL EXTENSIONS directive (see B7).

B4.7 RECEIVER MODE

Bits 0–2 of the SET RECEIVER PARAMETERS directive shall contain the receiver mode options. This field identifies the operating mode of the receiver:

a) '000' = Mission Specific;

b) '001' = Proximity-1 Protocol;

c) '010' = Mission Specific;

d) ‘011’ = Mission Specific;

e) ‘100’ = Mission Specific;

f) ‘101’ = Mission Specific;

g) ‘110’ = Reserved by CCSDS;

h) ‘111’ = Reserved by CCSDS.

B5 SET V(R) DIRECTIVE

B5.1 GENERAL

The SET V(R) directive shall consist of three fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) Spare (5 bits);

c) Receiver Frame Sequence Number (SEQ_CTRL_FSN) (8 bits).

NOTE – The structural components of the SET V(R) directive are shown in figure B-5.

Bit 0 Bit 15

Receiver Frame Sequence Number SEQ_CTRL_FSN 8 bits	Spare 5 bits	Directive Type 3 bits
0,1,2,3,4,5,6,7	8,9,10,11,12	13,14,15

Figure B-5: SET V(R) Directive

B5.2 DIRECTIVE TYPE

B5.2.1 Bits 13–15 of the SET V(R) directive shall contain the Directive Type.

B5.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘011’ to identify the SET V(R) directive.

B5.3 SPARE

Bits 8–12 of the SET V(R) directive shall contain spare bits, set to ‘all zero’.

B5.4 RECEIVER FRAME SEQUENCE NUMBER

Bits 0–7 of the SET V(R) directive shall contain the value of the Frame Sequence Number (SEQ_CTRL_FSN) to which the receiving unit of the partnered transceiver is to be set.

B6 REPORT REQUEST DIRECTIVE

B6.1 GENERAL

The REPORT REQUEST directive is the mechanism by which either (1) a status report, (2) a time-tag, or (3) a PLCW per PCID can be requested of a protocol node. It shall consist of seven fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) PCID 1 PLCW Request (1 bit);

c) PCID 0 PLCW Request (1 bit);

d) Time-Tag Request (3 bits);

e) Status Request (5 bits);

f) Spare (3 bits).

NOTE – The structural components of the REPORT REQUEST directive are shown in figure B-6.

Bit 0 Bit 15

Spare 3 bits	Status Report Request 5 bits	Time-Tag Request 3 bit	PCID 0 PLCW Request 1 bit	PCID 1 PLCW Request 1 bit	Directive Type 3 bits
0,1,2	3,4,5,6,7	8,9,10	11	12	13,14,15

Figure B-6: Report Request

B6.2 DIRECTIVE TYPE

B6.2.1 Bits 13–15 of the REPORT REQUEST directive shall contain the Directive Type.

B6.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘100’.

B6.3 PHYSICAL CHANNEL ID 1 PLCW REPORT REQUEST FIELD

Bit 12 of the REPORT REQUEST directive shall indicate whether a PLCW report for PCID 1 is required:

a) '1' = PLCW report is needed for PCID 1; transmit this report on the same PCID that the report request arrived on.

b) '0' = PLCW report is not required.

B6.4 PHYSICAL CHANNEL ID 0 PLCW REPORT REQUEST FIELD

Bit 11 of the REPORT REQUEST directive shall indicate whether a PLCW report for PCID 0 is required:

a) '1' = PLCW report is needed for PCID 0; transmit this report on the same PCID that the report request arrived on.

b) '0' = PLCW report is not required.

B6.5 TIME-TAG REQUEST FIELD

Bits 8–10 of the directive, if set to a value other than '000', shall indicate a request to the remote transceiver to initiate a protocol time tag exchange (see reference [6], section 5).

B6.6 STATUS REPORT REQUEST

B6.6.1 The value contained in bits 3–7 of the REPORT REQUEST directive shall indicate the type of status report desired.

B6.6.2 If set to '00000', a status report is not required.

B6.6.3 The types of status reports are reserved for CCSDS use.

B6.7 SPARES

Bits 0–2 of the REPORT REQUEST directive shall contain spare bits set to 'all zero'.

B7 SET PL EXTENSIONS

B7.1 OVERVIEW

The SET PL EXTENSIONS directive is the mechanism by which additional Physical Layer parameters can be enabled or disabled. This directive is transferred across the space link from the local transceiver to the remote transceiver. This directive is provided for compatibility between transceivers with extensions in addition to those discussed in this Recommended Standard**.**

B7.2 GENERAL

The SET PL EXTENSIONS directive shall consist of ten fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) R-S Code (1 bit);

c) Differential Mark Encoding (1 bit);

d) Scrambler (2 bits);

e) Mode Select (2 bits);

f) Data Modulation (2 bits);

g) Carrier Modulation (2 bits);

h) Rate Table (1 bit);

i) Frequency Table (1 bit);

j) Direction (1 bit).

NOTE – The structural components of the SET PL EXTENSIONS directive are shown in figure B-7.

Bit 0 Bit 15

Direction	Freq Table	Rate Table	Carrier MOD	Data MOD	Mode Select	Scrambler	Differential Mark Encoding	R-S Code	Directive Type
1 bit	1 bit	1 bit	2 bits	2 bits	2 bits	2 bits	1 bit	1 bit	3 bits
0	1	2	3,4	5,6	7,8	9,10	11	12	13,14,15

Figure B-7: SET PL EXTENSIONS

B7.3 DIRECTIVE TYPE

B7.3.1 Bits 13–15 of the SET PL EXTENSIONS directive shall contain the Directive Type.

B7.3.2 The 3-bit Directive Type field identifies the directive type and shall contain the binary value '110'.

B7.4 REED-SOLOMON CODE

B7.4.1 Bit 12 of the SET PL EXTENSIONS directive shall indicate which R-S Code is used:

a) '0' = R-S(204,188) code;

b) '1' = R-S(255,239) code.

B7.4.2 Neither of these R-S Codes is specified by CCSDS in other Recommended Standards and therefore they are not required for cross-support.

B7.5 DIFFERENTIAL MARK ENCODING

Bit 11 of the SET PL EXTENSIONS directive shall indicate whether Differential Mark Encoding is enabled:

a) '0' = No differential encoding;

b) '1' = Differential encoding enabled. The current data bit is exclusive ORed with the previously transmitted bit to determine the value of the current transmitted bit. When the current data bit is a '1', then the current encoder output bit level changes relative to the previous output value. If the data bit is a '0', then the current encoder output bit level remains constant relative to the previous output value. (See table below.)

Option b) is not required for cross-support except for those missions required to interoperate with NASA MRO.

B7.6 SCRAMBLER

B7.6.1 Bits 9–10 of the SET PL EXTENSIONS directive shall indicate if and what type of digital bit scrambling is used:

a) '00' = Bypass all bit scrambling;

b) '01' = CCITT bit scrambling enabled (see reference [J2]);

c) '10' = Bypass all bit scrambling;

d) '11' = IESS bit scrambling enabled (see reference [J3]).

B7.6.2 None of these Scrambler options are specified by CCSDS in other Recommended Standards and therefore are not required for cross-support.

B7.7 MODE SELECT

Bits 7–8 of the SET PL EXTENSIONS directive shall indicate the type of carrier suppression used:

a) '00' = Suppressed Carrier (Requires transmit side utilize Modulation Index of 90° and transmit/receive sides utilize Differential Mark Encoding/Decoding);

b) '01' = Residual Carrier;

c) '10' = Reserved;

d) '11' = Reserved.

Option a) is not required for cross-support except for those missions required to interoperate with NASA MRO.

B7.8 DATA MODULATION

Bits 5–6 of the SET PL EXTENSIONS directive shall indicate the type of data modulation used:

a) '00' = NRZ-L;

b) '01' = Bi-Phase-Level (Manchester);

c) '10' = Reserved;

d) '11' = Reserved.

Option a) is not required for cross-support except for those missions required to interoperate with NASA MRO.

B7.9 CARRIER MODULATION

Bits 3–4 of the SET PL EXTENSIONS directive shall indicate the type of carrier modulation to be used:

a) '00' = No Modulation;

b) '01' = PSK;

c) '10' = FSK;

d) '11' = QPSK.

Options c) and d) are not required for cross-support.

B7.10 RATE TABLE

Bit 2 of the SET PL EXTENSIONS directive shall indicate which set of data rates prior to encoding shall be used.

a) '0' = Default Set defined in the Data Rate Field of the SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS Directives in this annex;

b) '1' = Extended Physical Layer Data Rate Set defined below.

'0000' = 1000 b/s	'0100' = 16000 b/s	'1000' = 256000 b/s	'1100' = Reserved
'0001' = 2000 b/s	'0101' = 32000 b/s	'1001' =512000 b/s	'1101' = Reserved
'0010' = 4000 b/s	'0110' = 64000 b/s	'1010' = 1024000 b/s	'1110' = Reserved
'0011' = 8000 b/s	'0111' = 128000 b/s	'1011' = 2048000 b/s	'1111' = Reserved

Option a) is required for cross-support. Option b) is required for cross-support for data rates less than 2000 b/s and greater than 256000 b/s.

B7.11 CODED SYMBOL (R_{CS}) AND DATA RATE (R_D) TABLE:

Coded Symbol Rates (R_{cs})	Uncoded Data Rates (R_d) $R_d = R_{cs}$	Convolutionally Coded Data Rates (R_d) $R_d = .5 * R_{cs}$	LDPC Computed Data Rates (R_d) $R_d = .48484 * R_{cs}$
1000	1000	N/A	N/A
2000	2000	1000	969.6969697
4000	4000	2000	1939.393939
8000	8000	4000	3878.787879
16000	16000	8000	7757.575758
32000	32000	16000	15515.15152
64000	64000	32000	31030.30303
128000	128000	64000	62060.60606
256000	256000	128000	124121.2121
512000	512000	256000	248242.4242
1024000	1024000	512000	496484.8485
2048000	2048000	1024000	992969.697
4096000	N/A	2048000	1985939.394

B7.12 FREQUENCY TABLE

B7.12.1 General

Bit 1 of the SET PL EXTENSIONS directive shall indicate what set of frequencies is to be used:

a) '0' = Channels 0 – 7 defined in the Frequency Field of the SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS Directives and specifically in the Physical Layer;

b) '1' = Channels 8 – 15 defined in the Extended Physical Layer Frequency Set defined below.

B7.12.2 Forward Link (e.g., Orbiter as Initiator; Landed Asset as Responder)

In the context of the forward link, this 3-bit field shall define the receive frequency of the responder. Actual frequency assignments are given in the Physical Layer (see section 5).

'000'	'001'	'010'	'011'	'100'	'101'	'110'	'111'
Ch8F	Ch9F	Ch10F	Ch11F	Ch12F	Ch13F	Ch14F	Ch15F

B7.12.3 Return Link (e.g., Orbiter as Initiator; Landed Asset as Responder)

In the context of the return link, this 3-bit field shall define the transmit frequency of the responder. Actual frequency assignments are given in the Physical Layer (see section 5).

'000'	'001'	'010'	'011'	'100'	'101'	'110'	'111'
Ch8R	Ch9R	Ch10R	Ch11R	Ch12R	Ch13R	Ch14R	Ch15R

B7.12.4 Direction

Bit 0 of the SET PL EXTENSIONS directive shall indicate if the fields in this directive apply to the transmit or receive side of the transceiver:

a) '0' = transmit side;

b) '1' = receive side.

B7.13 REPORT SOURCE SPACECRAFT ID

B7.13.1 General

The REPORT SOURCE SPACECRAFT ID is the mechanism by which the local transceiver can provide status of its source spacecraft ID to the remote transceiver across the link. It shall consist of three fields, positioned contiguously in the following sequence (described from least significant bit, Bit 15, to most significant bit, Bit 0):

a) Directive Type (3 bits);

b) Reserved (3 bits);

c) Source Spacecraft ID (10 bits).

NOTE – The structural components of the REPORT SOURCE SPACECRAFT ID are shown in figure B-8.

Bit 0		Bit 15
Source Spacecraft ID 10 bits	Reserved 3 bits	Directive Type 3 bits
0,1,2,3,4,5,6,7,8,9	10,11,12	13,14,15

Figure B-8: Report Source Spacecraft ID

B7.13.2 Directive Type

B7.13.2.1 Bits 13–15 of the REPORT SOURCE SPACECRAFT ID status report shall contain the Directive Type.

B7.13.2.2 The 3-bit Directive Type field identifies the type of status report and shall contain the binary value ‘111’.

B7.13.3 Reserved

Bits 10–12 of the REPORT SOURCE SPACECRAFT ID status report shall contain reserved bits, set to ‘all zero’.

B7.13.4 Source Spacecraft ID

Bits 0–9 of the REPORT SOURCE SPACECRAFT ID status report shall contain the SCID of the source of the Transfer Frame.

ANNEX C

TYPE 2 SPDU: TIME DISTRIBUTION SPDU DATA FIELD

(NORMATIVE)

C1 GENERAL

C1.1 The TIME DISTRIBUTION SPDU data field is the container that describes both the type and value of the time entity for distribution.

C1.2 A single TIME DISTRIBUTION directive shall be contained within a TIME DISTRIBUTION SPDU.

C1.3 The format of the TIME DISTRIBUTION SPDU data field shall consist of four fields, positioned contiguously, in the following sequence:

a) TIME DISTRIBUTION Directive Type (1 octet);

b) Transceiver Clock (8 octets);

c) Send Side Delay (3 octets);

d) One-Way-Light-Time (3 octets).

C1.4 All time code fields in this directive shall comply with the CCSDS Unsegmented Time Code format (reference [3]).

NOTE – The structural components of the TIME DISTRIBUTION SPDU data field are shown in figure C-1.

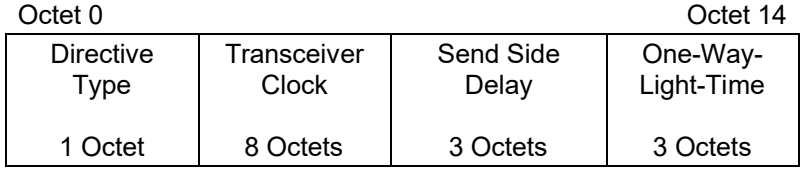


Figure C-1: Type 2 SPDU Data Field Contents

C2 TIME DISTRIBUTION DIRECTIVE

C2.1 GENERAL

C2.1.1 Octet 0 of the TIME DISTRIBUTION SPDU data field shall contain the TIME DISTRIBUTION Directive Type field indicating the function to be performed (if any) with the time contents.

C2.1.2 TIME DISTRIBUTION Types are:

a) '00000000' = NULL;

b) '00000001' = TIME TRANSFER;

c) all others = Reserved for CCSDS use.

C2.2 TRANSCEIVER CLOCK

C2.2.1 When the Time Distribution Type equals TIME TRANSFER,

a) octet 1 through octet 8 shall contain the value of the clock corresponding to when the trailing edge of the last bit of the ASM of the transmitted PLTU crosses the clock capture point within the transceiver;

b) this time code field shall be divided into 5 octets of coarse time and 3 octets of fine time. (See reference [3].)

C2.2.2 Otherwise, this field shall contain reserved bits, set to 'all zero'.

C2.3 SEND SIDE DELAY

C2.3.1 When the Time Distribution Type equals TIME TRANSFER,

a) octet 9 through octet 11 shall contain the delay time between the transceiver internal clock capture point and when the trailing edge of the last bit of the Sync-Marked Transfer Frame (SMTF) crossed the time reference point;

b) this time code field shall be divided into 1 octet of coarse time and 2 octets of fine time. (See reference [3].)

C2.3.2 Otherwise, this field shall contain reserved bits, set to 'all zero'.

C2.4 ONE WAY LIGHT TIME

C2.4.1 When the Time Distribution Type equals TIME TRANSFER, and when the mission has decided OWLT should be used,

a) octet 12 through octet 14 shall contain the calculated one way light time between the instance that the trailing edge of the last bit of the ASM of the transmitted SMTF crosses the time reference point of the initiator node to the time reference point of the destination node;

b) this time code field shall be divided into 1 octet of coarse time and 2 octets of fine time. (See reference [3].)

C2.4.2 Otherwise, this field shall contain reserved bits, set to 'all zero'.

ANNEX D

TYPE 4 SPDU—FIRST GENERATION LUNAR USE

(NORMATIVE)

Directive Name														
'000' = Link Establishment & Control Size = 56 bits	Link Direction	Demand/Query	Query Response	RNMD	Token	Duplex/Simplex	Frequency	Polarization	Mod Index	Modulation	Spares	Coding	Inst. Link SNR	Symbol Rate
3	1	1	1	1	1	3	5	1	3	4	2	6	8	16
'001' = Report Request Size = 16 bits	PCID 0 - PLCW Request	PCID 1 - PLCW Request	Time Tag Sample Request					Status Report Request						
3	1	1	6					5						
'010' = Set V(R) Size = 16 bits	Reserved						Frame Sequence Number							
3	5						8							
'011' = Report Source Spacecraft ID Size = 32 bits	Reserved										Source SCID			
3	13										16			
'100' = Reserved														
3														
'101' = Reserved														
3														
'110' = Reserved														
3														
'111' = Reserved														
3														

Figure D-1: Type 4 SPDU Data Field Contents

D1 GENERAL

D1.1 Type 4 SPDU directives shall be used for space link supervisory configuration and control of the transceiver and its operation at S-band.

D1.2 The SPDU data field shall contain 0 to 15 octets of supervisory data.

D1.3 Each directive shall perform a specific function.

D1.4 Each directive shall not exceed bits (15 octets × 8 bits) in length and shall be self-identified by the value in the directive name field (contained in bits 0 through 2 of the directive).

D1.5 Concatenation of the directives shall be allowed without intervening bits within the data field of the Type 4 SPDU as long as the total SPDU data field length does not exceed 120 bits.

NOTE – **Error! Reference source not found.** shows the SPDU Type 4 directive contents. The size of each field in bits is given below the field names.

D2 LINK ESTABLISHMENT & CONTROL DIRECTIVE

D2.1 OVERVIEW

The LINK ESTABLISHMENT & CONTROL directive is used to initiate communication via hailing, then move on to a working channel, and allow for follow on link control changes between partnered transceivers. A *caller* transceiver is the initiator of the link establishment process and manager of negotiation (if required) of the link session. A *responder* transceiver typically receives link establishment parameters from the caller. The caller initiates communication between itself and a responder on a prearranged communications (hailing) channel with predefined controlling parameters. As necessary, the caller and responder may negotiate the controlling parameters for the session at some level between fully controlled (demand) and completely adaptive (negotiated by query). These roles are invariant throughout the link session.

To accomplish hailing, this directive is transmitted back-to-back within the same transfer frame to establish both the forward and return link initial communication conditions. Similarly, once the link is established, the same approach is used to move onto the selected working channel. This directive is equivalent to the SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS or SET PL EXTENSIONS directives for SPDU Type 1 applications.

D2.2 SPECIFICATION

D2.2.1 General

The LINK ESTABLISHMENT & CONTROL directive shall consist of 15 fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to least significant bit 55): (see Figure D-2)

a) Directive Name (3 bits);

b) Link Direction (1 bit);

c) Demand/Query (1 bit);

d) Query Response (1 bit);

e) Remote No More Data (RNMD) (1 bit);

f) Token (1 bit);

g) Duplex/Simplex (3 bits);

h) Frequency (5 bits);

i) Polarization (1 bit);

j) Modulation Index (3 bits);

k) Modulation (4 bits);

l) Spares (2 bits);

m) Coding (6 bits);

n) Instantaneous Link SNR (8 bits);

o) Symbol Rate (16 bits);

NOTE – The structural components of the directive LINK ESTABLISHMENT & CONTROL directive are shown in Figure D-2.

Directive Name	Link Direction	Demand/ Query	Query Response	RNMD	Token	Duplex/ Simplex	Frequency	Polarization	Mod Index	Modulation	Spares	Coding	Inst. Link SNR	Symbol Rate
3 bits	1 bit	1 bit	1 bit	1 bit	1 bit	3 bits	5 bits	1 bit	3 bits	4 bits	2 bit	6 bits	8 bits	16 bits
0–2	3	4	5	6	7	8–10	11-15	16	17-19	20–23	24-25	26–31	32–39	40–55

Figure D-2: LINK ESTABLISHMENT & CONTROL Directive

D2.2.2 Directive Name

D2.2.2.1 Bits 0–2 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Directive Name.

D2.2.2.2 The 3-bit Directive Name field identifies the name of SPDU Type 4 protocol control directive and shall contain the binary value ‘000’ for the LINK ESTABLISHMENT & CONTROL directive.

D2.2.3 Link Direction

Bit 3 of the LINK ESTABLISHMENT & CONTROL directive shall indicate the link direction (Forward or Return link).

a) ‘0’ = Return Link:

Return is the link direction in which the responder transmits and the caller receives (e.g., typically a telemetry link). This directive sets the responder’s transmitter parameters.

b) ‘1’ = Forward Link:

Forward is the link direction in which the caller transmits and the responder receives (e.g., typically a command link). The caller (i.e., link initiator) is the node that transmits the Hail Directives. This directive sets the responder’s receiver parameters.

D2.2.4 Demand/Query

Bit 4 of the LINK ESTABLISHMENT & CONTROL directive shall contain either a Demand (caller expects the responder to accept the parameters in this directive ‘as is’) or as a Query to negotiate these link parameters specified in this directive between the caller and the responder. Used in hailing, moving onto a working channel, link control, and link termination.

a) ‘0’ = Demand (Command);

b) ‘1’ = Query (Link Negotiation).

D2.2.5 Query Response

Bit 5 of the LINK ESTABLISHMENT & CONTROL directive shall indicate whether the demand or the query was accepted or rejected by the responder.

a) ‘0’ = ACK;

b) ‘1’ = NACK.

D2.2.6 Remote No More Data (RNMD)

Bit 6 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Remote No More Data (RNMD) field as follows:

a) ‘0’ = No Change;

b) ‘1’ = Remote Node has No More Data to Send (RNMD).

NOTE – This field plays a role in link termination. This field either notifies the caller that either there is no change in the responder’s data state (i.e., ignore this field) or notifies the caller that the responder has no more data to send, in which case, the link may be terminated when the responder locally has no more data to send.

D2.2.7 Token

Bit 7 of the LINK ESTABLISHMENT & CONTROL directive shall contain the value of the Token field as follows:

a) ‘0’ = No Change;

b) ‘1’ = Transmit.

NOTE – This field either notifies the responder that there is no change in who has permission to transmit (i.e., ignore this field) or commands the responder to the transmit state.

D2.2.8 Duplex/Simplex

Bits 8–10 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Duplex/Simplex field as follows:

a) ‘000’ = No Change;

b) ‘001’ = Full Duplex;

c) ‘010’ = Half Duplex;

d) ‘011’ = Simplex Transmit;

e) ‘100’ = Simplex Receive;

f) ‘101’ = Reserved;

g) ‘110’ = Reserved;

h) ‘111’ = Reserved.

NOTE – This field either notifies the responder that there is no change in the responder’s Duplex state (i.e., ignore this field) or notifies the responder to change the directionality of communication accordingly.

D2.2.9 Frequency

D2.2.9.1 General

Bits 11–15 of the LINK ESTABLISHMENT & CONTROL directive shall be used to set the frequency of the partnered transceiver to the desired value (either receive or transmit frequency based upon the value of the Link Direction field defined in D2.2.3).

D2.2.9.2 Forward Link (i.e., Forward direction defined by the caller which transmits the Hail, i.e., initiator/link manager)

In the context of the forward link, this 5-bit field shall define the receive frequency of the responder. Actual frequency assignments per channel are defined in the Proximity-1 Physical Layer (see reference [5]).

00000	00001	00010	00011	00100	00101	00110	00111	01000	01001	01010	01011	01100	01101	01110	01111	10000	10001	10010	10011	10100	10101	10110	10111	11000	11001	11010	11011	11100	11101	11110	11111
Ch 0F	Ch 1F	Ch 2F	Ch 3F	Ch 4F	Ch 5F	Ch 6F	Ch 7F	Ch 8F	Ch 9F	Ch 10F	Ch 11F	Ch 12F	Ch 13F	Ch 14F	Ch 15F	Ch 16F	Ch 17F	Ch 18F	Ch 19F	Ch 20F	Ch 21F	Ch 22F	Ch 23F	Ch 24F	Ch 25F	Ch 26F	Ch 27F	Ch 28F	Ch 29F	Ch 30F	Ch 31F

NOTE: Forward Hailing Channel 0 and 9 are identified in red text.

B3.2.9.3 Return Link (i.e., Return direction defined by the responder which receives the Hail)

In the context of the return link, this 5-bit field shall define the transmit frequency of the responder. Actual frequency assignments per channel are defined in the Proximity-1 Physical Layer (see reference [5]).

00000	00001	00010	00011	00100	00101	00110	00111	01000	01001	01010	01011	01100	01101	01110	01111	10000	10001	10010	10011	10100	10101	10110	10111	11000	11001	11010	11011	11100	11101	11110	11111
Ch0R	Ch1R	Ch2R	Ch3R	Ch4R	Ch5R	Ch6R	Ch7R	Ch8R	Ch9R	Ch10R	Ch11R	Ch12R	Ch13R	Ch14R	Ch15R	Ch16R	Ch17R	Ch18R	Ch19R	Ch20R	Ch21R	Ch22R	Ch23R	Ch24R	Ch25R	Ch26R	Ch27R	Ch28R	Ch29R	Ch30R	Ch31R

NOTE: Return Hailing Channel 0 and 9 are identified in red text.

D2.2.10 Polarization

Bit 16 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Polarization field as follows:

a) ‘0’ = Left Hand Circular Polarization;

b) ‘1’ = Right Hand Circular Polarization.

D2.2.11 Modulation Index

Bits 17–19 of the LINK ESTABLISHMENT & CONTROL directive shall set the modulation index based upon the values below.

a) ‘000’ = 0 rad/pk (No Modulation);

b) ‘001’ = 0.4 rad/pk;

c) ‘010’ = 0.6 rad/pk;

d) ‘011’ = 0.8 rad/pk;

e) ‘100’ = π/3 rad/pk (60 degrees);

f) ‘101’ = 1.15 rad/pk;

g) ‘110’ = 1.3 rad/pk;

h) ‘111’ = 1.4 rad/pk.

D2.2.12 Modulation

Bits 20–23 of the LINK ESTABLISHMENT & CONTROL directive shall contain the modulation options based upon the values below.

a) ‘0000’ = PCM/PM/Bi-phase-L (filtered);

b) ‘0001’ = GMSK;

c) ‘0010’ = RESERVED BY CCSDS;

d) ‘0011’ = RESERVED BY CCSDS;

e) ‘0100’ = RESERVED BY CCSDS;

f) ‘0101’ = RESERVED BY CCSDS;

g) ‘0110’ = RESERVED BY CCSDS;

h) ‘0111’ = RESERVED BY CCSDS;

i) ‘1000’ = RESERVED BY CCSDS;

j) ‘1001’ = RESERVED BY CCSDS;

k) ‘1010’ = RESERVED BY CCSDS;

l) ‘1011’ = RESERVED BY CCSDS;

m) ‘1100’ = RESERVED BY CCSDS;

n) ‘1101’ = RESERVED BY CCSDS;

o) ‘1110’ = RESERVED BY CCSDS;

p) ‘1111’ = RESERVED BY CCSDS.

D2.2.13 Spares

Bits 24-25 of the LINK ESTABLISHMENT & CONTROL directive shall contain 2 spare bits reserved by the CCSDS.

D2.2.14 Coding

Bits 26–31 of the LINK ESTABLISHMENT & CONTROL directive shall contain the following coding options:

a) ‘000000’ = Uncoded;

b) ‘000001’ = LDPC(2048,1024);

c) ‘000010’ = Reserved by CCSDS;

d) ‘000011’ = Reserved by CCSDS;

e) ‘000100’ = Reserved by CCSDS;

f) ‘000101’= LDPC(6144,4096);

g) ‘000110’= Reserved by CCSDS;

h) ‘000111’= Reserved by CCSDS;

i) ‘001000’= Reserved by CCSDS;

j) ‘001001’= Reserved by CCSDS;

k) ‘001010’ = LDPC(8160,7136);

l) ‘001011’ through ‘111111’ = Reserved by CCSDS.

D2.2.15 Instantaneous Link SNR

Bits 32–39 of the LINK ESTABLISHMENT & CONTROL directive shall contain the link Signal-to-Noise (SNR) ratio i.e., Eb/No in dB Hz. This value is an integer number.

D2.2.16 Symbol Rate

Bits 40-55 of the LINK ESTABLISHMENT & CONTROL directive shall indicate the symbol rate in symbols per second. This value is a binary16-bit whose format is specified by the IEEE 754 standard for half-precision floating point numbers i.e., 1 sign bit (always set positive), 5 bits exponent, and 10 bits mantissa/significand. When this number is multiplied by 2^16, then the range of applicable symbol rates supported are between 1/256 symbol/sec and 2^31 (4,292,870,144) symbols/sec, with a precision of 0.1%. The sign bit is always transmitted as a positive value and the “all ones” i.e., ‘11111’ exponent value is excluded.

D2.3 REPORT REQUEST DIRECTIVE

D2.3.1 General

The REPORT REQUEST directive is the mechanism by which on of (1) a status report, (2) time-tag sample collection, or (3) a PLCW per PCID can be requested of a Proximity-1 node. It shall consist of five fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to the least significant bit, Bit 15):

a) Directive Name (3 bits);

b) PCID 0 PLCW Request (1 bit);

c) PCID 1 PLCW Request (1 bit);

d) Time-Tag Sample Request (6 bits);

e) Status Report Request (5 bits).

NOTE – The structural components of the REPORT REQUEST directive are shown in Figure D-3.

Bit 0				Bit 15
Directive Name	PCID 0 PLCW Request	PCID 1 PLCW Request	Time-Tag Sample Request	Status Report Request
3 bits	1 bit	1 bit	6 bits	5 bits
0–2	3	4	5–10	11–15

Figure D-3: Report Request Directive

D2.3.2 Directive Name

D2.3.2.1 Bits 0–2 of the REPORT REQUEST directive shall contain the Directive Type.

D2.3.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘001’.

D2.3.3 Physical Channel ID 0 PLCW Report Request Field

Bit 3 of the REPORT REQUEST directive shall indicate whether a PLCW report for PCID 0 is required:

a) ‘1’ = PLCW report is needed for PCID 0; transmit this report on the same PCID that the report request arrived on.

b) ‘0’ = PLCW report is not required.

D2.3.4 Physical Channel ID 1 PLCW Report Request Field

Bit 4 of the REPORT REQUEST directive shall indicate whether a PLCW report for PCID 1 is required:

a) ‘1’ = PLCW report is needed for PCID 1; transmit this report on the same PCID that the report request arrived on.

b) ‘0’ = PLCW report is not required.

D2.3.5 Time-Tag Sample Request Field

Bits 5–10 of this directive shall contain the Time-tag Sample Request field. When this field is non-zero, it notifies the recipient to capture the time and frame sequence number (associated with the Proximity timing service; see reference [6], section 5) for the next *n* frames received, where *n* is the number of Proximity Transfer Frames contained within the Time Sample Field.

D2.3.6 Status Report Request

D2.3.6.1 The value contained in bits 11–15 of the REPORT REQUEST directive shall indicate the type of status report desired.

D2.3.6.2 If set to '00000', a status report is not required.

D2.3.6.3 The types of status reports are reserved for CCSDS use as SPDU Type 3 directives.

D2.4 SET V(R) DIRECTIVE

D2.4.1 General

The SET V(R) directive used in the COP-1 (Reference X) and COP-P (Reference Y) procedures shall consist of three fields, positioned contiguously in the following sequence (described from most significant bit, Bit 0 to the least significant bit, Bit 15):

a) Directive Name (3 bits);

b) Spare (5 bits);

c) Receiver Frame Sequence Number (SEQ_CTRL_FSN) (8 bits).

NOTE – The structural components of the SET V(R) directive are shown in figure E-4.

Bit 0 Bit 15

Directive Name 3 bits	Spare 5 bits	Receiver Frame Sequence Number SEQ_CTRL_FSN 8 bits
0–2	3–7	8–15

Figure D-4: SET V(R) Directive

D2.4.2 Directive Name

D2.4.2.1 Bits 0–2 of the SET V(R) directive shall contain the Directive Name.

D2.4.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value '010' to identify the SET V(R) directive.

D2.4.3 Spare

Bits 3–7 of the SET V(R) directive shall contain spare bits, set to 'all zero'.

D2.4.4 Receiver Frame Sequence Number

Bits 8–15 of the SET V(R) directive shall contain the value of the Frame Sequence Number (SEQ_CTRL_FSN) to which the receiving unit of the partnered transceiver is to be set.

D2.5 REPORT SOURCE SPACECRAFT ID DIRECTIVE

D2.5.1 Overview

The REPORT SOURCE SPACECRAFT ID is the mechanism by which the local transceiver can provide status of its source spacecraft ID to the remote transceiver across the Proximity link. This directive is provided, because the verification test of the spacecraft ID performed by the protocol is by default based upon the destination spacecraft ID. There is however an option to include a test of the source spacecraft ID in the protocol as well. This directive allows the caller to query the responder for its Source Spacecraft ID.

D2.5.2 General

The REPORT SOURCE SPACECRAFT ID directive shall consist of three fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to the least significant bit, Bit 31):

a) Directive Name (3 bits);

b) Reserved (13 bits);

c) Source Spacecraft ID (16 bits).

NOTE – The structural components of the REPORT SOURCE SPACECRAFT ID are shown in Figure D-5.

Bit 0 Bit 31

Directive Name	Reserved	Source Spacecraft ID
3 bits	13 bits	16 bits
0–2	3–15	16–31

Figure D-5: Report Source Spacecraft ID

D2.5.3 Directive Name

D2.5.3.1 Bits 0–2 of the REPORT SOURCE SPACECRAFT ID status report shall contain the Directive Name.

D2.5.3.2 The 3-bit Directive Name field identifies the name of the Report Source Spacecraft ID directive and shall contain the binary value ‘011’.

D2.5.4 Reserved

Bits 3–15 of the REPORT SOURCE SPACECRAFT ID status report shall contain reserved bits, set to ‘all zero’.

D2.5.5 Source Spacecraft ID

Bits 16–31 of the REPORT SOURCE SPACECRAFT ID directive shall contain the SCID of the source of the Transfer Frame. The version 3 SCID for Proximity-1 transfer frames is 10 bits long and the version 4 SCID for USLP transfer frames is 16 bits long.

ANNEX E

TYPE 5 SPDU—SECOND GENERATION LUNAR USE

(NORMATIVE)

Directive Name																			
'000' = Link Establishment & Control Size = 104 bits	Spare	Link Direction	Directive Function	RNMD	Token	Duplex/Simplex	Polarization	Coherent/Noncoherent	Spare	Modulation	Modulation Index	Spare	Coding	Instantaneous Link SNR	Transceiver Mode	AOS Frame Length	Symbol Rate	Frequency
3	1	1	3	1	1	3	1	1	1	4	3	1	5	8	3	16	16	32
'001' = Report Request Size = 16 bits	PCID 0 - PLCW Request	PCID 1 - PLCW Request	Time Tag Sample Request				Status Report Request											
3	1	1	6				5											
'010' = Set V(R) Size = 16 bits	Reserved					Frame Sequence Number												
3	5					8												
'011' = Report Source Spacecraft ID Size = 32 bits	Reserved									Source SCID								
3	13									16								
'100' = PN Ranging																		
3																		
'101' = Reserved																		
3																		
'110' = Reserved																		
3																		
'111' = Reserved																		
3																		

Figure E-1: Type 5 SPDU Data Field Contents **Editors – Please update figure E-1 to include the picture of the PN Ranging directive under '100' in this figure – copy figure E-6 into it. **

E1 GENERAL

E1.1 Type 5 SPDU directives shall be used for space link supervisory configuration and control of the transceiver and its operation at S-band.

E1.2 The SPDU data field shall contain 0 to 15 octets of supervisory data.

E1.3 Each directive shall perform a specific function.

E1.4 Each directive shall not exceed 120 bits (15 octets × 8 bits) in length and shall be self-identified by the value in the directive name field (contained in bits 0 through 2 of the directive).

E1.5 Concatenation of the directives shall be allowed without intervening bits within the data field of the Type 5 SPDU as long as the total SPDU data field length does not exceed 120 bits.

NOTE – Figure E-1 shows the SPDU Type 5 directive contents. The size of each field in bits is given below the field names.

E2 LINK ESTABLISHMENT & CONTROL (LEC) DIRECTIVE

E2.1 OVERVIEW

The LINK ESTABLISHMENT & CONTROL (LEC) directive is used to initiate communication via hailing, then move on to a working channel, and allow for follow on link control changes between partnered transceivers. A *caller* transceiver is the initiator of the link establishment process and manager of negotiation (if required) of the link session. A *responder* transceiver typically receives link establishment parameters from the caller. The caller initiates communication between itself and a responder on a prearranged communications (hailing) channel with predefined controlling parameters. As necessary, the caller and responder may negotiate the controlling parameters for the session at some level between fully controlled (demand) and completely adaptive (negotiated by query). These roles are invariant throughout the link session.

To accomplish hailing, this directive is transmitted back-to-back within the same transfer frame to establish both the forward and return link initial communication conditions. Similarly, once the link is established, the same approach is used to move onto the selected working channel. This directive is equivalent to the SET TRANSMITTER PARAMETERS and SET RECEIVER PARAMETERS or SET PL EXTENSIONS directives for SPDU Type 1 applications.

E2.2 SPECIFICATION

E2.2.1 General

The LINK ESTABLISHMENT & CONTROL (LEC) directive shall consist of 19 fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to least significant bit 103): (see figure E-2)

a) Directive Name (3 bits);

b) Spare (1 bit);

c) Link Direction (1 bit);

d) Direction Function (3 bits);

e) Remote No More Data (RNMD) (1 bit);

f) Token (1 bit);

g) Duplex/Simplex (3 bits);

h) Polarization (1 bit);

i) Coherent/Non-coherent (1 bit);

j) Spare (1 bit);

k) Modulation (4 bits);

l) Modulation Index (3 bits);

m) Spare (1 bit);

n) Coding (5 bits);

o) Instantaneous Link SNR (8 bits);

p) Transceiver Mode (3 bits);

q) AOS frame length (16 bits);

r) Symbol Rate (16 bits);

s) Frequency (32 bits).

NOTE – The structural components of the directive LINK ESTABLISHMENT & CONTROL (LEC) directive are shown in figure E-2.

Bit 0 Bit 103

Directive Name	Spare	Link Direction	Directive Function	RNMD	Token	Duplex/ Simplex	Polarization	Coherent/Non-Coherent	Spare	Modulation	Mod Index	Spare	Coding	Inst. Link SNR	Transceiver Mode	AOS Frame Length	Symbol Rate	Frequency
3 bits	1 bit	1 bit	3 bits	1 bit	1 bit	3 bits	1 bit	1 bit	1 bit	4 bits	3 bits	1 bit	5 bits	8 bits	3 bits	16 bits	16 bits	32 bits
0–2	3	4	5-7	8	9	10–12	13	14	15	16–19	20–22	23	24–28	29–36	37–39	40–55	56–71	72–103

Figure E-2: LINK ESTABLISHMENT & CONTROL (LEC) Directive

E2.2.2 Directive Name

E2.2.2.1 Bits 0–2 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Directive Name.

E2.2.2.2 The 3-bit Directive Name field identifies the name of SPDU Type 5 protocol control directive and shall contain the binary value ‘000’ for the LINK ESTABLISHMENT & CONTROL directive.

E2.2.3 Spare

Bit 3 of the LINK ESTABLISHMENT & CONTROL directive shall contain 1 spare bit reserved by the CCSDS and shall contain the binary value ‘0’.

E2.2.4 Link Direction

Bit 4 of the LINK ESTABLISHMENT & CONTROL directive shall indicate the link direction (Forward or Return link).

a) ‘0’ = Return Link:

   Return is the link direction in which the responder transmits and the caller receives (e.g., typically a telemetry link). This directive sets the responder’s transmitter parameters.

b) ‘1’ = Forward Link:

   Forward is the link direction in which the caller transmits and the responder receives (e.g., typically a command link). The caller (i.e., link initiator) is the node that transmits the Hail Directives. This directive sets the responder’s receiver parameters.

E2.2.5 Directive Function

Bits 5 through 7 of the LINK ESTABLISHMENT & CONTROL (LEC) directive shall define the function this directive carries out. The options are: a) Demand directive (caller expects the responder to accept the parameters in this directive 'as is'); b) Query directive to allow the caller and responder to negotiate these link parameters between themselves; c) Query Response directive to indicate whether the query response was accepted or rejected by the responder (for hailing)/partnered node(in general). d) Report directive to provide status of the LEC parameters to the partnered node; e) Request directive to ask that the LEC parameters be sent to the partnered node. The LEC is used in hailing, moving onto a working channel, link control during data operations, and link termination.

a) '000' = Demand (Command);

b) '001' = Query Request (Link Negotiation);

c) '010' = Query Response (Accepted by the responder/partnered node - ACK);

d) '011'= Query Response (Rejected by the responder/partnered node - NACK);

e) '100'= Report (LEC parameters);

f) '101' = Request (LEC parameters);

g) '110' = Reserved by CCSDS;

h) '111' = Reserved by CCSDS.

E2.2.6 Remote No More Data (RNMD)

Bit 8 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Remote No More Data (RNMD) field as follows:

a) '0' = No Change;

b) '1' = Remote Node has No More Data to Send (RNMD).

NOTE – This field plays a role in link termination. This field either notifies the caller that either there is no change in the responder's data state (i.e., ignore this field) or notifies the caller that the responder has no more data to send, in which case, the link may be terminated when the responder locally has no more data to send.

E2.2.7 Token

Bit 9 of the LINK ESTABLISHMENT & CONTROL directive shall contain the value of the Token field as follows:

a) '0' = No Change;

b) ‘1’ = Transmit.

NOTE – This field either notifies the responder that there is no change in who has permission to transmit (i.e., ignore this field) or commands the responder to the transmit state.

E2.2.8 Duplex/Simplex

Bits 10–12 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Duplex/Simplex field as follows:

a) ‘000’ = No Change;

b) ‘001’ = Full Duplex;

c) ‘010’ = Half Duplex;

d) ‘011’ = Simplex Transmit;

e) ‘100’ = Simplex Receive;

f) ‘101’ = Reserved;

g) ‘110’ = Reserved;

h) ‘111’ = Reserved.

NOTE – This field either notifies the responder that there is no change in the responder’s Duplex state (i.e., ignore this field) or notifies the responder to change the directionality of communication accordingly.

E2.2.9 Polarization

Bit 13 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Polarization field as follows:

a) ‘0’ = Left Hand Circular Polarization;

b) ‘1’ = Right Hand Circular Polarization.

E2.2.10 Coherent/Non-coherent

Bit 14 of the LINK ESTABLISHMENT & CONTROL directive shall contain the transceiver coherent or non-coherent option as per below:

a) ‘0’ = Coherent;

b) ‘1’ = Non-coherent.

E2.2.11 Spare

Bit 15 of the LINK ESTABLISHMENT & CONTROL directive shall contain 1 spare bit reserved by the CCSDS and shall contain the binary value '0'.

E2.2.12 Modulation

Bits 16–19 of the LINK ESTABLISHMENT & CONTROL directive shall contain the modulation options based upon the values below.

a) '0000' = PCM/PM/Bi-phase-L (filtered);
b) '0001' = GMSK;
c) '0010' = OQPSK (filtered);
d) '0011' = BPSK (filtered);
e) '0100' = PCM/PSK/PM;
f) '0101' = PCM/PM/NRZ-L (filtered);
g) '0110' = RESERVED BY CCSDS;
h) '0111' = RESERVED BY CCSDS;
i) '1000' = RESERVED BY CCSDS;
j) '1001' = RESERVED BY CCSDS;
k) '1010' = RESERVED BY CCSDS;
l) '1011' = RESERVED BY CCSDS;
m) '1100' = RESERVED BY CCSDS;
n) '1101' = RESERVED BY CCSDS;
o) '1110' = RESERVED BY CCSDS;
p) '1111' = RESERVED BY CCSDS.

NOTE: Only options a) and b) are supported by Proximity-1 Physical Layer reference [5].

E2.2.13 Modulation Index

Bits 20–22 of the LINK ESTABLISHMENT & CONTROL directive shall set the modulation index based upon the values below.

a) '000' = 0 rad/pk (No Modulation);
b) '001' = 0.4 rad/pk;
c) '010' = 0.6 rad/pk;

d) ‘011’ = 0.8 rad/pk;

e) ‘100’ = π/3 rad/pk (60 degrees);

f) ‘101’ = 1.15 rad/pk;

g) ‘110’ = 1.3 rad/pk;

h) ‘111’ = 1.4 rad/pk.

NOTE: When using suppressed carrier modulation, this field is ignored.

E2.2.14 Spare

Bit 23 of the LINK ESTABLISHMENT & CONTROL (LEC) directive shall contain 1 spare bit reserved by the CCSDS and shall contain the binary value ‘0’.

E2.2.15 Coding

Bits 24–28 of the LINK ESTABLISHMENT & CONTROL directive shall contain the following coding options:

a) ‘00000’ = Uncoded;

b) ‘00001’ = LDPC(2048,1024);

c) ‘00010’ = LDPC(8192,4096);

d) ‘00011’ = LDPC(32768, 16384);

e) ‘00100’ = Reserved by CCSDS;

f) ‘00101’= LDPC(6144,4096);

g) ‘00110’= LDPC(24576, 16384);

h) ‘00111’= Reserved by CCSDS;

i) ‘01000’= Reserved by CCSDS;

j) ‘01001’= LDPC(20480, 16384);

k) ‘01010’ = LDPC(8160,7136);

l) ‘01011’ = Convolutional Code(7,1/2);

m) ‘01100’ through ‘11111’ = Reserved by CCSDS.

NOTE: (1) Only options a, b, f, and k are supported by the Proximity-1 C&S sublayer, reference [4]. For options c, d, g, j, and l, see reference [8].

NOTE: (2) The uncoded option with suppressed carrier modulation and without randomization cannot guarantee sufficient bit transitions resulting in an unreliable link.

E2.2.16 Instantaneous Link SNR

Bits 29–36 of the LINK ESTABLISHMENT & CONTROL directive shall contain the link Signal-to-Noise Ratio (SNR), E_s/N_0, in dB. The unit of the step size is defined by the LINK_SNR_STEP managed parameter. See Annex F. This value is a 2s complement signed integer multiplied by the step size.

NOTE: If the step size is 0.1dB, 0x01 = 0.1dB, 0xff = -0.1dB, 0x81 = -12.7dB, 0x80 = unavailable or invalid.

E2.2.17 Transceiver Mode

Bits 37–39 of the LINK ESTABLISHMENT & CONTROL directive shall contain the Transceiver Mode options. This field identifies the data link layer protocol operating in the transceiver:

a) ‘000’ = Proximity-1;

b) ‘001’ = USLP;

c) ‘010’ = AOS Sliced;

d) ‘011’ = AOS Non-Sliced;

e) ‘100’ = Reserved by CCSDS;

f) ‘101’ = Reserved by CCSDS;

g) ‘110’ = Reserved by CCSDS;

h) ‘111’ = Reserved by CCSDS.

NOTE: See reference [8] for the two applications: 1) sliced (Stream of Sync-Marked AOS Transfer Frames and 2) non-sliced AOS frames.

E2.2.18 AOS Frame Length

Bits 40–55 of the LINK ESTABLISHMENT & CONTROL directive shall contain the 16-bit AOS Transfer Frame Length.

NOTE – Since AOS is a fixed-length transfer frame data link layer protocol, the partnered transceiver needs to know it in order to configure the link. Missions not implementing CCSDS 732.0 set this field to zero.

E2.2.19 Symbol Rate

E2.2.19.1 Bits 56–71 of the LINK ESTABLISHMENT & CONTROL directive shall indicate the symbol rate in symbols per second. This value is a binary16-bit whose format is specified by the IEEE 754 standard for half-precision floating point numbers i.e., 1 sign bit (always set positive), 5 bits exponent, and 10 bits mantissa/significand.

E2.2.19.2 To derive the value of this field, the symbol rate in symbols per second shall be divided by 2^{16} and then converted to IEEE 754 half-precision floating point.

E2.2.19.3 To derive the symbol rate value from the field value, the opposite operation shall be done; that is, the floating point value shall be multiplied by 2^{16}.

E2.2.19.4 The sign bit is always transmitted as a positive value and the 'all ones' that is, '11111' exponent value is excluded.

NOTES

1 Symbol Rate assignment must take into consideration the actual spectrum available.

2 The default hailing symbol rate in S-band is 2,000 symbols/s, which is LDPC (2048,1024) encoded.

3 Symbol rate values in the range from 1,000 symbols/s to 4,096,000 symbols/s are represented with a precision of 0.1% from the defined Proximity-1 channel symbol rates as measured at the output of the transmitter.

4 Minimum and maximum symbol rates are based upon the selected Proximity-1 Physical Layer [5] and Coding & Synchronization Sublayer [4] parameter options.

E2.2.20 Frequency

Bits 72–103 of the LINK ESTABLISHMENT & CONTROL directive shall be used to set the frequency of the partnered transceiver to the desired value (either receive or transmit frequency based upon the value of the Link Direction field defined in E2.2.4). The format of this value is a binary32-bit specified by the IEEE 754 standard for single precision floating point numbers, that is, 1 sign bit (always set positive), 8 bits exponent, and 24 bits mantissa/significand.

NOTE – Frequency channel assignments including hailing channels are defined in the Proximity-1 Physical Layer Blue Book (reference [5]).

E2.3 REPORT REQUEST DIRECTIVE

E2.3.1 General

The REPORT REQUEST directive is the mechanism by which on of (1) a status report, (2) time-tag sample collection, or (3) a PLCW per PCID can be requested of a Proximity-1 node. It shall consist of five fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to the least significant bit, Bit 15):

f) Directive Name (3 bits);

g) PCID 0 PLCW Request (1 bit);

h) PCID 1 PLCW Request (1 bit);

i) Time-Tag Sample Request (6 bits);

j) Status Report Request (5 bits).

NOTE – The structural components of the REPORT REQUEST directive are shown in figure E-3.

Bit 0 Bit 15

Directive Name	PCID 0 PLCW Request	PCID 1 PLCW Request	Time-Tag Sample Request	Status Report Request
3 bits	1 bit	1 bit	6 bits	5 bits
0–2	3	4	5–10	11–15

Figure E-3: Report Request Directive

E2.3.2 Directive Name

E2.3.2.1 Bits 0–2 of the REPORT REQUEST directive shall contain the Directive Type.

E2.3.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘001’.

E2.3.3 Physical Channel ID 0 PLCW Report Request Field

Bit 3 of the REPORT REQUEST directive shall indicate whether a PLCW report for PCID 0 is required:

c) ‘1’ = PLCW report is needed for PCID 0; transmit this report on the same PCID that the report request arrived on.

d) ‘0’ = PLCW report is not required.

E2.3.4 Physical Channel ID 1 PLCW Report Request Field

Bit 4 of the REPORT REQUEST directive shall indicate whether a PLCW report for PCID 1 is required:

c) ‘1’ = PLCW report is needed for PCID 1; transmit this report on the same PCID that the report request arrived on.

d) ‘0’ = PLCW report is not required.

E2.3.5 Time-Tag Sample Request Field

Bits 5–10 of this directive shall contain the Time-tag Sample Request field. When this field is non-zero, it notifies the recipient to capture the time and frame sequence number (associated with the Proximity timing service; see reference [6], section 5) for the next *n* frames received, where *n* is the number of Proximity Transfer Frames contained within the Time Sample Field.

E2.3.6 Status Report Request

E2.3.6.1 The value contained in bits 11–15 of the REPORT REQUEST directive shall indicate the type of status report desired.

E2.3.6.2 If set to ‘00000’, a status report is not required.

E2.3.6.3 The types of status reports are reserved for CCSDS use as SPDU Type 3 directives.

E2.4 SET V(R) DIRECTIVE

E2.4.1 General

The SET V(R) directive used in the COP-1 (Reference X) and COP-P (Reference Y) procedures shall consist of three fields, positioned contiguously in the following sequence (described from most significant bit, Bit 0 to the least significant bit, Bit 15):

d) Directive Name (3 bits);

e) Spare (5 bits);

f) Receiver Frame Sequence Number (SEQ_CTRL_FSN) (8 bits).

NOTE – The structural components of the SET V(R) directive are shown in figure E-4.

Bit 0 Bit 15

Directive Name 3 bits	Spare 5 bits	Receiver Frame Sequence Number SEQ_CTRL_FSN 8 bits
0–2	3–7	8–15

Figure E-4: SET V(R) Directive

E2.4.2 Directive Name

E2.4.2.1 Bits 0–2 of the SET V(R) directive shall contain the Directive Name.

E2.4.2.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘010’ to identify the SET V(R) directive.

E2.4.3 Spare

Bits 3–7 of the SET V(R) directive shall contain spare bits, set to ‘all zero’.

E2.4.4 Receiver Frame Sequence Number

Bits 8–15 of the SET V(R) directive shall contain the value of the Frame Sequence Number (SEQ_CTRL_FSN) to which the receiving unit of the partnered transceiver is to be set.

E2.5 REPORT SOURCE SPACECRAFT ID DIRECTIVE

E2.5.1 Overview

The REPORT SOURCE SPACECRAFT ID is the mechanism by which the local transceiver can provide status of its source spacecraft ID to the remote transceiver across the Proximity link. This directive is provided, because the verification test of the spacecraft ID performed by the protocol is by default based upon the destination spacecraft ID. There is however an option to include a test of the source spacecraft ID in the protocol as well. This directive allows the caller to query the responder for its Source Spacecraft ID.

E2.5.2 General

The REPORT SOURCE SPACECRAFT ID directive shall consist of three fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to the least significant bit, Bit 31):

d) Directive Name (3 bits);

e) Reserved (13 bits);

f) Source Spacecraft ID (16 bits).

NOTE – The structural components of the REPORT SOURCE SPACECRAFT ID are shown in figure E-5.

Bit 0 Bit 31

Directive Name	Reserved	Source Spacecraft ID
3 bits	13 bits	16 bits
0–2	3–15	16–31

Figure E-5: Report Source Spacecraft ID

E2.5.3 Directive Name

E2.5.3.1 Bits 0–2 of the REPORT SOURCE SPACECRAFT ID status report shall contain the Directive Name.

E2.5.3.2 The 3-bit Directive Name field identifies the name of the Report Source Spacecraft ID directive and shall contain the binary value ‘011’.

E2.5.4 Reserved

Bits 3–15 of the REPORT SOURCE SPACECRAFT ID status report shall contain reserved bits, set to ‘all zero’.

E2.5.5 Source Spacecraft ID

Bits 16–31 of the REPORT SOURCE SPACECRAFT ID directive shall contain the SCID of the source of the Transfer Frame. The version 3 SCID for Proximity-1 transfer frames is 10 bits long and the version 4 SCID for USLP transfer frames is 16 bits long.

E2.6 PN RANGING DIRECTIVE

E2.6.1 Overview

Below are some of the assumptions and requirements associated with the PN RANGING directive.

- The PN RANGING directive should support one-way (pseudorange) and two-way ranging.
- Only a single PN ranging code sequence will be specified for Prox-1 use
- Prox-1 has a TIME DISTRIBUTION directive which can be used for time correlation and transfer.
- The PN RANGING directive will follow the SPDU Type 5 format for second generation lunar use.

It is assumed that the link has already been established prior to sending the PN RANGING directive, i.e., the link direction, antenna polarization, frequency channel, etc. has been agreed upon between the two radios. To start the Prox-1 ranging session, the initiator sends the PN RANGING directive to establish the ranging parameters to be used with the responder. These parameters include chip rate, PN code type, PN ranging mode (coherent, non-coherent, regenerative, non-regenerative), ranging mod index, and the epoch time tag for the start of the PN sequence. The PN RANGING directive can also be used to request the status of the PN ranging configuration, calibration delay, or ranging observables from the responder prior or after the ranging session.

E2.6.2 General

The PN RANGING directive is the mechanism by which the PN ranging for a Proximity-1 node is initiated and configured. It can also be used to request a status report on the PN ranging configuration of a Proximity-1 node.

This directive assumes that the link between the nodes has already been established using the LINK ESTABLISHMENT AND CONTROL directive. As such, the forward and return directions and channel carrier frequencies have already been established for each node, and are not repeated as part of this directive.

The PN RANGING directive shall consist of six fields, positioned contiguously in the following sequence (described from the most significant bit, Bit 0, to the least significant bit):

a) Directive Name (3 bits);

b) PN Ranging Mode Type (2 bits);

c) Ranging Code (2 bits);

d) Chip Rate (31 bits);

e) Ranging Modulation Index (3 bits)

f) PN Epoch Time-Tag (48 bits);

g) Status Report Request (5 bits);

h) Spares (2 bits).

NOTE – The structural components of the PN RANGING directive are shown in figure E-.

Bit 0 Bit 95

Directive Name 3 bits	Mode Type 2 bits	Ranging Code 2 bits	Chip Rate 31 bits			Ranging Mod Index 3 bits	PN Epoch Time Tag 48 bits		Status Report Request 5 bits	Spares 2 bits
			k	*l*	m		Day	ms of day		
0-2	3-4	5-6	7-9	10-23	24-37	38-40	41-56	57-88	89-93	94-95

Figure E-6: PN RANGING Directive

E2.6.3 Directive Name

E2.6.3.1 Bits 0-2 of the PN RANGING directive shall contain the Directive Type.

E2.6.3.2 The 3-bit Directive Type field identifies the type of protocol control directive and shall contain the binary value ‘100’ for PN RANGING.

E2.6.4 Mode Type PN Ranging Field

Bits 3-4 of the PN RANGING directive shall indicate the PN ranging type:

e) ‘00’ = Ranging Off

f) ‘01’ = One-way Ranging (pseudorange)

g) ‘10’ = Two-way Non-Regenerative Ranging (turnaround ranging)

h) ‘11’ = Two-way Regenerative Ranging

E2.6.5 Ranging Code

Bits 5-6 of the PN RANGING directive shall indicate the PN ranging code type:

a) '00' = Maximum Length (PN18) code

b) '01' = T2B

c) '10' = T4B

d) '11' = Reserved

NOTE: Only option a is supported by the Proximity-1 Physical Layer reference [5]. For options b and c, reference [9] applies.

E2.6.6 Chip Rate PN Ranging Field

Bits 7-37 of the PN RANGING directive shall indicate the transmit PN Chip rate. The chip rate is dependent on the forward link carrier frequency, and is defined by three parameters (*l*, *k*, and *m*). Bits 7-9 indicate the value of *k*, and Bits 10-23 indicate the valve of *l,* and Bits 24-37 indicate the value of *m*. The formula to calculate the S-band chip rate is shown below:

$$F_{chip} = 2\,F_{clock} = \frac{m}{l} \cdot \frac{f_{s-band}}{128 \cdot 2^k}$$

Ideally, the values of *l*, *k*, and *m* should be chosen such that multiples of the PN code sequence align with the second boundary, and the forward and return coherent link frequencies align closely with the nominal Prox-1 channel center frequencies. This can be accomplished by choosing values of *l* and *m* using factors of the PN code length, while *k* is selected to increase or decrease the chip rate by factors of 2.

Table 1 shows the recommended values for *k, l,* and *m* for an assumed maximum length PN ranging code of 2^18-1 = 262143 chips. For this PN code length, *m* is recommended to be 9709 (= 7 x 19 x 73) and the corresponding values of *l* and *k* for the frequency channels 1-8 are shown in the table. Other values for *m*, *l*, and *k* are possible depending on the ranging needs.

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Table 1. Recommended Values for *m*, *l*, and *k*

Channel No.	***$f_{s\text{-}band}$ (Forward Link)***	***f_{return} (Return Link)***	***m***	***l***	***k***	***F_{chip} (kchips/s)***	***PN Code Period (sec)***
1	2085.765120	2265.084293	9709	9430	6	262.143	1
					5	524.286	0.5
					4	1048.572	0.25
					3	2097.144	0.125
2	2086.649856	2266.045092	9709	9434	6	262.143	1
					5	524.286	0.5
					4	1048.572	0.25
					3	2097.144	0.125
3	2087.534592	2267.005892	9709	9438	6	262.143	1
					5	524.286	0.5
					4	1048.572	0.25
					3	2097.144	0.125
4	2088.419328	2267.966691	9709	9442	6	262.143	1
					5	524.286	0.5
					4	1048.572	0.25
					3	2097.144	0.125
5	2095.718400	2275.893285	9709	9475	6	262.143	1
					5	524.286	0.5

					4 3	1048.572 2097.144	0.25 0.125
6	2096.824320	2277.094284	9709	9480	6 5 4 3	262.143 524.286 1048.572 2097.144	1 0.5 0.25 0.125
7	2097.709056	2278.055083	9709	9484	6 5 4 3	262.143 524.286 1048.572 2097.144	1 0.5 0.25 0.125
8	2098.593792	2279.015883	9709	9488	6 5 4 3	262.143 524.286 1048.572 2097.144	1 0.5 0.25 0.125

E2.6.7 Ranging Modulation Index PN Ranging Field

Bits 38-40 of the PN RANGING directive shall indicate the PN ranging modulation index.

a) ‘000’ = 8.75 degrees
b) ‘001’ = 17.5 degrees
c) ‘010’ = 35 degrees
d) ‘011’ = 45 degrees
e) ‘100’ = 60 degrees
f) ‘101’ = 70 degrees
g) ‘110’ = Reserved

h) ‘111’ = Reserved

E2.6.8 Epoch Time PN Ranging Field

E2.6.8.1 The value contained in bits 41–88 of the PN RANGING directive shall indicate the epoch of the PN ranging code. The epoch is the time tag of when the leading edge of first chip of the transmit PN sequence crosses the clock capture point (defined by the implementation) within the transceiver. Similar to the Proximity-1 timing services defined in CCSDS 211.0-B-6, the reference point for all timing calculation shall be defined by the enterprise.

E2.6.8.2 The PN range code epoch shall be transmitted using the CCSDS Day Segmented (CDS) time code format defined in CCSDS 301.0-B-4. Bits 39-54 presents the number of days from 1958 January 1 starting with 0. Bits 55-86 represent the milliseconds of the day. Ideally the start of the PN range code will align with an integer number of seconds. Since this time code format is UTC-based, the leap second correction must be made.

E2.6.9 Status Report Request

E2.6.9.1 The value contained in bits 89–93 of the PN RANGING directive shall indicate the type of ranging status report desired.

a) ‘00000’ = No status report is required;

b) ‘00001’ = Ranging configuration report;

c) ‘00010’ = Ranging delay calibration report;

d) ‘00011’ = Position, Velocity, and Time report;

e) ‘00100’ = Ranging observable report;

f) Other values = Reserved.

E2.6.9.2 The types of status reports are reserved for CCSDS use as SPDU Type 3 directives.

E2.6.10 Spares

The value contained in bits 94-95 of the PN RANGING directive shall be reserved by the CCSDS and set to ‘00’.

ANNEX F

MANAGEMENT INFORMATION BASE PARAMETERS

(NORMATIVE)

This table lists each MIB parameter in the document along with how it is used and in what layer or sublayer it is used. Values for the Layer column are: P = Physical, C = C&S, F = Frame, M = MAC, D = Data Services, I = I/O, U = USLP. Parameter Definitions are provided where they are referenced in the specification.

Parameter	Use	Layer
Acquisition_Idle_Duration	Mandatory. Used in full-duplex, half-duplex, and simplex session establishment and COMM_CHANGE. Session static. (See 5.1.3.4.)	P, M
Carrier_Loss_Timer_Duration	Mandatory. Used in full- and half-duplex operations. Session static. (See 5.1.3.6.)	D
Carrier_Only_Duration	Mandatory. Used in full-duplex, half-duplex, and simplex session establishment and COMM_CHANGE. Session static. (See 5.1.3.3.)	P, M
Comm_Change_Lifetime	Mandatory. Used in the COMM_CHANGE persistent activity. Session static. (See 5.1.3.10.)	M
Comm_Change_Notification	Mandatory. Used in the COMM_CHANGE persistent activity. Session static. (See 5.1.3.9.)	M
Comm_Change_Response	Mandatory. Used in the COMM_CHANGE persistent activity. Session static. (See 5.1.3.8.)	M
Comm_Change_Waiting_Period	Mandatory. Used in the COMM_CHANGE persistent activity. Session static. (See 5.1.3.7.)	M
Hail_Lifetime	Mandatory. Used in the hailing persistent activity. Session static. (See 5.1.3.14.)	M
Hail_Notification	Mandatory. Used in the hailing persistent activity. Session static. (See 5.1.3.13.)	M

Parameter	Use	Layer
Hail_Response	Mandatory. Used in the hailing persistent activity. Session static. (See 5.1.3.12.)	M
Hail_Wait_Duration	Mandatory. Used in the hailing persistent activity. Session static. (See 5.1.3.11.)	M
Hailing_Channel	Mandatory. Channel assignment used in the hailing persistent activity during link establishment. Session static. (See 5.1.3.15.)	P, M
Hailing_Data_Rate	Mandatory. Data rate used in the hailing persistent activity during link establishment. Session static. (See 5.1.3.15.)	P, M
Hailing_Symbol_Rate	Mandatory. Symbol rate used in the hailing persistent activity during link establishment. Session static. (See 5.1.3.15.)	P, M
Interval_Clock	Mandatory. A frequency (e.g., 100 Hz) that is used for interval timing. Session static. (See reference [4], subsection 5.2.)	C
Link_SNR_Step	Mandatory. Used to establish the step size in fractions of a dB reporting the instantaneous link SNR by the radio.	P
Local_PCID	Optional. Used to set the value of the local receiver's PCID.	M
Local_Spacecraft_ID	Mandatory. Used as a frame validation check when Source-or-Destination ID equals *source*. Session static. (See reference [6] subsection 3.2.2.9.3.)	M
Maximum_Failed_Token_Passes	Optional. Half-duplex. Defines the maximum number of times the transceiver is allowed to cycle through S50->S56->S58->S62->E50->S51->S52->S50 before E83 is triggered forcing the transceiver into S80 (reconnect).	D
Frame Version in use	3 (Proximity-1) or 4 (USLP)	F, U

Parameter	Use	Layer
Maximum_Frame_Length	Mandatory. Defines the maximum size Transfer Frame transferred between nodes. Link efficiency at various data rates may require varying frame lengths. (See reference [6] subsection 3.2.3.1 for Version-3 Transfer Frames and reference [7], subsection 5.2 for Version-4 Transfer frames.) The maximum value allowed for this parameter depends on the Frame Version in use.	F, U
Maximum_Packet_Size	Mandatory if packets are used. Maximum size of a packet in octets. Used in the segmentation process. Session static. (See reference [6] subsection 4.4.2.1.)	I
Persistence_Wait_Time	Mandatory. Defines the maximum amount of time the initiating transceiver stays in persistence until either (1) it receives an acknowledgement from the remote transceiver that the COMM_CHANGE was acted upon, or (2) the wait timer times out. (See table 5-7, 'Full Duplex Communication Change State Table'.)	M
PLCW_Repeat_Interval	Mandatory. Used in COP-P. Session static. (See 5.1.3.19.)	D
Receive_Duration	Mandatory. Used in half-duplex data services. Session static. (See 5.1.3.18.)	D
Drop_Carrier_Duration	Mandatory. Used in full- and half-duplex data services. The caller drops the carrier for this defined amount of time. Used to force the responder to lose carrier lock and therefore transition to State S2: Waiting for Hail. Session static. (See 5.3.2.)	D
Remote_PCID	Mandatory. Used to identify the powered receiver used by the remote transceiver.	M
Remote_Spacecraft_ID	Mandatory. Used to address one or several remote spacecraft, as opposed to the local spacecraft. Session dynamic. (See reference [6] subsection 3.2.2.9.3.)	F, M, D, I

Parameter	Use	Layer
Resync_Local	Mandatory. If Resync_Local equals *false*, it is the responsibility of the local controller to decide how synchronization will be re-established. Otherwise, if *true*, the Sender Node's FOP-P forces synchronization by requesting a SET V(R) persistent activity. Session static. (See 6.2.3.2, 'SET V(R) Persistent Activity'.)	D
Resync_Lifetime	Mandatory. Used in the FOP-P SET V(R) persistent activity. Session static. (See 6.2.3.2.)	M, D
Resync_Notification	Mandatory. Used in the FOP-P SET V(R) persistent activity. Session static. (See 6.2.3.2)	M, D
Resync_Response	Mandatory. Used in the FOP-P SET V(R) persistent activity. Session static. (See 6.2.3.2.)	M, D
Resync_Waiting_Period	Mandatory. Used in the FOP-P SET V(R) persistent activity. Session static. (See 6.2.3.2.)	M, D
Send_Duration	Mandatory. Used in half-duplex data services. Session static. (See 5.1.3.17.)	D
Source_Destination_ID	Mandatory. '0'= Source SCID; '1'=Destination SCID. Session Static. (See reference [6] subsection 3.2.2.9.)	M
Synch_Timeout	Mandatory. Defines the value to which the SYNCH_TIMER is initialized or reinitialized. Session static. (See 6.2.2.)	D
Tail_Idle_Duration	Mandatory. Used in full-duplex, half-duplex, and simplex session establishment and COMM_CHANGE. Session static. (See 5.1.3.5.)	P, M
Test_Source	Mandatory. Used in the verification of the spacecraft ID when the Source-or-Destination ID is *source*. Session static. (See 5.1.3.2.)	F

Parameter	Use	Layer
Transmission_Window	Mandatory. Sets the maximum size of the transmission window for the COP-P. Session static. (See 6.2.3.3, note 3.)	D

ANNEX G

NOTIFICATIONS TO VEHICLE CONTROLLER

(NORMATIVE)

This table summarizes all of the conditions throughout the document under which the vehicle controller is notified from within the protocol.

Number	Condition	Reference
1	RESULT OF PERSISTENT ACTIVITY Notification of the success or failure of a persistent activity.	(See 4, 'Persistence'.)
2	STATE CONTROL STATUS Status of the State Control Variables.	(See state tables contained in section 5.)
3	INVALID FRAME SOURCE When the SCID field and RECEIVING_SCID_BUFFER disagree, and Test_Source is *true*, then a session violation has occurred and the vehicle controller shall be notified.	(See 5.6, 'RECEIVING OPERATIONS'.)
4	TIMING SERVICES INSTANCE At the end of receiving the SET CONTROL PARAMETERS (*time sample*) directives, the recipient transceiver notifies its vehicle controller that time tags and frame sequence numbers are available.	(See reference [6] section 5, 'Proximity-1 Timing Services'.)
5	NO CARRIER RECEIVED—HALF DUPLEX	State table 5-9, Event 50.
6	NO DATA TRANSFERRED—HALF DUPLEX	State table 5-9, Event 45, E50
7	SENDER EXCEEDED PRESCRIBED TRANSMISSION PERIOD—HALF DUPLEX	State table 5-9, Event 44.
8	COP-P LOSS OF SYNCHRONIZATION When FOP-P detects out-of-synchronization condition (SYNCH_TIMER Expires).	(See 6.2.3.3, 'FOP-P State Table', Event SE4.)
9	CARRIER_LOSS_TIMER UNDERFLOWS	State table 5-8, Event 27.

Number	Condition	Reference
10	END OF SESSION (# octets received) Caller notifies the vehicle controller of the number of octets received during the session.	State table 5-8, Events E26, E27, E28. State table 5-11, Events E57, E58, E61. State table 5-12, Event E73.
11	CARRIER ONLY RECEIVED Caller notifies vehicle controller that carrier only was detected immediately after the 'Wait for Hail Response' state terminated.	State table 5-6, Event E8. State table 5-9, Event E36.

ANNEX H

DEFAULT HAILING PARAMETERS FOR SPACE ENTERPRISES

(NORMATIVE)

H1 OVERVIEW

This normative annex documents the default hailing (session establishment) parameters used for hailing a partnered transceiver for specific Space Enterprises, for example, Mars, Lunar. This annex provides the parameter values to carry out both a demand and a negotiated hail.

H2 HAILING CHANNEL PARAMETERS

H2.1 UHF-BAND HAILING CHANNEL PARAMETERS

Hailing-channel parameters for UHF-Band (used at Mars) shall be:

- Hailing Channel Number: Channel 1 forward and return link;[1]
- Hailing Symbol Rate: 8,000 symbols/second
- Coding: Uncoded
- Modulation: Bi-Phase-L
- Polarization: Right Hand Circular
- Transceiver Mode: Proximity-1
- Coherency: Non-coherent

H2.2 S BAND HAILING CHANNEL PARAMETERS

Hailing-channel parameters for S Band (used at the Moon) shall be:

- Default Hailing Channel Number: 0;[2]
- Optional Hailing Channel Number: 9;[2]
- Hailing Coded Symbol Rate: 2000 symbols/second TBD;

[1] Hailing is performed between transceivers that are pre-configured. Therefore, it is nominally performed on the hailing channel. However, if transceivers are compatibly configured, hailing can occur on an agreed-to channel. Several first generation transceivers in the Mars Enterprise at UHF are fixed frequency and use Channel 0 instead.

[2] Specific frequency channel assignments are defined in the Proximity-1 Space Link Protocol - Physical Layer (reference [5]).

- Coding: LDPC (n=2048,k=1024) rate 1/2 code defined in reference [4];
- Modulation: Bi-phase-L;
- Polarization:
  - Left Hand Circular for Default Hailing Channel 0;
  - Right Hand Circular for Optional Hailing Channel 9;
- Transceiver Mode: USLP;
- Coherency: Non-coherent;

A1.1 KA BAND HAILING CHANNEL PARAMETERS

Hailing-channel parameters for KA Band (used at the Moon) shall be:

- [TBD].

ANNEX I

SECURITY, SANA, AND PATENT CONSIDERATIONS

(INFORMATIVE)

I1 SECURITY CONSIDERATIONS

I1.1 SECURITY BACKGROUND

It may be required that security services be applied to the payloads carried by IP datagrams over CCSDS space links. The specification of such security services is outside the scope of this document but is discussed in the following subsections.

If there is a reason to believe that non-authorized entities might be able to view or obtain the payload data, and if there is a need to ensure that non-authorized entities not be able to view or obtain the data, then confidentiality needs to be applied. If there is a need to ensure that the payload data has not been modified in transit without such modification being recognized, then integrity needs to be applied. If the authenticity of the source of the payload data is required (e.g., the payload contains a command), then authentication needs to be applied. It is possible for a single datagram to require all three security services to ensure that the payload is not disclosed, not altered, and authentic.

I1.2 SECURITY CONCERNS

As stated in the previous subsection, various security services might need to be applied to the IP datagram depending on the threat, the mission security policy (or policies), and the desire of the mission planners. While these security concerns are valid, they are outside the scope of this document. This document assumes that either upper or lower layers of the OSI model will provide the security services. That is, if authenticity at the granularity of a specific user is required, it is best applied at the Application Layer. If less granularity is required, it could be applied at the Network or Data Link Layers. If integrity is required, it can be applied at either the Application, Network, or Data Link Layer. If confidentiality is required, it can be applied at either the Application Layer, the Network Layer, or the Data Link Layer. Reference [J4] provides more information regarding the choice of service and where it can be implemented.

I1.3 POTENTIAL THREATS AND ATTACK SCENARIOS

Without authentication, unauthorized commands or software might be uploaded to a spacecraft or data retrieved from a source masquerading as the spacecraft. Without integrity, corrupted commands or software might be uploaded to a spacecraft. Without integrity, corrupted telemetry might be retrieved from a spacecraft, and the result could be that an incorrect course of action is taken. Sensitive or private information might be disclosed to an eavesdropper if confidentiality is not applied to the data.

I1.4 CONSEQUENCES OF NOT APPLYING SECURITY

The security services are out of scope of this document and should be applied at layers above or below those specified in this document. However, should there be a requirement for authentication, and if it is not implemented, unauthorized commands or software might be loaded onto a spacecraft. If integrity is not implemented, erroneous commands or software might be loaded onto a spacecraft, potentially resulting in the loss of the mission. If confidentiality is not implemented, data flowing to or from a spacecraft might be visible to unauthorized entities, resulting in disclosure of sensitive or private information.

I2 SANA CONSIDERATION

The current issue of this Recommended Standard does not require any action from SANA. Existing SANA registries created in support of previous issues of this Recommended Standard should continue to be maintained.

I3 PATENT CONSIDERATIONS

No patents are known to apply to this Recommended Standard. Information concerning patent rights and licensing for LDPC coding is contained in annex B of reference [2].

ANNEX J

INFORMATIVE REFERENCES

(INFORMATIVE)

NOTE – References [J1]-[J3] define Physical Layer techniques that are not part of the Proximity-1 Physical Layer specification. They are included here so that transceivers with an extended Physical Layer can interoperate.

[J1] *Digital Video Broadcasting (DVB); Framing Structure, Channel Coding and Modulation for 11/12 GHz Satellite Services*. ETSI EN 300 421 V1.1.2 (1997-08). Sophia-Antipolis: ETSI, 1997.

[J2] *A 48/56/64 kbit/s Data Circuit-Terminating Equipment Standardized for Use on Digital Point-to-Point Leased Circuits*. ITU-T Recommendation V.38. Geneva: ITU, 1996.

[J3] *Performance Characteristics for Intermediate Data Rate Digital Carriers Using Convolutional Encoding/Viterbi Encoding*. Rev. 10. IESS 308. Washington, DC: INTELSAT, 2000.

[J4] *The Application of Security to CCSDS Protocols*. Issue 3. Report Concerning Space Data System Standards (Green Book), CCSDS 350.0-G-3. Washington, D.C.: CCSDS, March 2019.

[J5] *Proximity-1 Space Link Protocol—Rationale, Architecture, and Scenarios*. Issue 2. Report Concerning Space Data System Standards (Green Book), CCSDS 210.0-G-2. Washington, D.C.: CCSDS, December 2013.

ANNEX K

ABBREVIATIONS AND ACRONYMS

(INFORMATIVE)

Term	Meaning
ASM	attached synchronization marker
C&DH	command and data handling
C&S	coding and synchronization
CCSDS	Consultative Committee for Space Data Systems
COP-P	communication operations procedure-P
CRC	cyclic redundancy check
ETSI	European Telecommunications Standards Institute
FARM-P	frame acceptance and reporting mechanism-P
FIFO	first in, first out
FOP-P	frame operations procedure-P
FSK	frequency shift keying
FSN	frame sequence number
I/O	input/output
ITU	International Telecommunications Union
LCCD	local COMM_CHANGE directive
LDPC	low density parity check
LNMD	LOCAL_NO_MORE_DATA
MAC	medium access control
MIB	management information base
MSB	most significant bit
NRZ-L	non-return to zero-level
OSI	Open Systems Interconnection
OWLT	one way light time

PCID	physical channel ID
PDU	protocol data unit
P-frame	supervisory/protocol frame
PLCW	protocol link control word
PSK	phase shift keyed
PVN	packet version number
QoS	quality of service
QPSK	quadrature phase shift keying
RF	radio frequency
R-S	Reed-Solomon
RNMD	REMOTE_NO_MORE_DATA
RSL	received signal level
RX	receive/receiver
SANA	Space Assigned Numbers Authority
SCID	spacecraft identifier
SCPS-NP	Space Communications Protocol Standards-Network Protocol
SDU	service data unit
SMTF	Sync-Marked Transfer Frame
SNR	signal to noise ratio
SPDU	supervisory protocol data unit
SS	sub-state
TX	transmit/transmitter
U-frame	user data frame
USLP	Unified Space Data Link Protocol